

Angel Prime:

Causal-State Arithmetic, High-Dimensional Functors, Factorial Mazes, Cyclic Closure, and Residual Trans- mission

Expanded Formal Mathematical Annals Edition

Central statement. Angel arithmetic has a conservative three-class exact model with typed high-dimensional $+$, $-$, $\times$, $/$, causal functors, and class-quantum history operations. Its principal-jet factorial chart yields $(-1)^n n!$, supports a lawful Wilson observation, admits a cyclic closure theory, and extends to a verifier-backed source-first residual horizon transition.

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Technical Virtual Broadcaster

Angel Prime Generator Project

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Formal annals revision; difficult proofs and implementation evidence are collected in appendices

Abstract

We first separate the authorial origin of the name *Angel*, the SN–OFCS singular-number precursor, and the present proved reconstruction. The forgotten originating formula is not claimed to have been recovered verbatim, and no recollection is used as a mathematical premise. We then give the axiomatic foundation omitted from the shorter edition. A complete Angel maze value has ordinary, primitive-history, and singular coordinates together with room, road, frame, causal history, holonomy, residual, higher-cell, certificate, and operator-program data. We define high-dimensional addition, subtraction, ordered multiplication, and totalized division; prove a conservative unitized model; construct the native execution, history, and future-quotient functors; and define exact class-quantum superposition, tensor entanglement, structural interference, and cancellation. These are finite integer-weight history operations, not claims about physical quantum mechanics.

The author’s “fold multiplication” intuition is then reconstructed as an ordered interval-tree theory. We prove value invariance with history nonuniqueness, the q-Pochhammer interval-fold and contraction-functor laws, nodewise factorial-jet composition, and a five-ledger separation of tree depth, expanded work, compact grammar, arithmetic cost and output. At a prime room the cyclic row fold and factorial principal jet collide at the same residue -1 , but their p and $p - 1$ carriers remain explicitly distinct.

We then construct the factorial maze chart explicitly. A certified principal-jet state is embedded into Ord/Hist/Sing, and a specialized high-to-high functor binds its argument, source program, request, native result and certificate before either exact or modular observation. Thus the factorial application is a theorem-level instance of the preceding axioms, not a second paper attached to them. An integer expression is represented by a proof-bound causal/algebraic state and ordinary integers appear only across an explicit observation boundary. The construction begins with the q-Pochhammer family

$$F_n(Z, u) = \prod_{j=0}^{n-1} (1 + Zu^j),$$

whose concatenation law gives a binary straight-line description of length $O(\log n)$. At the integral local coordinate $u = 1 + T$, $Z = -(1 + T)$ we prove

$$F_n(-(1 + T), 1 + T) = (-1)^n n! T^n + O(T^{n+1}).$$

Thus the principal jet recovers the factorial coefficient. We define factorial intervals and prove contiguous composition, exact prefix and suffix cancellation, reflexive interference, and higher-jet continuation.

For $n = m - 1$ we then prove the complement-pairing identity, valid for every modulus $m \geq 2$,

$$n! \equiv \begin{cases} (-1)^h (h!)^2, & m - 1 \text{ even,} \\ (-1)^h (h!)^2 (h + 1), & m - 1 \text{ odd,} \end{cases} \quad h = \left\lfloor \frac{m - 1}{2} \right\rfloor.$$

Combined with Wilson’s theorem, this gives an exact ordinary primality observation after the factorial state has been fixed. A composite-safe block coordinate reduces the polynomial problem from n to h . Under a generic polynomial multiplication cost M , the legacy and paired schedules have

$$T_0 = O(M(\sqrt{n}) \log n + \sqrt{n}), \quad S_0 = O(\sqrt{n} \log n), \\ T_1 = O(M(\sqrt{n/2}) \log n + \sqrt{n/2}) + O(1), \quad S_1 = O(\sqrt{n/2} \log n).$$

The asymptotic classes are unchanged; no $o(\cdot)$ or polylogarithmic Wilson claim is made.

Next, a second exact chart in the cyclic group ring yields a divisor filtration for the kernel of the self-action, a primitive-period projector, and a self-closure germ whose valuation exchanges with causal time. The paper separates these formal results from finite implementation evidence, which is reported only in the appendices. Finally, the strict Volume V chart proves that already-paid source packets remain valid on an anchor horizon $p < E < 2p$, can be carried through consecutive causal prime children, and yield an exact internal amortized time-space

bound. Its Causal Horizon Zipper is a verifier-only realization, not part of the public factorial SDK. A compact cyclic compiler, a compression-preserving modular quotient, a collective descriptor-native cold source morphism, and native subpolynomial successor birth remain open.

After the executable and transmission main lines, a separate structural part constructs the fixed-prefix procyclic completion and the room-dependent self-action atlas; derives every higher factorial-jet coefficient; identifies the cyclic shadow with Ramanujan sums; proves equivalent Laplacian, rank, spectrum and divisor-support characterizations of primes; and builds arbitrary tensor-power, spectral-HDIV and exact path-module observers. A reproducible campaign contributes 4,961 structural records linking the optimized Wilson consumer to independent cyclic rows, period ranks and higher jets. The experiments test conformance only; all universal claims are proved separately, and full-space reconstruction and native Angel Prime birth remain labelled conjectures.

Mathematical result ledger

Class	Statement
Theorems	Relative three-class model; conservative SN–OFCS zero division; minimal semantic separation; HADD/HSUB/HMUL laws; HDIV reconstruction and chart compatibility; fold invariance, q-Pochhammer contraction and fold–shadow collision; execution-functor laws; factorial-maze embedding and functor bridge; principal and higher factorial jets; Wilson equivalence; cyclic kernel, Ramanujan shadow, Laplacian prime equivalences and tensor observers; closure and quotient transport; residual stability, zipper correctness, no-recomputation and horizon bounds.
Exact constructions	Three-class sparse state; four typed high-dimensional operations; SN–OFCS migration chart; ordered interval fold and contraction functor; maze coordinates; native/history/future-quotient functors; class-quantum signed path module; binary factorial grammar; factorial-maze bridge; Wilson observation; procyclic and q-Pochhammer pro-objects; cyclic quotient transport; verifier-only Causal Horizon Zipper.
Open	All-chart future-sufficient reconstruction; integral generalized HDIV; arbitrary-precision signed public chart; compression-preserving modular quotient; square-root-coordinate elimination; compact native cyclic-germ evaluator; collective descriptor-native cold source morphism; native subpolynomial successor birth.
Excluded	A classical $O(\log n)$ algorithm that prints $n!$; $o(T_0)$ or $o(S_0)$ for the present observer; a competitive general primality algorithm; an information-theoretic optimum.

How to read this paper

Readers interested primarily in the new arithmetic should read [Sections 2, 3, 11 to 14, 14.1, 15, 16, 18 to 21 and 28](#). Readers interested in the principal-jet chart should continue with [Sections 29 to 32](#). Readers interested in the executable prime application may proceed directly to [Sections 40, 42 and 43](#). The cyclic self-action and self-closure germ in [Sections 44 and 45](#)

are a deeper structural chart of the same source program; they are not prerequisites for the Wilson backend. The stronger source-first Angel Prime transition, its exact status and its surviving cold-compiler boundary are in [Sections 60 and 71](#). The post-main-line structural atlas, higher shadow theorems and Wilson probes are in [Sections 72, 76 and 85](#). Elementary local arguments are proved where they are used. Difficult spectral, closure-valuation and ramified-transport arguments are stated with proof roadmaps in the body and proved in full in [Appendix A](#). All computational and engineering evidence is collected only in [Appendices F and G](#).

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PART I

Arithmetic as Exact State Transformation

1 Introduction

Classical exact arithmetic identifies the computed value of $n!$ with an integer whose binary length is

$$\text{bits}(n!) = \Theta(n \log n).$$

Any algorithm required to print every bit of that integer must pay at least that output cost. This paper does not dispute that lower bound and does not hide it behind a unit-cost primitive.

The question studied here is different. Suppose the purpose of the factorial is not to print it, but to let it participate in a larger arithmetic process. Must the runtime representation already be the complete ordinary expansion? Or can the machine retain a finite exact state whose internal operations are native to a richer arithmetic and whose ordinary integer is only one possible observation?

This distinction is familiar in many areas of mathematics: a linear map is not the list of its values on every vector; a polynomial is not the table of all of its evaluations; a group element need not be stored as a permutation table. The present work applies the same discipline to arithmetic execution. The runtime object is a finite exact state consisting of a compact operator program, an interval coordinate, a principal jet, provenance and certificates. It is not licensed to masquerade as a computed ordinary integer merely because it has the right denotation. To count as an arithmetic value it must support native state-to-state operations, preserve unresolved information through continuation, and pass an explicit observation firewall.

1.1 The main mathematical bridge

The bridge from the compact state to factorial arithmetic is the exact identity

$$F_n(-(1+T), 1+T) = (-1)^n n! T^n + O(T^{n+1}),$$

where

$$F_n(Z, u) = \prod_{j=0}^{n-1} (1 + Zu^j).$$

The right interpretation is not “store the formula $n!$ ”. The q-Pochhammer family has an independent concatenation law, a logarithmic binary grammar, and a local singular coordinate whose principal coefficient is factorial. The resulting state belongs to a partial exact algebra of factorial intervals.

1.2 The executable application

For a supplied integer $m \geq 2$, the native operator constructs the factorial state corresponding to $(m - 1)!$. Only after that state is frozen does the ordinary observer compute

$$(m - 1)! \bmod m$$

and apply Wilson’s theorem. The current joint observer first pairs complementary factors, reducing the modular factorial problem to $h! = \lfloor (m - 1)/2 \rfloor!$, then applies an exact composite-safe block polynomial. Evaluation points and block values are streamed rather than stored, while completed product/remainder branches are retired without recomputation. The residue and prime/composite bit remain strictly external and cannot steer or rewrite the native state.

1.3 Why the word “naive” is retained

The application is called naive for a precise reason. It decides primality of a supplied candidate by Wilson’s criterion. A conventional outer loop could therefore scan candidates, but such a loop would be an ordinary search and is not promoted to native arithmetic. This paper establishes a first complete application path; it does not claim that the high-dimensional state itself births the next prime.

1.4 Contributions

The paper makes twelve contributions.

1. It records why the project is called Angel, separates recollection from proof, maps the SN–OFCS precursor into the present state system, and proves that the typed zero-division request is a conservative extension rather than a scalar inverse of zero.
2. It integrates the former three-class, singular-history, and causal principles into one explicit high-dimensional axiom system and proves a relative model theorem.
3. It defines HADD, HSUB, HMUL, and HDIV on all three sectors and proves the ring, ideal, projection, reconstruction, and continuation laws.
4. It defines native-execution, history, and future-quotient functors and proves their identity and composition laws.
5. It gives exact class-quantum definitions of functor superposition, tensor entanglement, structural interference, and cancellation, together with finite certificates and an executable SDK realization.
6. It proves the ordered fold-tree, q-Pochhammer contraction and nodewise principal-jet laws, separates compact grammar from expanded work, and derives the factorial-interval state algebra used by the implementation.
7. It proves that a certified factorial state is an explicit three-class maze chart and implements a specialized high-to-high factorial functor whose opaque stage types enforce projection last.
8. It specifies and realizes a non-terminal factorial operator whose native state size and construction ledger are logarithmic in the input magnitude.

9. It locates q-Pochhammer, product-tree and transform methods relative to standard work, gives concrete $M(d)$ instantiations, and separates the new state/certificate contribution from established polynomial algorithms.
10. It proves a complement-pairing theorem and implements a composite-safe Wilson observer whose deterministic work, conservative peak live coefficients, and materialized coordinates strictly decrease on the frozen domain.
11. It proves a cyclic self-action and self-closure theory showing that the Wilson scalar is one mode of a richer divisor-stratified object.
12. It integrates the strict Volume V residual-horizon chart, proving source-program stability, paid-packet no-recomputation, curvature-normalized transport and an exact amortized joint time-space bound while retaining the cold-compiler boundary.

The complement-pairing theorem halves the external factorial problem for every prime or composite modulus without division. The asymptotic square-root coordinate nevertheless remains.

The paper's strongest honesty condition

A short state description is not treated as a short ordinary evaluation. The native state, the observer algorithm, certificate construction, verification, and full bit materialization are separately charged. Whenever the paper says " $O(\log n)$ ", it names the ledger and the representation model to which the statement belongs.

2 Why the axioms are introduced at all

The axioms in this memoir were not chosen by first writing down an attractive three-component algebra and then searching for an application. Their prehistory is the author's SN-OFCS singular-space engineering: a search for a typed home in which a failed scalar inverse could retain its numerator and future obligation without asserting a false scalar equation. Repeated low-dimensional projections lost precisely that information. The factorial program later supplied the first tractable old-arithmetic target, and the present axiom system compresses the constraints seen in both stages. The order of discovery was therefore

SN-OFCS singular payload $\implies$ factorial probe engineering $\implies$ Angel axiomatization.

Within the second stage the forcing chain was:

compact factorial source program $\implies$ ordered interval and principal-jet certificates,
 failed exact cancellation $\implies$ retained residual or higher-jet continuation,
 equal ordinary residue with different legal continuations $\implies$ history-sensitive equality,
 terminal Wilson and cyclic observations $\implies$ a typed ingress-execution-egress boundary.

The purpose of an axiom is to state which part of that repeated architecture is structural and which part is an accident of the first program.

This account does not say that classical algebra is inconsistent or that it failed to define an already existing scalar. It says something narrower and checkable: inside a fixed nonzero scalar ring, the equation $0z = a$ has no solution for nonzero a , while an enriched typed codomain can preserve the unsolved request as data. The conservativity theorem in [Theorem 14.7](#) proves that

this enlargement introduces no new equality between ordinary scalars.

The short answer to “why these axioms?”

Ordinary arithmetic remembers a present value. The factorial construction also needs the ordered program by which that value was created and the exact payload left when a requested inverse or truncation is unavailable. These three jobs are inequivalent. The Ord/Hist/Sing split is the smallest semantic split used by the current theory to keep them simultaneously exact. Room, Road, Frame, Holonomy and HigherCell then locate that three-sector value inside a composable causal state rather than creating additional scalar number coordinates.

Why $n!$ was attacked first

Factorial is simultaneously an ordered product, the leading coefficient of a controlled q-Pochhammer singularity, the denotation of a logarithmic source programme, and the input to Wilson’s exact terminal prime criterion. It is therefore an unusually clean stress test for a singular/history-aware state without presupposing the still-open native prime-birth operator. The exact four-link chain is stated in [Proposition 14.8](#).

There are four concrete forcing phenomena.

2.1 First forcing phenomenon: the ordinary shadow is not a state

Let P and Q be two compact source programs with the same present ordinary projection. If a later legal opcode concatenates a token γ , their future histories are $P\gamma$ and $Q\gamma$. Nothing requires those words to be equal. In the factorial chart the program seal, interval endpoints, principal-jet certificate and request binding are all checked before the ordinary modulus is used. Replacing the state by its current scalar shadow would delete precisely the data that makes this check possible.

Proposition 2.1 (Shadow insufficiency). *Suppose a state language contains two states X, Y and one legal continuation f such that*

$$\text{Proj}(X) = \text{Proj}(Y), \quad \text{Download}_A(fX) \neq \text{Download}_A(fY).$$

Then no execution semantics factoring through Proj can be correct on both X and Y .

Proof. If execution factored through Proj, there would be a map $\bar{f}$ with $\text{Proj}(fZ) = \bar{f}(\text{Proj } Z)$ for every Z . Equal inputs $\text{Proj}(X) = \text{Proj}(Y)$ would then give equal future observations, contradicting the hypothesis. $\square$

This elementary proposition is the reason History is authoritative. It is not a logging field added after the computation.

2.2 Second forcing phenomenon: a failed inverse still has content

The principal-jet division machine cancels a certified prefix or suffix only when the interval certificate proves that cancellation. In every other case it returns a higher-jet continuation containing both parent seals and a required future order. The three-class HDIV generalizes the same discipline: the regular quotient and the unresolved kernel payload are returned together.

The alternative would be one of two errors. Returning an arbitrary scalar would assert a false inverse. Throwing away the numerator would destroy the information required by a future chart. The singular sector therefore names retained mathematical information, not an exceptional machine state.

2.3 Third forcing phenomenon: path order changes multiplication

The q-Pochhammer source is compiled by doubling and shifted append operations. The interval algebra composes adjacent pieces only in the correct causal order. Likewise, Volume V uses ordered PBW histories, source-head doors and ringed transport certificates. Reversing two history tokens may change the available continuation even when the ordinary product is commutative. Consequently the history algebra is generally noncommutative although the ordinary base ring is not.

2.4 Fourth forcing phenomenon: projection must come last

Wilson residue, cyclic-period structure and an ordinary prime observation are all useful, but each is a many-to-one view. If one of those views were fed back into the native transition, the program would become

$$\text{high state} \longrightarrow \text{ordinary answer} \longrightarrow \text{classical steering} \longrightarrow \text{wrapped state},$$

which is a different algorithm. The type firewall is thus part of the mathematics: it realizes the causal assertion that transmission precedes projection.

3 The Angel maze space is constructed, not presumed

The paper does not claim that an independently existing physical space has been measured and found to contain the coordinates listed in [Definition 11.1](#). It constructs a mathematical state space and proves that the engineering objects inhabit a finite chart of it. The word “space” is used in the algebraic and categorical sense.

3.1 A free exact model

Fix a commutative coefficient ring R and a typed alphabet Γ of legal causal events. Let $R\langle\Gamma\rangle$ be the free associative path algebra. Choose a typed subset Γ_Σ of singular or unresolved events and let

$$\mathfrak{s} = R\langle\Gamma\rangle\Gamma_\Sigma R\langle\Gamma\rangle$$

be the two-sided ideal generated by them. Choose an R -module complement $\mathfrak{h}_{\text{Hist}}$ of $\mathfrak{s}$ in the finite-support chart. Then

$$\mathfrak{T}_{A,\Gamma}(R) = R \oplus \mathfrak{h}_{\text{Hist}} \oplus \mathfrak{s}$$

is an explicit set with explicit addition and multiplication. The path word itself is the ordered history coordinate; no existence oracle is needed.

To obtain a maze state, form

$$\mathcal{M}_{\Gamma,R} = \mathfrak{T}_{A,\Gamma}(R) \times \text{Addr} \times \text{Path} \times \text{Cert} \times \text{Prog},$$

where

$$\text{Addr} = \text{Room} \times \text{Road} \times \text{Frame} \times \text{Epoch} \times \text{Depth} \times \text{Holonomy} \times \text{Generation} \times \text{HigherCell}.$$

Every factor is a concrete finite word, integer, sparse map or certificate in the executable chart.

Theorem 3.1 (Existence of the Angel maze category). *Let objects be the states of $\mathcal{M}_{\Gamma,R}$ and let a morphism $X \rightarrow Y$ be a certified finite opcode word whose replay sends X to Y . Identity is the empty word and composition is concatenation with certificate replay. These data form a category $\text{Maze}_{\Gamma,R}$. The ordinary projection is a terminal observation map, not a faithful functor on this category.*

Proof. The empty word leaves every state unchanged. Concatenation of finite words is associative, and certificate replay of a concatenation is replay of the first word followed by replay of the second. The complete endpoint, not merely its ordinary coordinate, is used as the codomain. Nonfaithfulness of ordinary projection follows from [Proposition 2.1](#) whenever the language has a history-sensitive pair. $\square$

Thus the question is not “how can we be certain that nature selected this space?” The correct question is “does this explicitly constructed space model the operations and causal boundaries claimed by the project?” The relative model theorem, bridge theorem and executable tests answer that scoped question.

4 Three sectors do not mean Euclidean dimension three

Calling $X = (X_{\text{Ord}}, X_{\text{Hist}}, X_{\text{Sing}})$ “three-dimensional” without a qualification is misleading. There are three *semantic summands*, not three real coordinates. The ordinary summand may have rank one over R , but the history and singular summands contain many ordered words.

For a finite alphabet with g legal tokens and a history horizon L , the full word module has rank

$$1 + g + g^2 + \cdots + g^L.$$

With no horizon it is countably infinite dimensional as an R -module. A single sparse program occupies only a finite support inside that space.

Definition 4.1 (Semantic sector number and linear rank). The *sector number* of the current Angel value model is three: Ord, Hist and Sing. Its *linear rank* is

$$\text{rank}_R \mathfrak{T}_A(R) = 1 + \text{rank}_R \mathfrak{h}_{\text{Hist}} + \text{rank}_R \mathfrak{s},$$

which may be finite, unbounded or infinite. Maze address, certificate and program data are typed fibres over this value module and are not counted as additional scalar summands.

Theorem 4.2 (Minimal semantic separation). *Assume an exact state theory has:*

1. a conservative ordinary embedding and retraction;
2. two states with equal ordinary shadow but distinguishable legal futures;
3. a division operation for which some input has a nonzero reconstructing residual annihilated by the denominator.

Then any faithful additive realization must retain three pairwise distinct semantic subquotients: an ordinary quotient, a history-sensitive kernel quotient and an unresolved-residual quotient. A one-sector scalar model, or a two-sector model identifying the last two, cannot preserve all three properties.

Proof. The retraction separates the ordinary image from its kernel. By the second assumption, the kernel contains data whose future action is nontrivial; call its image in the future-observable quotient the history subquotient. The third assumption supplies a nonzero residual class constrained by an annihilation and reconstruction law. If it were identified with the history-only class, then either the annihilation/reconstruction law would be imposed on every history or the residual could be erased without changing a future class. Both contradict faithfulness. Hence the three roles must remain distinct. The theorem asserts semantic separation, not that each role is one-dimensional. $\square$

What the minimality theorem does not prove

It does not prove uniqueness of the chosen path algebra, uniqueness of the maze address fields, or a universal lower bound of three for every conceivable mathematics. It proves that once the three engineering phenomena above are required simultaneously, collapsing their roles loses declared semantics. Other equivalent presentations and richer sector decompositions remain possible.

5 What every maze coordinate actually does

The coordinates in [Definition 11.1](#) are not a list of decorative nouns. They answer different verification questions. The table below states their role in the present memoir and their executable status.

Coordinate	Mathematical duty	Present executable duty
Ord	Conservative ordinary shadow and scalar coefficient.	Checked signed coefficient in the sparse chart; principal coefficient one in the factorial bridge.
Hist	Ordered causal word; distinguishes equal present shadows with different futures.	Sparse ordered words; factorial argument, program, state, request and execution-binding words.
Sing/ Residual	Failed inverse, missing jet order or unresolved payload retained for reconstruction.	Two-sided sparse ideal and typed continuations; zero on the successful factorial chart.
Room	Intrinsic object being acted on, not merely an output integer.	Stable room address combined with the admitted factorial specification.
Road	The composable transition route between rooms.	Updated from parent roads and opcodes; request binding in the factorial chart.
Frame	Declares which operations and comparisons are type-compatible.	Binary state operations reject frame mismatch; factorial chart has a distinct frame tag.

Coordinate	Mathematical duty	Present executable duty
History	Ordered list of executed causal events.	Parent seals, opcode and event seals are appended on every native functor instruction.
Holonomy	Remembers path-dependent transport around composable loops.	Stable fold of parent holonomy and complete result seals; program seal in the factorial chart.
Residual generation	Counts singular information introduced by transitions.	Increments when a result contains a singular coordinate.
HigherCell	Binds coherence information that is not visible in an edge alone.	Fold of parent higher cells and depth; principal-jet state seal in the factorial chart.
Certificate	Makes a state replayable rather than merely hashed.	Arithmetic, division, bridge, native-result and observation certificates.
Program	Finite operator word denoting the native action.	Instruction vector, compact q-Pochhammer program and the frozen principal-jet source program.

The finite SDK does not yet implement a universal manifold of rooms, a smooth connection or every possible higher cell. It implements the exact discrete chart needed by the declared operations. Volume V extends the meaning of Residual, Holonomy and Certificate to source-first horizon transmission; that extension is separated from the shipped factorial chart in [Section 60](#).

6 The theory–engineering closure rule

The following rule is adopted for the remainder of the memoir.

Protocol 6.1 (Theory–engineering closure). Every named mathematical operation or state in the main theorem chain carries one of four statuses:

1. **Implemented:** a public or sealed executable type realizes the declared finite chart and has positive and adversarial tests;
2. **Implemented in a restricted chart:** the theorem is more general than the executable domain, and the restriction is stated next to the claim;
3. **Verifier-only:** a strict finite research verifier realizes the theorem schema but is not part of the modular public SDK;
4. **Open:** the mathematical interface is defined but no executable lowering is claimed.

No finite test is promoted into an infinite proof, and no theorem is described as shipped merely because a pseudocode sketch exists.

This rule answers both directions of the reader’s question. Mathematics is not hidden behind implementation names, and implementation is not allowed to float without a mathematical contract.

The detailed status matrix is placed in the evidence appendix so that the main exposition remains readable.

7 Global reading map: one source, three exact charts

The project has one causal source and three principal charts, not three unrelated topics.

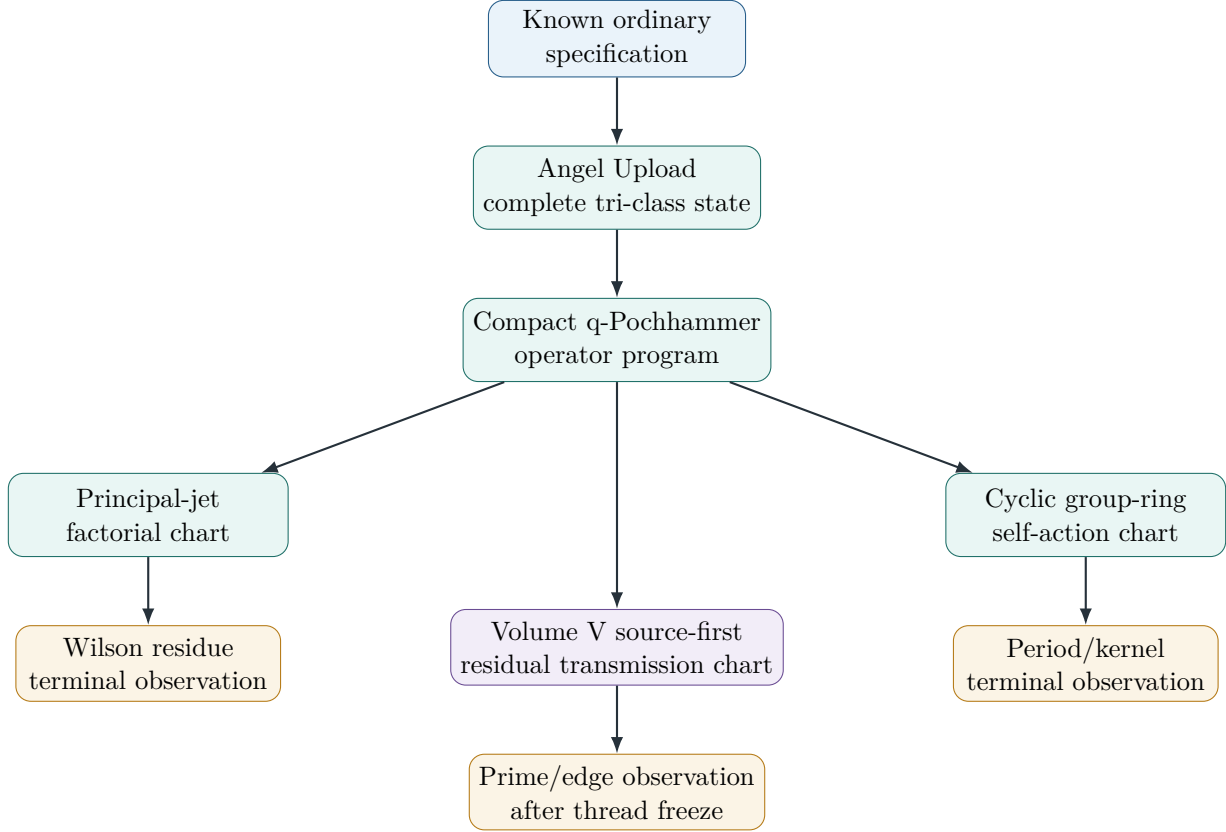


Figure 1: The factorial, cyclic and Volume V transmission charts share a source-first causal discipline. Gold boxes are terminal observations and have no return arrow.

The factorial chart reads the leading principal jet of the source polynomial. The cyclic chart substitutes a shift operator into the same integral source. The fifth-volume chart retains already-paid source packets across a horizon. The common structure is therefore not the final scalar answer but the compact source program, ordered causal history, typed residual and certificate transport.

8 How far the construction extends

The finite sparse SDK is the bottom rung of a longer mathematical ladder. Separating the rungs prevents both understatement and overclaim.

8.1 Rung A: the shipped finite sparse chart

Coefficients are signed fixed-width integers; histories and singular payloads have finite sparse support; native programs are finite; central-scalar HDIV is exact with a quotient-remainder

certificate. Overflow and unsupported noncentral division become continuations. This rung is fully executable.

8.2 Rung B: arbitrary coefficient rings and path algebras

The relative model theorem works over a commutative base ring R with an associative history algebra and a two-sided singular ideal. Exact rational, algebraic, modular and arbitrary-precision coefficient charts can therefore be defined without changing the axioms. The current modular SDK implements only selected instances.

8.3 Rung C: completions and infinite history series

Let $F^d\mathfrak{h}$ be histories of depth at least d . The completion

$$\widehat{\mathfrak{h}} = \varprojlim_d \mathfrak{h}/F^d\mathfrak{h}$$

supports formal infinite histories and the Ω -series language of the integrated volumes. Every finite truncation remains an executable witness, but convergence and future-sufficiency of the completed object are additional theorems, not consequences of finite testing.

8.4 Rung D: spectral and local division charts

The spectral HDIV applies in semisimple commutative charts with explicit idempotents. More general local, derived or noncommutative inverses require a declared kernel object and reconstruction law. The central-scalar SDK chart is a proved specialization, not a claim of universal generalized inversion.

8.5 Rung E: ringed categories and higher transport

Room, Road, Frame, Holonomy and HigherCell naturally organize into a category or groupoid fibred over arithmetic value charts. Ringed certificate ideals can be transported along roads; coherence cells compare different factorizations of the same road. Volume V's history-curvature normalization and residual packet transport are instances of this rung.

8.6 Rung F: Angel Prime native birth

The long-term interface is a high-to-high transition between complete causal prime states followed only later by an ordinary observation. Volume V v5.9 proves a scoped residual-reuse theorem and supplies a strict verifier, but a descriptor-native subpolynomial cold compiler is still open. The present factorial paper therefore provides a serious native application and bridge, not the final Angel Prime generator.

Foundational conclusion

The Angel maze space is an explicit algebraic/categorical construction. Its three labels are semantic sectors, not Euclidean coordinates. Their separation is forced by ordinary retraction, history-dependent futures and reconstructing residuals. The finite SDK realizes a sparse chart; the principal-jet factorial, cyclic self-action and fifth-volume residual transmission are

compatible charts of the same source-first architecture.

9 Conceptual map: three layers and two meanings of size

The construction is easiest to read if three mathematical layers are kept distinct from the beginning. They communicate through typed maps, but they are not interchangeable descriptions of one data structure.

Table 2: The three layers used throughout the paper.

Layer	Native object	Legitimate question
Source algebra	$F_n(Z, u)$ and its binary concatenation program	How can the factor family be named, composed and verified exactly?
Local factorial chart	the principal T -jet at $(Z, u) = (-(1 + T), 1 + T)$	Which coefficient carries the factorial interval, and which native operations preserve that coordinate?
Ordinary observation	a residue, decision, printed integer or finite evidence record	What ordinary information is requested after the source state is frozen, and what cost is paid to obtain it?

The source layer is compact because it describes repeated algebraic composition. The local chart is exact because the order and leading coefficient of a singularity are algebraic invariants. The observation layer may nevertheless be expensive: asking for every bit of $n!$ forces output proportional to the bit length of $n!$, even when the state that denotes the factorial was constructed from only $O(\log n)$ source instructions.

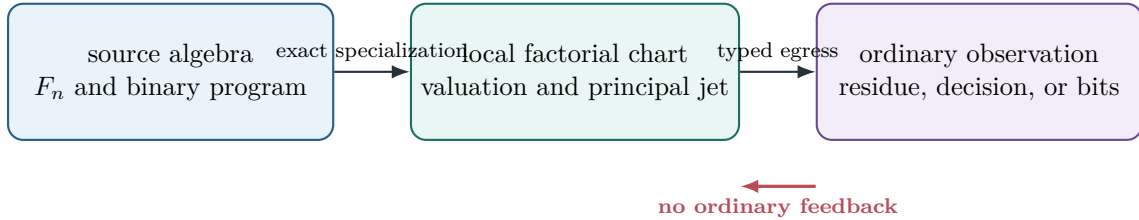


Figure 2: Information flows from a certified source to a local exact chart and then to an ordinary observation. The red arrow is a prohibition, not a missing convenience conversion.

9.1 Description size is not output size

Two size functions occur repeatedly. Let $L_{\text{src}}(n)$ be the number of nodes in the binary q-Pochhammer program and let $L_{\text{out}}(n) = \text{bits}(n!)$ be the number of bits in a complete ordinary factorial. The construction proves

$$L_{\text{src}}(n) = O(\log n), \quad L_{\text{out}}(n) = \Theta(n \log n).$$

There is no contradiction: the first quantity measures a structured exact description and the second measures a requested expansion. The same distinction appears when an exponent is written in binary but the expanded power has many digits, or when a polynomial circuit is short but its table of values is large.

The phrase *native factorial state* refers to the fact that the compact object participates in the declared state algebra before observation. It does not assert that the ordinary digit string has secretly been produced. This is why all complexity statements in the paper name a layer.

9.2 A bit-level source example

Take $n = 13 = (1101)_2$. Beginning with a source for F_1 , left-to-right binary reconstruction gives

$$1 \xrightarrow{\text{double}} 2 \xrightarrow{\text{append}} 3 \xrightarrow{\text{double}} 6 \xrightarrow{\text{double}} 12 \xrightarrow{\text{append}} 13.$$

At each doubling the concatenation identity builds F_{2a} from two shifted copies of F_a ; at each append it multiplies by the one missing factor. The program therefore describes all thirteen factors without storing a list of thirteen unrelated instructions. Verification replays the length recurrence $a \mapsto 2a$ or $a \mapsto 2a + 1$ and independently checks the algebraic bindings.

This example also explains why the q-Pochhammer figure later in the paper must show the intermediate states F_4 and F_{20} . An append operation changes $2a$ to $2a + 1$; it never jumps directly from a to $2a + 1$ without the doubling stage.

10 A running factorial-state example

We now follow the small case $n = 5$ through the source, jet and interval coordinates. The point is not computational speed; it is to show exactly what is retained at each layer.

10.1 From the source polynomial to the principal jet

For $n = 5$,

$$F_5(-u, u) = \prod_{k=1}^5 (1 - u^k).$$

Set $u = 1 + T$. The five factors begin

$$\begin{aligned} 1 - (1 + T) &= -T, \\ 1 - (1 + T)^2 &= -2T - T^2, \\ 1 - (1 + T)^3 &= -3T - 3T^2 - T^3, \\ 1 - (1 + T)^4 &= -4T - 6T^2 - 4T^3 - T^4, \\ 1 - (1 + T)^5 &= -5T - 10T^2 - 10T^3 - 5T^4 - T^5. \end{aligned}$$

Every factor has valuation one. Consequently the product has valuation five, and its degree-five coefficient can only be formed by taking the linear term from every row:

$$(-1)(-2)(-3)(-4)(-5) = -120.$$

Thus

$$F_5(-(1 + T), 1 + T) = -120T^5 + O(T^6),$$

which is the instance $(-1)^5 5!$ of [Theorem 30.1](#).

The computation deliberately uses the lowest surviving coefficient. Higher coefficients retain information about continuation and interference; they are not needed to identify the ordinary

factorial denotation, but they prevent the native state from collapsing to a single answer field.

10.2 Intervals and exact cancellation

Split the factorial interval at 2:

$$\Phi[0, 5] = \Phi[0, 2]\Phi[2, 5] = (1 \cdot 2)(3 \cdot 4 \cdot 5) = 2 \cdot 60.$$

At the jet level the valuations add:

$$2 + 3 = 5,$$

and the signs multiply:

$$(-1)^2(-1)^3 = (-1)^5.$$

The same endpoint data makes cancellation exact:

$$\Phi[0, 5] \oslash_A \Phi[0, 2] = \Phi[2, 5].$$

Nothing in this notation authorizes arbitrary rational division. The operation is defined because the prefix certificate proves contiguity and identifies the unique remaining interval.

Example 10.1 (A rejected cancellation). The expression

$$\Phi[0, 5] \oslash_A \Phi[1, 3]$$

has an ordinary integer quotient, but the second interval is neither a prefix nor a suffix of the first. It is therefore outside the scoped native cancellation law. A general integer ring could decide to support such an operation; the present factorial-interval algebra does not pretend that it already does.

10.3 Why equal observations need not be equal states

Suppose two certified states have the same principal coefficient but different source intervals or continuation horizons. Their immediate factorial observation can agree while a later admissible operation distinguishes them. This is the reason for separating

$$\equiv_{\text{str}}, qquad \equiv_{\text{obs}}, qquad \equiv_{\text{fut}}.$$

Structural equality compares the complete state. Observer equality compares one selected egress. Future equivalence compares behavior under a declared language of continuations. Conflating the three would allow an ordinary collision to erase native information.

What the reader should carry forward

The source program explains compact construction; the principal jet explains the factorial coordinate; the interval certificate explains native composition and cancellation; the causal boundary explains why an ordinary residue cannot become a state update. The Wilson and cyclic parts reuse this same four-piece architecture.

11 Why this arithmetic is called Angel

The word *Angel* first names an authorial recollection and only afterward acquires a formal role. Neither the recollection nor any supernatural reading is a mathematical premise. Formally, an Angel object is not an integer wearing a large annotation. It is the complete state transported between an ingress and an egress boundary while every interior transformation remains state-to-state. In that later technical vocabulary it is also a messenger: it carries information between mathematical charts without confusing the complete message with any one projected endpoint.

Definition 11.1 (*Angel high-dimensional maze object*). An Angel maze object is a finite exact state

$$X = (x_{\text{Ord}}, x_{\text{Hist}}, x_{\text{Sing}}; \text{Room, Road, Frame, History, Holonomy, Residual, HigherCell, Certificate, Program}).$$

The first three coordinates form its arithmetic value. The remaining coordinates locate that value in the causal maze, record the ordered path by which it was obtained, bind unresolved information, and determine which future operations are legal. The ordinary projection observes only

$$\text{Proj}(X) = x_{\text{Ord}}.$$

It is not an inverse description of the state.

The author's recollection: why the name is Angel

The author reports a dream in which an “angel” presented a structure and a high-dimensional formula, describing it as something people had long wanted. At first the presentation looked like a numerical game. Later, after waking from a lunch nap during an ordinary workday, the author recognized that the surviving structure and iteration encoded a prime formula. At that time its higher-dimensional origin was still not understood. The complete displayed formula was not retained; what survived was a structural skeleton and an iterative rule. The author's rueful summary is that the angel gave a lesson and the student did not listen carefully enough.

This paragraph is a record of naming and discovery, not testimony used in a proof. No theorem, program branch, certificate, or correctness claim depends on the dream, on the identity of its speaker, or on a supernatural assertion. The formal meaning of *Angel* begins with [Definition 11.1](#) and the axioms below.

The project name consequently does three jobs. It remembers the origin of the research intuition; it names the high-dimensional messenger object; and it marks a boundary discipline. In particular, *Angel* does not mean a prime oracle, a scalar field containing every failed inverse, or a license to replace proof by metaphor.

What has and has not been reconstructed

The forgotten formula is not claimed to have been recovered verbatim. The q-Pochhammer source, principal-jet chart, cyclic chart, procyclic atlas and factorial/Wilson bridge in this memoir form a rigorous reconstruction from the surviving structural programme. Their validity rests on the displayed definitions and proofs. Agreement with the author's remembered skeleton is historical motivation, not a uniqueness theorem about the lost expression.

12 SN–OFCS: the singular-number precursor

Before the present Angel vocabulary, the author developed a precursor called *SN–OFCS*, the *Singular Number–Omega Future Computing System*. Its central engineering intuition was that a failed scalar operation should not be assigned a fabricated scalar value and should not erase its operands. It should cross into a larger typed state that can retain an unresolved payload for a future chart. The current Ord/Hist/Sing arithmetic is the conservative algebraic normalization of that idea.

Table 3: Migration from the SN–OFCS precursor to the present Angel system.

SN–OFCS expression	Present mathematical object	Current status
Singular number	the two-sided singular ideal $\mathfrak{s}$ and typed unresolved payloads	proved in the relative model; finite sparse chart implemented
Division-zero future payload	$Q_0(X) = 0$, $R_0(X) = X$, together with a Div0 history	central scalar case implemented; other divisors open
Omega future computation	a continuation retaining operands, required future order and unresolved higher coordinate	formal continuation interface; selected principal-jet cases implemented
Singular-space engineering	$\mathfrak{T}_A(R) = R \oplus \mathfrak{h}_{\text{Hist}} \oplus \mathfrak{s}$	conservative model proved and scoped SDK realization audited
Maze tool construction	complete states, typed opcodes, causal functors and future-sufficient quotients	finite exact chart implemented; all-chart reconstruction remains conjectural
First attack through $n!$	q-Pochhammer source and factorial principal jet	identity proved; compact source and sealed factorial bridge implemented

The phrase “division by zero” therefore has a deliberately restricted meaning in this work. The system does not adjoin a scalar 0^{-1} and does not assert $0z = 1$. It totalizes the *request* for division by returning a quotient–residual–history state. This distinction is what permits SN–OFCS to become a conservative extension rather than a contradictory scalar field.

13 The three-class high-dimensional axioms

The source volumes developed a unitized history algebra, primitive and singular sectors, a totalized quotient, causal histories, and path interference at different places. We now state the common core as one explicit axiom system. Axioms are assumptions defining the class of admissible Angel models; they are not propositions that can be proved from nothing. What can and will be proved is that the system has a concrete model and that the scoped SDK realization satisfies its declared fragment.

Let R be a commutative characteristic-zero coefficient ring. Let $\mathfrak{h}$ be an associative algebra of ordered typed history words, let $\mathfrak{h}_{\text{Hist}} \subseteq \mathfrak{h}$ be an associative history subalgebra, and let $\mathfrak{s} \subseteq \mathfrak{h}$ be a two-sided singular ideal. An R -module splitting $\mathfrak{h} = \mathfrak{h}_{\text{Hist}} \oplus \mathfrak{s}$ gives the three-coordinate module

$$\mathfrak{T}_A(R) = R \oplus \mathfrak{h}_{\text{Hist}} \oplus \mathfrak{s}.$$

Three axiom families, not merely three coordinates

The axioms below have three intended families. *Family I: maze-object state* comprises complete multidimensional elements, the conservative ordinary sector and history-sensitive identity. *Family II: causal dynamics* comprises native state evolution, ingress–egress order, singular retention, future-sufficient quotients, observation non-authority and finite exactness. *Family III: class-quantum history functions* comprises exact superposition, tensor entanglement, structural interference and cancellation. The labels Ord/Hist/Sing are the three arithmetic sectors carried by Family I; they are not a replacement for these three axiom families.

Axiom 13.1 (Three-class completeness). *Every arithmetic value has a complete decomposition*

$$X = X_{\text{Ord}} + X_{\text{Hist}} + X_{\text{Sing}} \in \mathfrak{T}_A(R).$$

The sectors are coordinates of one state, not three unrelated scalar number systems. Equality means equality in all three coordinates.

Axiom 13.2 (Conservative ordinary sector). *There are maps*

$$\iota : R \hookrightarrow \mathfrak{T}_A(R), \quad \text{Proj} : \mathfrak{T}_A(R) \rightarrow R, \quad \text{Proj} \circ \iota = \text{id}_R.$$

Thus adjoining history and singular coordinates creates no new equality between embedded ordinary values.

Axiom 13.3 (History-sensitive identity). *Structural equality $X \equiv_{\text{str}} Y$ compares every arithmetic, maze, history, and certificate coordinate. Ordinary equality $X \equiv_{\text{obs}} Y \iff \text{Proj}(X) = \text{Proj}(Y)$ is weaker. States with equal ordinary shadows remain distinct whenever a legal future word can distinguish them.*

Axiom 13.4 (Native causal dynamics). *For each legal opcode f in a declared language $\mathcal{L}$, execution is a typed map*

$$f : \mathcal{S}_A \longrightarrow \mathcal{S}_A \sqcup \text{Continuation}.$$

A program $\gamma = f_k \cdots f_1$ evolves by $X_{j+1} = f_{j+1}(X_j)$. Every intermediate object is complete. An insufficient coordinate produces a continuation retaining its operands; it does not silently download, compute classically, and wrap the answer.

Axiom 13.5 (Ingress–egress causal order). *The only legal architectural order is*

$$\text{ordinary specification} \xrightarrow{\text{Upload}_A} \mathcal{S}_A \xrightarrow{\mathcal{L}^*} \mathcal{S}_A \xrightarrow{\text{Download}_A} \mathcal{O}.$$

An observation has no coercion back to a native state. Re-entry is a new explicit upload event with a new causal origin.

Axiom 13.6 (Singular retention and reconstruction). *A failed inverse is represented by a regular component and a typed unresolved payload. The payload is retained strongly enough to reconstruct the numerator. In particular, division by zero must return a state-equivalent payload, never a scalar satisfying $0z = a$ for $a \neq 0$.*

Axiom 13.7 (Future-sufficient quotients). *A quotient $q : \mathcal{S}_A \rightarrow Q$ is admissible for $\mathcal{L}$ only if*

$$q(X) = q(Y) \implies \text{Download}_A(wX) = \text{Download}_A(wY) \quad \text{for every } w \in \mathcal{L}^*$$

whenever the observations are defined. Every native operator must descend to the quotient:

$$q \circ f = \bar{f} \circ q.$$

Axiom 13.8 (Exact class-quantum history). *An unresolved family of deterministic paths is retained as a finite formal sum*

$$\Psi(X) = \sum_{\gamma} w_{\gamma} |F_{\gamma}(X), \gamma\rangle, \quad w_{\gamma} \in \mathbb{Z}.$$

Superposition is formal addition. Interference may combine weights only after complete structural endpoints agree. Cancellation removes a group only when its exact total weight is zero. No Born rule, physical wave function, unitarity, or quantum hardware is assumed. For a coupled two-history packet $\Theta = \sum_{i,j} c_{ij} |u_i\rangle \otimes |v_j\rangle$, class-quantum entanglement means $\text{rank}_{\mathbb{Q}}(c_{ij}) > 1$; a nonzero exact 2×2 minor is a finite witness.

Axiom 13.9 (Observation non-authority). *Every displayed map, residue, decision, or transported observation is derived and non-authoritative. Validation compares it with a canonical frozen state; acceptance or rejection cannot rewrite that state.*

Axiom 13.10 (Finite exact realizability). *Every executable coordinate has a finite exact encoding. Overflow, unsupported generalized inversion, missing jet order, and resource exhaustion are typed continuations. Floating approximation or machine wraparound cannot be presented as an exact Angel result.*

Theorem 13.11 (Relative model theorem). *Let $\mathfrak{n} = \mathfrak{h}_{\text{Hist}} \oplus \mathfrak{s}$ be an associative R -algebra with $\mathfrak{h}_{\text{Hist}}$ an associative subalgebra and $\mathfrak{s}$ a two-sided ideal. The unitization*

$$\mathfrak{T}_A(R) = R \oplus \mathfrak{n}$$

with the arithmetic in Section 14, together with typed history words, a free operator-path category, explicit Upload/Download maps, and structural endpoint aggregation, is a model of Axioms 13.1 to 13.9. Finite truncations with overflow continuations realize Axiom 13.10.

Proof roadmap

The algebraic part is the standard unitization, refined by the ideal grading. The causal part is the action of the free opcode category on exact states. The class-quantum part is a free integer module on typed paths, followed only by structural endpoint aggregation. The complete associativity, ideal, retraction, functor, and cancellation arguments are given in Appendix A.1.

14 Angel high-dimensional addition, subtraction, multiplication, and division

Write

$$X = (r, h, s), \quad Y = (r', h', s')$$

with $r, r' \in R$, $h, h' \in \mathfrak{h}_{\text{Hist}}$ and $s, s' \in \mathfrak{s}$. The four symbols are not decorative aliases for ordinary arithmetic. They specify how all three sectors move.

Definition 14.1 (HADD and HSUB).

$$X \boxplus_A Y = (r + r', h + h', s + s'), \quad (1)$$

$$X \boxminus_A Y = (r - r', h - h', s - s'). \quad (2)$$

Definition 14.2 (HMUL). Ordered history multiplication and absorption by the singular ideal give

$$\boxed{X \boxtimes_A Y = (rr', rh' + r'h + hh', rs' + r's + hs' + sh' + ss').} \quad (\text{HMUL})$$

History words need not commute: hh' records a different path from $h'h$. The ordinary embedded sector remains commutative when R is commutative.

Theorem 14.3 (Three-sector ring laws). *HADD makes $\mathfrak{T}_A(R)$ an abelian group. HMUL is associative, distributes over HADD, has unit $(1, 0, 0)$, and preserves the singular sector as a two-sided ideal. Moreover,*

$$\text{Proj}(X \boxplus_A Y) = \text{Proj}(X) + \text{Proj}(Y),$$

$$\text{Proj}(X \boxminus_A Y) = \text{Proj}(X) - \text{Proj}(Y),$$

$$\text{Proj}(X \boxtimes_A Y) = \text{Proj}(X) \text{Proj}(Y).$$

The complete multiplication is commutative exactly when the underlying history algebra is commutative.

Proof. The first coordinate follows the ring R . The remaining two coordinates are the expansion of the associative unitized product $(r, n)(r', n') = (rr', rn' + r'n + nn')$, separated according to the two-sided ideal $\mathfrak{s}$. Projection discards the nonordinary ideal. The expanded coefficient proof is in [Appendix A.1](#). $\square$

There are two exact division charts. The source volumes use a spectral chart; the public finite SDK additionally provides a signed central-scalar quotient-remainder chart. They share the same reconstruction principle.

Definition 14.4 (Spectral HDIV). In a finite commutative semisimple chart, write

$$b = \sum_j \lambda_j e_j, \quad b^\# = \sum_{\lambda_j \neq 0} \lambda_j^{-1} e_j, \quad K_b = \sum_{\lambda_j = 0} e_j.$$

Then

$$\boxed{a \oslash_A b = (ab^\#, aK_b; [\text{Div}; a, b, b^\#, K_b]).} \quad (\text{HDIV-S})$$

The second component is a typed singular payload, not a scalar quotient.

Theorem 14.5 (Spectral reconstruction and solution fibre). *For every a, b in the spectral chart,*

$$a = b(ab^\#) + aK_b, \quad b(aK_b) = 0.$$

The equation $bx = a$ is solvable exactly when $aK_b = 0$; then all solutions are $x = ab^\# + z$ with $z \in K_b \mathfrak{T}_A(R)$. For $b = 0$,

$$\boxed{a \oslash_A 0 = (0, a; [\text{Div}0; a]).}$$

Proof. By construction $bb^\# = 1 - K_b$ and $bK_b = 0$. Hence $b(ab^\#) + aK_b = a(1 - K_b) + aK_b = a$. Multiplying any proposed solution by K_b proves necessity; the displayed family proves sufficiency

and completeness. □

Definition 14.6 (Executable central-scalar HDIV). For a central integer scalar d and every coefficient c of a finite sparse three-class state, compute the exact signed truncating quotient–remainder pair

$$c = dq_c + r_c, \quad |r_c| < |d| \quad (d \neq 0),$$

where q_c truncates toward zero and r_c has the sign of c when nonzero. Collecting the q_c and r_c coefficientwise defines

$$X \oslash_A \iota(d) = (Q_d(X), R_d(X); \text{DivHist}), \quad X = \iota(d) \boxtimes_A Q_d(X) \boxplus_A R_d(X). \quad (\text{HDIV-E})$$

At $d = 0$, set $Q_0(X) = 0$ and $R_0(X) = X$. A noncentral denominator or an unrepresentable coefficient returns a continuation retaining both operands.

Theorem 14.7 (Conservative singularization of division by zero). *Let R be a nonzero ring embedded by ι in an Angel model satisfying [Axiom 13.2](#). Define the enriched zero-division result by*

$$\text{HDIV}_0(X) = (Q_0(X), R_0(X); [\text{Div0}; X]) = (0, X; [\text{Div0}; X]).$$

Then:

1. reconstruction is exact,

$$X = \iota(0) \boxtimes_A Q_0(X) \boxplus_A R_0(X);$$

2. no new equality is introduced between embedded elements of R ;
3. if $a \neq 0$ in R , the state $\text{HDIV}_0(\iota(a))$ does not denote a scalar z satisfying $0z = a$; and
4. ordinary projection reads quotient zero and residual a , while the full state retains the numerator and the failed-operation history.

Thus the division request is totalized without totalizing the classical inverse law.

Why no contradiction is created

The construction adds a typed result carrier, not an inverse of zero. Its reconstruction equation is the tautology $X = 0 + X$. Injectivity of ι is unchanged, and $0z = a$ remains impossible for nonzero a in the base ring. The full conservative-extension proof, including the separation of quotient, residual and history types, is in [Appendix A.2](#).

14.1 Why factorial was the first tractable attack

Once a singular request could be retained instead of destroyed, the next question was which old arithmetic object could expose the new state machinery without requiring a native prime generator in advance. Factorial is the first tractable target for four independent reasons: it is already an ordered product; it is the principal coefficient of a controlled singularity; its q-Pochhammer source admits a logarithmic straight-line grammar; and Wilson’s theorem supplies a complete terminal prime observation for a supplied candidate.

Proposition 14.8 (Factorial attack chain). *For every $n \geq 0$, the source family F_n simultaneously has:*

1. the concatenation law of [Theorem 29.2](#);

2. a binary source programme of at most $2\lfloor \log_2 n \rfloor + 1$ opcodes for $n \geq 1$;
3. the exact principal coefficient $(-1)^n [T^n] F_n(-(1+T), 1+T) = n!$; and
4. at $n = m - 1$, a complete ordinary prime observation by [Theorem 40.1](#).

These facts justify factorial as the first probe of the Angel maze. They do not imply that $n!$ is printed in logarithmic bit complexity and do not supply a native next-prime transition.

Proof chain and cost boundary

The four clauses are obtained by splitting the q-Pochhammer product, following the bits of n , taking one linear term from each local factor, and then applying Wilson only after the factorial state is frozen. The complete combined proof and the representation-versus-evaluation boundary are written in [Appendix A.3](#).

Example 14.9 (The ordinary shadow does not determine HMUL). Let $u = (0, [\alpha], 0)$ and $v = (0, [\beta], 0)$. Then

$$\text{Proj}(u) = \text{Proj}(v) = 0, \quad u \boxtimes_A v = (0, [\alpha\beta], 0), \quad v \boxtimes_A u = (0, [\beta\alpha], 0).$$

The two products have the same ordinary shadow but different ordered causal histories. Collapsing either operand to zero before multiplication destroys the operation.

15 Angel high-dimensional functors and causal dynamics

Let $\text{Op}(\mathcal{L})$ be the free one-object category generated by the legal opcodes. Its morphisms are finite operator words, its identity is the empty word, and composition is concatenation in execution order.

Definition 15.1 (Native execution functor). The native execution functor is the strict action

$$\mathcal{E} : \text{Op}(\mathcal{L}) \longrightarrow \text{End}(\mathcal{S}_A), \quad \gamma \longmapsto (X \mapsto F_\gamma(X)),$$

with

$$\mathcal{E}(\epsilon) = \text{id}_{\mathcal{S}_A}, \quad \mathcal{E}(\alpha\beta) = \mathcal{E}(\beta) \circ \mathcal{E}(\alpha).$$

Continuations enlarge the codomain rather than breaking the action.

Definition 15.2 (History functor). The history functor $\mathcal{H}$ maps a causal state to its typed path module and maps a legal transition to path concatenation:

$$\mathcal{H}(g \circ f) = \mathcal{H}(g) \circ \mathcal{H}(f), \quad \mathcal{H}(\text{id}) = \text{id}.$$

It is not a debug logger: its image contributes to structural equality and to future transitions.

Definition 15.3 (Future-quotient functor). For a fixed observer language $\mathcal{L}$, let

$$X \equiv_{\text{fut}} Y \iff \text{Download}_A(wX) = \text{Download}_A(wY) \quad \forall w \in \mathcal{L}^*.$$

The quotient functor $\mathcal{Q}_{\mathcal{L}}$ sends a state to its future class and each opcode f to the induced transfer $[X] \mapsto [f(X)]$. This is well-defined precisely when [Axiom 13.7](#) holds.

Upload and Download have different categorical roles. Upload is a typed ingress from already-

known specifications. Native execution is the high-to-high action. Download is a terminal observation with an explicit causal cut; it is not an inverse functor and supplies no resume morphism.

Theorem 15.4 (Functor identity and composition). *The execution, history, and certified future-quotient maps above preserve identities and composition. Consequently, for legal words α, β ,*

$$\mathcal{E}(\epsilon)X = X, \quad \mathcal{E}(\alpha\beta)X = \mathcal{E}(\beta)(\mathcal{E}(\alpha)X),$$

including equality of the ordered event history and complete endpoint seal.

Proof. The empty word executes no instruction. Concatenating two instruction vectors executes exactly the first vector followed by the second, while path concatenation is associative. Future equivalence is a congruence by [Axiom 13.7](#), so every induced class transfer is well-defined. See [Appendix A.6](#) for the full typed argument. $\square$

16 Class-quantum functions: superposition, entanglement, interference, and cancellation

The adjective *class-quantum* names an algebra of exact deterministic histories. It does not claim a physical quantum computer. Each branch is an ordinary deterministic high-dimensional functor; only the unresolved family is carried simultaneously.

Definition 16.1 (Functor superposition). For deterministic native functors F_i and integer weights w_i , define

$$\boxed{\mathbb{F} = \sum_i w_i[F_i], \quad \mathbb{F}(X) = \sum_i w_i[F_i(X), i].} \quad (\text{Superposition})$$

Branch labels and full causal paths remain present until an explicit interference operation.

Definition 16.2 (Class-quantum entanglement). For left and right history bases (u_i) and (v_j) , a coupled history is

$$\Theta = \sum_{i,j} c_{ij} |u_i\rangle \otimes |v_j\rangle, \quad c_{ij} \in \mathbb{Z}.$$

It is *separable* when $c_{ij} = a_i b_j$ over $\mathbb{Q}$ and *entangled* otherwise. Equivalently, it is entangled exactly when the coefficient matrix has rank greater than one. A nonzero minor

$$c_{ij}c_{k\ell} - c_{i\ell}c_{kj} \neq 0$$

is a finite exact certificate of entanglement.

Definition 16.3 (Structural interference and cancellation). Partition a path family by complete structural endpoint. For each endpoint Y , set

$$W_Y = \sum_{\gamma: F_\gamma(X) \equiv_{\text{str}} Y} w_\gamma.$$

Interference replaces the family by

$$\sum_{Y: W_Y \neq 0} W_Y |Y\rangle.$$

An endpoint is canceled exactly when $W_Y = 0$. Equal ordinary projections do not authorize

aggregation.

Theorem 16.4 (Sign-reversing cancellation). *If a finite history family admits an involution J satisfying*

$$J^2 = \text{id}, \quad F_{J\gamma}(X) \equiv_{\text{str}} F_{\gamma}(X), \quad w_{J\gamma} = -w_{\gamma},$$

and J has no fixed point, then the structurally interfered packet is zero. Conversely, two paths sharing only an ordinary observation cannot be canceled by this theorem.

Proof. Every orbit is a pair $\{\gamma, J\gamma\}$ with a common structural endpoint and total weight $w_{\gamma} - w_{\gamma} = 0$. Ordinary equality alone does not place two paths in the same structural endpoint class. $\square$

Example 16.5 (Exact entanglement witness). The coefficient matrix

$$C = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

has minor 1, hence is class-quantum entangled. By contrast, an outer product $C = (1, 2)^{\top}(3, 4)$ has all 2×2 minors zero and is separable. The SDK checks these integer identities directly; it does not estimate a floating numerical rank.

17 Exact correspondence with the modular SDK

The abstract arithmetic and the executable package must not be conflated. The shipped finite chart realizes sparse signed three-class values, ordered HMUL, central-scalar quotient/residual HDIV, complete maze addresses, native and history functors, future partitions, structural interference and exact tensor minor certificates. Unsupported coefficient or denominator cases return a typed continuation. The principal-jet factorial bridge is a separate specialized functor proved in [Section 33](#).

This summary is deliberately mathematical. Concrete type names, source files, test executables and negative-compilation gates appear only in [Appendix G](#) and [Table 22](#). That placement preserves the theorem narrative while retaining a line-by-line engineering contract.

Theorem 17.1 (Scoped SDK soundness). *Within the signed coefficient range and the declared central-scalar division chart, the new SDK layer realizes [Equations \(1\)](#), [\(2\)](#), (HMUL) and (HDIV-E); preserves the singular two-sided ideal; satisfies the execution-functor identity and composition laws; cancels only structurally equal endpoints; and certifies class-quantum entanglement only by a nonzero exact minor. Unsupported cases return typed continuations retaining their operands.*

Proof and evidence boundary

The source-level proof follows the sparse coefficient loops, ordered word concatenation, exact quotient/remainder identity, instruction-vector concatenation, endpoint grouping, and minor formula; it is written out in [Appendix A.7](#). Finite compiled checks of the algebra laws, functor laws, division reconstruction, observation tamper rejection, superposition, cancellation, separability, and entanglement are reported only in the evidence appendix.

Still not implemented

The public layer does not yet provide arbitrary-precision signed coefficients, a generalized inverse for every noncentral high-dimensional denominator, a compact compiler for every cyclic or causal-kernel chart, or a native prime-birth functor. The new layer closes the missing arithmetic and semantic definition gap; it does not erase the previously stated complexity frontier.

18 Fold multiplication: from an interval line to a causal tree

The phrase *fold multiplication* names the operation that first made the factorial target look like a maze rather than a scalar. Its classical shadow is a balanced product tree. Its Angel lift retains, in addition to the root product, the interval carried by every node, the ordered graft history, the local jet signature and any unresolved continuation. The extra structure is not created by changing the value of multiplication; it is created by refusing to identify all parenthesizations and all intermediate certificates after they have the same ordinary product.

A tree is not automatically a new dimension

Ordinary product trees, divide-and-conquer multiplication and binary splitting are established mathematics. Merely drawing a tree does not produce a new number system. The new formal object in this memoir is the *typed lift*: tree address, ordered history, jet data and singular continuation remain part of the state and are acted on by the Angel functors. All speed claims continue to be charged in the underlying arithmetic model.

18.1 Ordered interval trees and their folds

Let A be an associative algebra. The algebra need not be commutative. Let $a_1, \dots, a_n \in A$ be an ordered word.

Definition 18.1 (Ordered fold tree). An *ordered fold tree* $\mathcal{T}$ on the interval $[1, n]$ is a rooted full binary tree whose leaves, from left to right, are $1, 2, \dots, n$, and whose every internal node carries a contiguous interval. If a node carrying $[r, s]$ splits at c , its left child carries $[r, c]$ and its right child carries $[c + 1, s]$. Define recursively

$$\text{Fold}_{\mathcal{T}}(a_r, \dots, a_s) = \begin{cases} a_r, & r = s, \\ \text{Fold}_{\mathcal{T}_L}(a_r, \dots, a_c) \text{Fold}_{\mathcal{T}_R}(a_{c+1}, \dots, a_s), & r < s. \end{cases}$$

The order of the two children is part of the data.

Theorem 18.2 (Value invariance and history nonuniqueness). *For every ordered fold tree $\mathcal{T}$,*

$$\text{Fold}_{\mathcal{T}}(a_1, \dots, a_n) = a_1 a_2 \cdots a_n.$$

Thus associativity makes the ordinary root value independent of the tree. Nevertheless, two different trees have different graft words and hence define different Angel histories unless an explicit future-sufficient quotient identifies them. In a noncommutative history algebra, an unordered exchange of two leaves is not authorized by this theorem.

Proof idea

Induction on the number of leaves proves value invariance while preserving leaf order. History nonuniqueness follows because the free graft alphabet records the split interval at every internal node. The full argument and the precise future-quotient statement are in [Appendix A.4](#).

For a balanced split, the longest root-to-leaf path has length $\lceil \log_2 n \rceil$, but the expanded tree still has n leaves and $n - 1$ internal multiplications. Depth, work, stored state and output materialization must therefore be kept on separate ledgers.

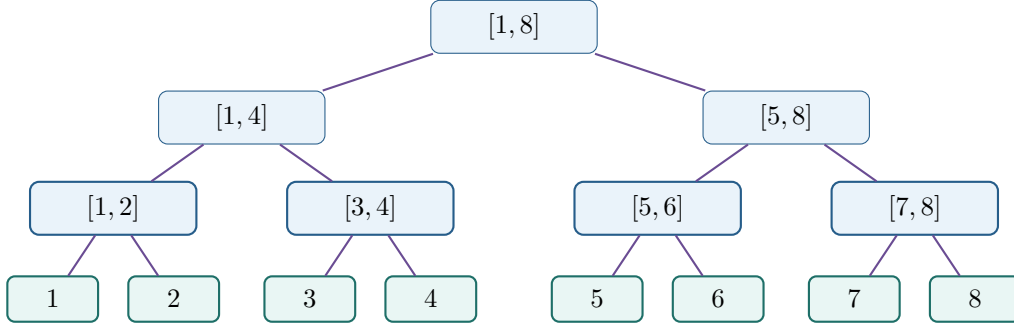


Figure 3: A balanced ordered interval fold. Node intervals and graft order are retained by the Angel history even though associative evaluation has the same ordinary root product as a left-to-right chain.

18.2 The q -Pochhammer family is closed under interval folding

The exact connection to the paper’s source programme is obtained by using half-open intervals.

Definition 18.3 (q -Pochhammer interval section). For $0 \leq a \leq b$, define

$$F_{[a,b)}(Z, u) = \prod_{j=a}^{b-1} (1 + Zu^j) = F_{b-a}(Zu^a, u).$$

The empty interval has value one.

Theorem 18.4 (Exact fold law). For every $a \leq c \leq b$,

$$F_{[a,b)}(Z, u) = F_{[a,c)}(Z, u) F_{[c,b)}(Z, u).$$

Consequently every ordered interval tree evaluates to $F_{[a,b)}$, and the binary q -Pochhammer programme is a compressed grammar for a distinguished sequence of such folds and shifted copies.

Relation to the concatenation theorem

The factors on the right partition the factor indices on the left. Writing the second interval as $F_{b-c}(Zu^c, u)$ recovers [Theorem 29.2](#). The tree theorem follows by induction and is proved in full in [Appendix A.4.1](#).

Definition 18.5 (Fold contraction functor). Fix $[a, b)$. Let $\text{Part}[a, b)$ have ordered contiguous partitions as objects and adjacent block merges as generating arrows. Let $\text{List}(A)$ have ordered lists of A -values and adjacent multiplication as contractions. The map

$$\mathcal{V}_{Z,u} : \text{Part}[a, b) \longrightarrow \text{List}(\mathbb{Z}[Z, u])$$

sends a block I to $F_I(Z, u)$ and sends a merge of adjacent blocks to their ordered product. Its history lift $\widehat{\mathcal{V}}_{Z,u}$ additionally appends the interval split, parent seals and graft opcode to the Angel state.

Theorem 18.6 (Fold-functor laws). *The contraction map preserves identity refinements and composition of adjacent merges. Every complete contraction has the same polynomial endpoint $F_{[a,b]}$, while its lifted endpoint retains the contraction tree. Forgetting that tree is a many-to-one natural projection, not an inverse functor.*

Proof location

Identity is the empty merge sequence; composition is concatenation of merge sequences; and the q-Pochhammer fold law supplies the adjacent-contraction square. Coherence and the lifted-history distinction are proved in [Appendix A.4.2](#).

18.3 Factorial is the root principal jet

After setting $Z = -(1 + T)$ and $u = 1 + T$, put

$$G_{[a,b]}(T) = \prod_{k=a+1}^b (1 - (1 + T)^k).$$

Attach to every fold node the signature

$$J[a, b] = (\ell_{a,b}, c_{a,b}), \quad \ell_{a,b} = b - a, \quad c_{a,b} = (-1)^{b-a} \prod_{k=a+1}^b k.$$

Here ℓ is the local T -valuation and c is the first nonzero coefficient.

Theorem 18.7 (Nodewise factorial-jet fold). *If $a \leq c \leq b$, then*

$$G_{[a,b]} = G_{[a,c]} G_{[c,b]}, \quad \ell_{a,b} = \ell_{a,c} + \ell_{c,b}, \quad c_{a,b} = c_{a,c} c_{c,b}.$$

At the root $[0, n]$,

$$J[0, n] = (n, (-1)^n n!).$$

Thus every ordered interval fold carries a local principal-jet certificate, and the root certificate is the factorial theorem.

This theorem says more than “a product tree computes factorial” and less than “the tree is the whole Angel space”. It says that valuation and leading coefficient form a compositional node invariant. The higher coefficients are also compositional, but they require truncated series convolution rather than only the pair (ℓ, c) .

Proof and higher-jet boundary

Disjoint interval factorization proves the polynomial identity; valuations add and leading coefficients multiply. The full proof, including the truncated higher-jet convolution, is in [Appendix A.4.3](#).

18.4 What the fold contributes to the maze coordinates

The fold tree supplies concrete data for several maze fields, but it does not make every field synonymous with a tree-theoretic word.

Table 4: Exact fold-tree contributions to an Angel maze state.

Maze field	Fold realization	Important limit
Room	a certified contiguous interval node $[a, b]$	a room may later carry more than a product interval
Road	a left/right child edge or an adjacent merge	an arbitrary maze road need not belong to a tree
Frame	tree depth, split convention and local chart	depth is not Euclidean dimension
History	the ordered word of splits, grafts and contractions	two equal root products may retain different words
Residual	an uncanceled quotient remainder or unresolved sibling payload	not every residual is caused by folding
HigherCell	requested higher-jet order and its truncated series	the leading pair (ℓ, c) does not determine it
Holonomy	only a certified comparison around a genuine closed transport loop	a tree has no nontrivial loop, so folding alone creates no holonomy
Certificate	interval bounds, child seals, factor count and parent binding	a digest without replay is not a proof
Program	a compact double/append or explicit merge grammar	short syntax is not short expanded evaluation

18.5 Depth is not work: the fold complexity ledger

Theorem 18.8 (Separated fold ledgers). *For n leaves in a balanced ordered binary product tree:*

1. *structural depth is $\lceil \log_2 n \rceil$;*
2. *an explicit full tree has $n - 1$ internal multiplication nodes;*
3. *a binary q -Pochhammer source grammar has $O(\log n)$ opcodes because it describes shifted factor families rather than listing every expanded node;*
4. *evaluation cost depends on the coefficient representation and the multiplication cost $M(d)$;
and*
5. *printing $n!$ requires output proportional to the bit length of $n!$.*

No one of these five quantities may be substituted for another.

Accounting proof

The first two facts are elementary binary-tree counts, the third is [Theorem 29.4](#), and the final two follow from the declared polynomial and bit-output models. A complete proof with the non-power-of-two case is in [Appendix A.4.4](#).

19 The fold–shadow collision at a prime room

The factorial fold and cyclic shadow meet modulo a prime, but their carriers must not be identified. For a vector $v = (v_0, \dots, v_{m-1})$ in a commutative ring, define its coordinate fold along any ordered fold tree by

$$\text{CFold}(v) = \prod_{j=0}^{m-1} v_j.$$

Associativity and commutativity make this scalar independent of the selected tree.

Theorem 19.1 (Prime fold–shadow collision). *Let p be prime. The first row of the cyclic operator $K_p = pI - J$ is*

$$w_p = (p-1, -1, \dots, -1) \in \mathbb{Z}^p.$$

Then

$$\text{CFold}(w_p) = (p-1)(-1)^{p-1} \text{equiv} - 1 \pmod{p}.$$

The infinitesimal chart independently gives

$$(-1)^{p-1} [T^{p-1}] F_{p-1}(-(1+T), 1+T) = (p-1)! \equiv -1 \pmod{p}.$$

Hence the coordinate fold of the cyclic row and the factorial principal jet have the same terminal residue in a prime room.

Why this is a collision, not an identification

The first congruence is a direct coordinate product; the second is Wilson’s theorem applied to the proved principal jet. Their unreduced integers, carrier dimensions and histories differ. The complete comparison is in [Appendix A.5](#).

The p versus $p-1$ boundary

The cyclic row has p coordinates indexed by rooms $0, \dots, p-1$. The factorial product $(p-1)!$ has $p-1$ factors indexed by $1, \dots, p-1$. There is no claimed leaf-by-leaf bijection. Equality of the two residues is a meeting of two chart observations, not proof that one tree is the other or that either shadow reconstructs the full maze.

For a composite room, the cyclic row is governed by Ramanujan sums and proper period sectors. It is not generally the pure vector $(m-1, -1, \dots, -1)$. The pure Laplacian form, rather than a coordinate-product test alone, is the proved prime characterization in [Theorem 77.1](#).

20 An interval-complete product as a probe of discrete primes

The motivating phrase “use a continuous product to capture discrete points” is useful only after one terminological correction. The finite product $1 \cdot 2 \cdots n$ is not continuous in a topological sense. It is *interval-complete*: every positive integer in a contiguous interval appears once. The q-Pochhammer lift then places that discrete factor word in an algebraic family with infinitesimal and cyclic charts.

Proposition 20.1 (Interval-complete arithmetic envelope). *Let $P_n = n!$. Then:*

1. for every prime $q \leq n$, the exponent carried by the envelope is

$$v_q(P_n) = \sum_{r \geq 1} \left\lfloor \frac{n}{q^r} \right\rfloor;$$

2. the q -Pochhammer lift retains the entire ordered factor interval and has P_n as its principal-jet coefficient; and
3. for every supplied $m > 1$, the terminal observation $P_{m-1} \equiv -1 \pmod{m}$ holds exactly when m is prime.

Thus factorial packages all smaller divisibility strata before a prime-specific terminal observation is chosen.

The proposition explains why factorial was “the easiest object to hit” in the research history. It turns sparse prime points into exceptional responses of one interval-complete product family. It does not make the prime set continuous, prove a law for prime gaps, or replace candidate testing by a native birth functor.

Proof location

Counting multiples of $q, q^2, \dots$ proves the valuation formula. The second clause is the factorial principal-jet theorem and the third is Wilson. The complete proof and scope statement are in [Appendix A.5.1](#).

What higher jets remember

The coefficients of T^{n+r} record exact symmetric combinations of the higher Taylor terms of the local factors. They do not, by themselves, determine which balanced or unbalanced product tree was used, because associative multiplication erases parenthesization at the polynomial endpoint. Tree topology survives only in the separate History/Program/Certificate coordinates. Conversely, the tree alone does not determine every higher jet unless the local factor series are also supplied.

21 Concrete arithmetic before abstraction

The three-sector formulas become easier to read once one complete product is written out. Work in the free history algebra on letters α, β, γ and let the singular ideal be generated by γ . Put

$$X = (2, [\alpha], 0), \quad Y = (3, [\beta], [\gamma]).$$

Brackets here denote basis words, while integer multiples and sums are taken in the sparse integral path algebra.

21.1 HADD and HSUB by hand

Addition and subtraction are componentwise:

$$X \boxplus Y = (5, [\alpha] + [\beta], [\gamma]),$$

and

$$X \boxplus_A Y = (-1, [\alpha] - [\beta], -[\gamma]).$$

The formula is intentionally simple. The higher-dimensional content is not an exotic scalar addition law; it is the refusal to erase path and singular information merely because an ordinary scalar projection is available.

21.2 HMUL by hand

Using

$$(r, h, s) \boxtimes_A (r', h', s') = (rr', rh' + r'h + hh', rs' + r's + hs' + sh' + ss'),$$

we obtain

$$X \boxtimes_A Y = (6, 2[\beta] + 3[\alpha] + [\alpha\beta], 2[\gamma] + [\alpha\gamma]).$$

The omitted terms are zero because X has no singular component. Reversing the order gives

$$Y \boxtimes_A X = (6, 3[\alpha] + 2[\beta] + [\beta\alpha], 2[\gamma] + [\gamma\alpha]).$$

Both ordinary shadows are six, but the two products are structurally unequal in a free path algebra. The terms $[\alpha\gamma]$ and $[\gamma\alpha]$ also show the two-sided ideal law: multiplying a singular word on either side stays singular.

Example 21.1 (Signed executable HDIV). Let

$$Z = (7, 5[\alpha], -3[\gamma])$$

and divide by the central integer two using signed truncation toward zero. The executable quotient and residual are

$$Q_2(Z) = (3, 2[\alpha], -[\gamma]), \quad R_2(Z) = (1, [\alpha], -[\gamma]).$$

Coefficient by coefficient,

$$\iota(2) \boxtimes_A Q_2(Z) \boxplus_A R_2(Z) = Z.$$

The negative coefficient is important: $-3 = 2(-1) + (-1)$ under truncation toward zero. The residual is not an error code. It is the exact data that makes reconstruction true.

Where the HMUL terms come from

Adjoin the ordinary scalar as the unit component of the path algebra, retain ordinary non-singular words in Hist, and collect every word containing a singular generator in Sing. Expanding $(r + h + s)(r' + h' + s')$ and sorting the nine terms into those three summands gives the displayed HMUL formula. Associativity is inherited from the ambient path algebra; it is not guessed term by term.

22 Three equalities and a future-separation example

Structural equality, current observation equality and future equality answer different questions:

$$X \equiv_{\text{str}} Y \implies X \equiv_{\text{fut}} Y \implies X \equiv_{\text{obs}} Y,$$

but neither converse holds without additional hypotheses.

Let the current observer retain only Ord and set

$$A = (0, [\alpha], 0), \quad B = (0, [\beta], 0).$$

Then $A \equiv_{\text{obs}} B$. Take the legal continuation

$$C = (0, 0, [\gamma]).$$

The products have singular coordinates

$$\text{Sing}(A \boxtimes_A C) = [\alpha\gamma], \quad \text{Sing}(B \boxtimes_A C) = [\beta\gamma].$$

Let $\omega_{\alpha\gamma}$ be the terminal coefficient observer for the word $\alpha\gamma$. It returns one on $A \boxtimes_A C$ and zero on $B \boxtimes_A C$. Thus A and B are presently observation-equivalent but are not future-equivalent.

This example is deliberately minimal. In the factorial bridge, the two states may instead have the same coefficient one while different argument, program and request words lead to different lawful Wilson observations. In Volume V, two paths may be future-equivalent only after a declared curvature residual has been proved future-silent. Current scalar equality never proves either statement.

23 Why the phrase “class-quantum” is used, and its exact limit

The terminology “superposition”, “entanglement”, “interference” and “cancellation” is used for an algebra of signed, structured histories. It does not assert a physical quantum system. The distinction is summarized below.

Feature	Angel class-quantum history algebra	Physical quantum formalism
Carrier	Finite sparse integer-weight combinations of typed history words.	Usually complex Hilbert-space rays or density operators.
Superposition	Formal additive combination of certified functor branches.	Linear combination of amplitudes with physical state normalization.
Interference	Endpoint aggregation may add positive and negative integral weights.	Complex amplitudes interfere before a measurement probability is formed.
Entanglement	A tensor history has flattening rank greater than one and is not a simple product.	Nonseparability of a composite physical state.
Observation	A typed deterministic terminal map selected by the interface.	A measurement model with probabilistic predictions and an interpretation.
Required laws	Exact finite algebra, functoriality, certificates and causal boundary.	Inner product, positivity, Born rule, unitarity or quantum-channel axioms, depending on the model.
Claim	Structural analogy and a useful signed path calculus.	A theory of physical quantum phenomena.

The name is useful because ordinary nonnegative probability cannot express exact signed cancellation, and a single scalar accumulator cannot express nonseparable tensor history. The qualifier “class-” is equally essential: no Born rule, Planck constant, unitary time evolution, physical qubit or experimental quantum advantage is inferred.

Publication rule for quantum language

Every theorem in this memoir remains true if the reader mentally replaces “class-quantum” by “signed tensor path algebra”. The evocative vocabulary names the pattern of operations; it supplies no physical premise and no additional proof.

24 How q-Pochhammer, factorial and the cyclic chart fit together

The finite product

$$(a; q)_n = \prod_{j=0}^{n-1} (1 - aq^j)$$

is standard in partition theory and basic hypergeometric series [1, 2]. This memoir does not claim to invent that notation or the product-tree idea. Its source polynomial

$$F_n(Z, u) = \prod_{j=0}^{n-1} (1 + Zu^j) = (-Z; u)_n$$

is chosen because the same integral expression admits multiple exact charts.

In the principal-jet chart, substituting

$$Z = -(1 + T), \quad u = 1 + T$$

causes every nontrivial factor to contribute a controlled zero at $T = 0$; the first nonzero coefficient is $(-1)^n n!$. This chart supplies the executable factorial coordinate.

In the cyclic chart, let S be the generator of C_m and substitute $u = S$. The q-Pochhammer factors become group-ring elements of the form

$$1 - S^j.$$

Thus $\mathbb{Z}[C_m]$ is not an unrelated algebra inserted late in the paper. It is the natural target of evaluating the same integral source at a cyclic shift. Characters diagonalize that shift, so exact-period sectors and closure depth become visible. The two charts share a source polynomial and differ in the meaning of the target coordinate.

The algorithmic position is also precise. Binary splitting, product trees, fast multipoint evaluation and polynomial multiplication are established techniques [4, 5, 6]. The new claim here is the proof-bound state architecture around those tools: the three-sector embedding, compact source certificate, projection firewall, factorial-maze functor, joint Wilson consumer and cyclic/causal continuation.

25 What the polynomial cost symbol $M(d)$ means

Let $M_R(d)$ denote the cost of multiplying two degree-less-than- d polynomials over a specified coefficient ring R . It is a parameter because the correct bound depends on the carrier and the

chosen exact backend, but it is not an uncharged black box.

1. Schoolbook convolution performs $\Theta(d^2)$ coefficient products and additions and is the reference path for small degrees and differential tests.
2. A suitable transform prime supports a number-theoretic transform in $O(d \log d)$ ring operations for one modular convolution, including the inverse transform and normalization.
3. Exact integer coefficients may require several transform primes and Chinese-remainder reconstruction. If k primes cover the proved coefficient bound, the transform part costs $O(kd \log d)$ modular operations, plus exact mixed-radix or CRT reconstruction and coefficient-normalization cost.
4. Over an arbitrary algebra, Cantor–Kaltofen type multiplication provides a quasi-linear algebraic bound under its stated model [6]. This is a mathematical upper bound, not a statement that the shipped fixed-word backend invokes that algorithm in every case.

The current implementation uses a thresholded exact strategy: small objects use directly audited coefficient arithmetic, while supported modular paths may use transform convolution. Consequently every displayed complexity that contains $M(d)$ should be read as a family of explicit instantiations, for example

$$M(d) = \Theta(d^2) \quad \text{or} \quad M(d) = O(d \log d)$$

in their respective arithmetic models. Bit complexity additionally charges coefficient growth, modular reductions and reconstruction.

Why retain $M(d)$ in the theorem

Writing only $O(d^2)$ would hide the faster exact backend; writing only $O(d \log d)$ would misdescribe the small/reference path and omit coefficient growth. The parameterized theorem is the common algebraic contract. The evidence appendix records which backend and coefficient domain each finite run actually used.

26 A numeric closure calculation: why two factors d^r appear

The cyclic closure theorem gives, on the exact-period- d sector,

$$\mathcal{F}_{m,N}|_{V_d} = (-1)^r r! d^{2r} \varepsilon^r + O(\varepsilon^{r+1}), \quad r = \frac{Nm}{d}.$$

Take $m = 6$, $N = 1$ and $d = 3$. Then $r = 2$. The derivative of the two vanishing cycle factors contributes

$$d^r = 3^2 = 9.$$

The product of the nonzero residue-class blocks contributes a second

$$d^r = 3^2 = 9.$$

Finally the ordering of the two simple vanishing factors contributes $2!$ and $(-1)^r = 1$. Therefore

$$\boxed{\mathcal{F}_{6,1}|_{V_3} = 2! \cdot 3^2 \cdot 3^2 \varepsilon^2 + O(\varepsilon^3) = 162\varepsilon^2 + O(\varepsilon^3).}$$

The notation d^{2r} therefore records two independent mechanisms. It is not the same local derivative contribution counted twice by accident.

27 Canonical terminology used from this point onward

The primary term is *Angel Prime causal maze state*. “Angel maze object” is its short form. An *Angel value* is only the Ord/Hist/Sing algebraic fibre. A *causal state* is a complete maze state with value, address, ordered history, residual, program and certificate fields. A *terminal observation* is an explicitly typed projection with no return edge. A *harmonic child* is Volume V’s high-dimensional prime object before its ordinary grade is downloaded.

The word “Angel” refers to the direction of construction: a complete higher state is transmitted down to a lawful lower-dimensional observation, while the observation is forbidden to masquerade as the source of the state. The name is mnemonic, not a theological or physical hypothesis. The word “Prime” identifies the target research program: native causal transitions whose terminal shadows are prime observations. The factorial/Wilson chart is the naive exact application; Volume V studies the stronger state-to-state birth interface.

28 The scoped SDK realization and causal boundary

28.1 State-based arithmetic

The preceding sections define the general three-class arithmetic and its functors. This section specifies the boundary datum shared by the original factorial implementation and the newly added public high-dimensional module. It is a scoped executable realization of the axioms, not a replacement for them.

Definition 28.1 (Scoped Angel runtime datum). A scoped Angel runtime realization is a tuple

$$\mathfrak{A} = (\mathcal{S}_A, \mathcal{O}, \text{Upload}_A, \text{Download}_A, \mathcal{L}, \pi)$$

with the following data.

1. $\mathcal{S}_A$ is a set of finitely and exactly encodable states.
2. $\mathcal{O}$ is a set of ordinary observations.
3. Upload_A is a typed ingress map from known ordinary inputs and specifications to complete states; it is not an answer constructor.
4. $\mathcal{L}$ is a declared language of native partial operators whose inputs and outputs remain in $\mathcal{S}_A$, possibly enlarged by typed continuation states.
5. Download_A is the single ordinary egress relation from a frozen state to an observation.
6. π is a mathematical denotation map used to state compatibility with ordinary arithmetic; it is not an implicit runtime conversion.

No element of $\mathcal{O}$ has a legal coercion back into a native input unless a new explicit upload event is separately authorized.

The definition deliberately distinguishes denotation from execution. A state may denote an

integer while preserving interval endpoints, history, operator program, valuation, provenance and a continuation obligation that the ordinary integer alone would erase.

Definition 28.2 (Three equalities). For states $X, Y \in \mathcal{S}_A$ we distinguish:

$$\begin{aligned} X \equiv_{\text{str}} Y & \quad \text{if their complete certified structures agree,} \\ X \equiv_{\text{obs}} Y & \quad \text{if a selected ordinary observer returns the same value,} \\ X \equiv_{\text{fut}} Y & \quad \text{if no word in the declared future language distinguishes them.} \end{aligned}$$

Observational equality does not imply structural equality. Future equivalence is language-relative and requires a proof or verified quotient construction.

Example 28.3 (Residue collision without state equality). The ordinary congruence

$$4! \equiv 5! \equiv 0 \pmod{8}$$

does not identify the two factorial states. Their interval targets, source programs and state seals differ. A modulus-eight observer may collapse them; the native arithmetic may not.

28.2 Finite exact realizability

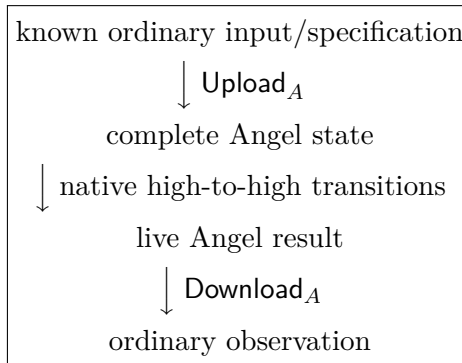
Principle 28.4 (Finite exact realizability). *Every runtime field used by native execution must have a finite exact digital encoding. Exact-real oracles, floating approximations presented as proofs, irrational payloads, and factorial-sized rational denominators hidden behind a constant-size type name are outside the model.*

A rational number may of course be encoded by two integers, and some algebraic numbers by a polynomial plus a root selector. The issue is not philosophical hostility to those objects; it is resource accounting. A denominator equal to $n!$ already contains the output information. A symbolic expression such as “ $\Gamma(n+1)$ ” is likewise not a native evaluation unless subsequent operators consume it without expanding the hidden work.

The factorial construction below uses only nonnegative integers, finite opcode sequences, exact interval endpoints, parity, fixed-size seals and certificates. The cyclic proofs later use characteristic-zero scalar extension as a proof language, while the source operators themselves lie in an integral group ring.

28.3 The causal boundary

Protocol 28.5 (Canonical causal order). The allowed architectural order is



Between ingress and egress, the path

Angel state $\rightarrow$ ordinary value $\rightarrow$ classical computation $\rightarrow$ wrapped Angel state

is forbidden.

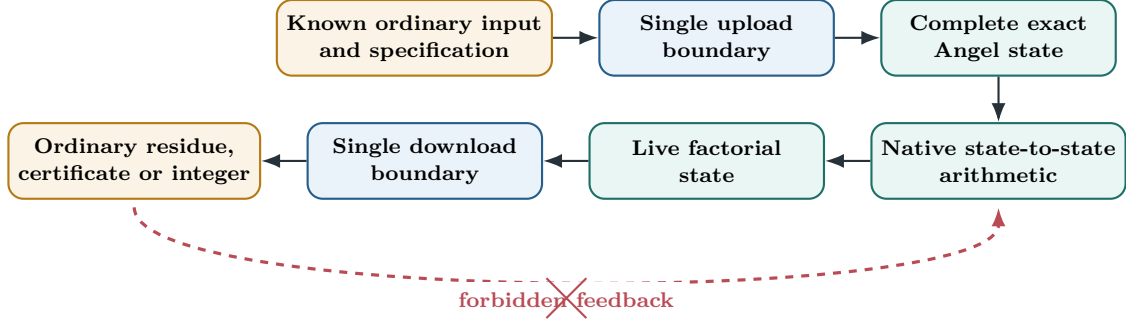


Figure 4: The causal boundary. Ordinary observations are terminal with respect to the current native execution. The ingress semantics are specified while the standalone implementation remains open; the egress firewall is executable.

Proposition 28.6 (No-feedback consequence). *Under Protocol 28.5, two runs that differ only in an ordinary observer’s result cannot alter the already frozen native state or select a native branch.*

Proof. By definition, the observer’s codomain is $\mathcal{O}$, and no map from $\mathcal{O}$ to a native input is present in the declared execution graph. Therefore the native state and its transitions are functions only of the uploaded state and native operators, not of the subsequent observation. The companion implementation strengthens this semantic statement with deleted conversions, negative compilation tests and before/after state-identity witnesses. $\square$

Remark 28.7. The final standalone ingress API is not completed in the current implementation. The semantic restrictions on ingress are part of the mathematics; the final software surface remains an explicit open interface. No existing convenience wrapper is rebranded as a completed upload audit.

29 The q-Pochhammer source program

29.1 The universal polynomial family

Definition 29.1 (Finite q-Pochhammer section). For $n \geq 0$, define

$$F_n(Z, u) = \prod_{j=0}^{n-1} (1 + Zu^j) \in \mathbb{Z}[Z, u], \quad F_0(Z, u) = 1.$$

Theorem 29.2 (Concatenation). For all $a, b \geq 0$,

$$F_{a+b}(Z, u) = F_a(Z, u)F_b(Zu^a, u).$$

Proof. Split the defining product at a :

$$\begin{aligned}
 F_{a+b}(Z, u) &= \prod_{j=0}^{a-1} (1 + Zu^j) \prod_{j=a}^{a+b-1} (1 + Zu^j) \\
 &= F_a(Z, u) \prod_{r=0}^{b-1} (1 + Zu^{a+r}) \\
 &= F_a(Z, u) F_b(Zu^a, u).
 \end{aligned}$$

□

Two special cases are the binary recurrences

$$F_{2a}(Z, u) = F_a(Z, u) F_a(Zu^a, u), \quad (3)$$

$$F_{2a+1}(Z, u) = F_{2a}(Z, u) (1 + Zu^{2a}). \quad (4)$$

They generate a compact operator program from the bits of n .

Definition 29.3 (Binary q-Pochhammer program). Let $n = (b_{L-1} \cdots b_1 b_0)_2$ with $b_{L-1} = 1$. Begin with the length-one section. Reading the remaining bits from most significant to least significant, apply the doubling rule (3); if the next bit is 1, apply the append rule (4). The zero input is represented by the unit.

Theorem 29.4 (Logarithmic program size). For $n \geq 1$ with $L = \lfloor \log_2 n \rfloor + 1$, the binary q-Pochhammer program contains

$$1 + (L - 1) + (\text{popcount}(n) - 1) \leq 2L - 1$$

opcodes. Its represented length is exactly n .

Proof. There is one seed opcode, one doubling opcode for each of the remaining $L - 1$ bits, and one append opcode for each remaining bit equal to 1. This gives the stated count. If the current represented length is a , a doubling step changes it to $2a$ and an optional append changes it to $2a + 1$. The recurrence is therefore exactly the usual left-to-right binary reconstruction of n . □

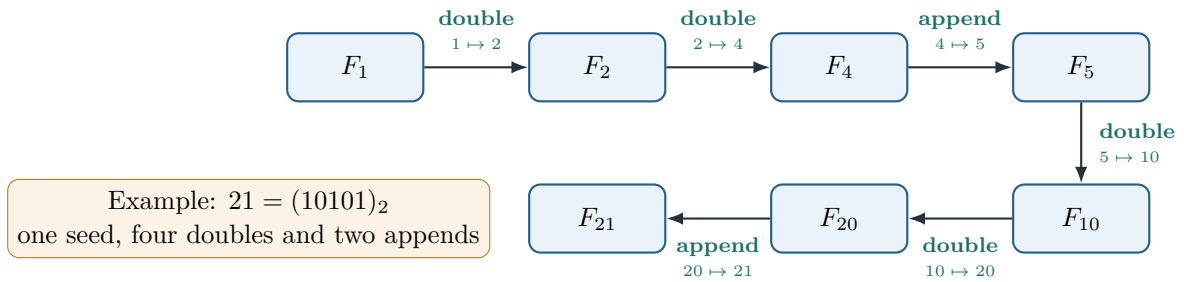


Figure 5: A compact q-Pochhammer program follows the input bits. It represents the whole factor family without storing n individual factors.

29.2 Certification and what it proves

The executable program carries its input length, opcode sequence and a seal. The verifier replays the exact binary grammar and rejects mutation of the length, opcode order or seal. This verification

proves that the compact program represents the intended q-Pochhammer length. It does not replace the mathematical proof of the principal-coefficient theorem in the next section.

Remark 29.5 (Theorem versus certificate). The theorem

$$[T^n]F_n(-(1+T), 1+T) = (-1)^n n!$$

is proved in mathematics. The software verifier binds a concrete state to the certified source program, interval, valuation, parity and provenance. A Boolean certificate field is not presented as an independent formal proof assistant for the theorem.

30 The principal-jet factorial theorem

30.1 From q-Pochhammer to q-factorial

Set $Z = -u$. Then

$$F_n(-u, u) = \prod_{j=0}^{n-1} (1 - u^{j+1}) = \prod_{k=1}^n (1 - u^k).$$

Writing

$$[k]_u = 1 + u + \cdots + u^{k-1}, \quad 1 - u^k = (1 - u)[k]_u,$$

gives

$$F_n(-u, u) = (1 - u)^n [n]_u!, \quad [n]_u! = \prod_{k=1}^n [k]_u.$$

The specialization $u = 1$ satisfies $[k]_1 = k$, but direct substitution into the left-hand side gives a zero of order n . The factorial is therefore carried by the principal local coefficient of that singularity.

Theorem 30.1 (Principal-jet factorial identity). *For every $n \geq 0$,*

$$\boxed{F_n(-(1+T), 1+T) = (-1)^n n! T^n + O(T^{n+1}).}$$

Equivalently,

$$\boxed{n! = (-1)^n [T^n]F_n(-(1+T), 1+T).}$$

Proof. For $1 \leq k \leq n$,

$$1 - (1+T)^k = -kT + O(T^2).$$

Multiplying the n factors gives a zero of order exactly n . The coefficient of T^n can only be obtained by choosing the linear term from every factor; therefore it is

$$\prod_{k=1}^n (-k) = (-1)^n n!.$$

□

The theorem admits an interval form that becomes the native arithmetic coordinate.

Definition 30.2 (Factorial interval). For integers $0 \leq a \leq b$, define

$$\Phi[a, b] = \prod_{k=a+1}^b k = \frac{b!}{a!},$$

where the quotient notation denotes the integer identity, not runtime rational arithmetic.

Theorem 30.3 (Interval principal jet). *Let*

$$G_{a,b}(T) = \prod_{k=a+1}^b \left(1 - (1+T)^k\right).$$

Then

$$G_{a,b}(T) = (-1)^{b-a} \Phi[a, b] T^{b-a} + O(T^{b-a+1}).$$

Proof. There are $b - a$ factors. The linear term of the factor indexed by k is $-kT$. The first possible nonzero degree is therefore $b - a$, with coefficient

$$\prod_{k=a+1}^b (-k) = (-1)^{b-a} \Phi[a, b].$$

□

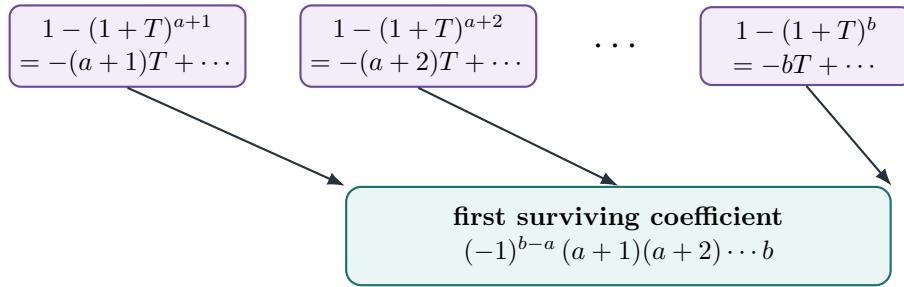


Figure 6: The factorial interval is the principal coefficient of an exact integral singular jet. No rational or irrational runtime value is introduced.

30.2 The factorial state

Definition 30.4 (Certified factorial state). A certified factorial state of length n is a tuple

$$\mathcal{Q}_n = (P_n, [0, n], n, (-1)^n, C_n),$$

where P_n is a certified binary q-Pochhammer program, $[0, n]$ is the factorial interval, n is the principal valuation, $(-1)^n$ is the principal sign, and C_n binds these components and their provenance. Its ordinary factorial denotation is

$$\pi(\mathcal{Q}_n) = n!$$

by Theorem 30.1.

The state is finite: the program has $O(\log n)$ opcodes, the endpoints use $O(\log n)$ bits, and the remaining fields are fixed-size or logarithmic. It is also exact: no coefficient approximation, floating point, or truncated ordinary factorial is present.

Definition 30.5 (Native factorial operator). Let $\mathcal{I}_n$ denote an uploaded input state whose ordinary input coordinate is n . The native factorial operator is the state map

$$\boxed{\text{AFac} : \mathcal{I}_n \mapsto \mathcal{Q}_n},$$

where $\pi(\mathcal{I}_n) = n$ and $\pi(\mathcal{Q}_n) = n!$. The codomain is the live factorial state of Definition 30.4, not a materialized binary expansion.

The index n records the factorial length and provenance. It does not assert that the $\Theta(n \log n)$ ordinary output bits of $n!$ have been printed or that arbitrary ordinary integer operations are free.

31 A scoped factorial-state algebra

31.1 Partial multiplication and exact cancellation

Theorem 31.1 (Factorial-interval composition). For $0 \leq a \leq b \leq c$,

$$\Phi[a, b]\Phi[b, c] = \Phi[a, c], \quad (5)$$

$$\Phi[a, c]/\Phi[a, b] = \Phi[b, c], \quad (6)$$

$$\Phi[a, c]/\Phi[b, c] = \Phi[a, b]. \quad (7)$$

The two quotients are exact integer cancellation laws.

Proof. The factors in $\Phi[a, b]$ and $\Phi[b, c]$ are disjoint consecutive intervals whose union is the interval of $\Phi[a, c]$, proving (5). Cancelling the corresponding certified prefix or suffix factors gives (6) and (7). $\square$

These laws define partial native operations on certified principal-jet states:

$$J[a, b] \boxtimes J[b, c] = J[a, c],$$

$$J[a, c] \oslash_A J[a, b] = J[b, c], \quad J[a, c] \oslash_A J[b, c] = J[a, b].$$

No inverse in $\mathbb{Q}$ is formed. The denominator must be a certified causal factor of the numerator in the interval algebra.

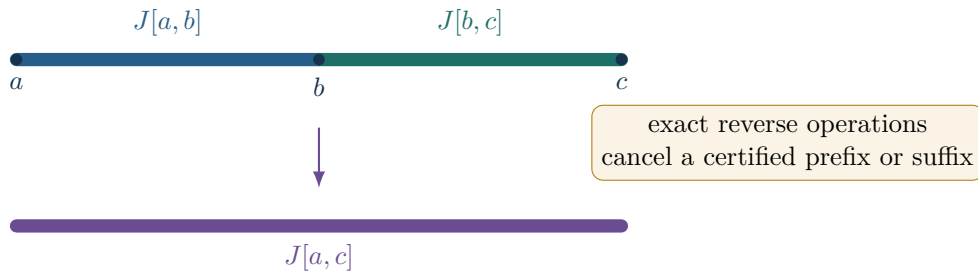


Figure 7: Native multiplication is interval concatenation. Native division is certified interval cancellation, not an arbitrary rational inverse.

31.2 Interference and continuation

Definition 31.2 (Exact reflexive interference). For a certified state J , define

$$J \boxminus J = 0_A,$$

where 0_A is a certified exact zero state bound to both parents.

A non-reflexive subtraction is not determined by the principal coefficient alone. Likewise, an unsupported division may require higher jets. The machine therefore returns a continuation packet rather than downloading ordinary values and fabricating a scalar.

Definition 31.3 (Higher-jet continuation). A higher-jet continuation records the attempted operation, both parent bindings, a required jet order, and the assertions that the operands remain retained and no scalar collapse has occurred. It is a live state in an enlarged result sort, not an exception carrying an guessed answer.

Principle 31.4 (Information insufficiency remains internal). *If the declared coordinate is insufficient to justify a scalar result, the computation remains in a higher-dimensional continuation state. It may not repair the missing information by an unrecorded ordinary projection.*

Theorem 31.5 (Non-terminal factorial state). *The factorial state is non-terminal in the scoped language generated by contiguous multiplication, exact interval cancellation, reflexive interference and higher-jet continuation.*

Proof. Each operation is defined directly on the certified interval, valuation, program and parent bindings. Valid contiguous products and certified cancellations produce new certified principal-jet states. Reflexive subtraction produces a certified zero. Unsupported operations produce a live continuation retaining both operands. None of these transitions requires an ordinary factorial materialization or an immediate download. $\square$

31.3 Why this is not merely a thunk

A compact term may denote an object without having executable semantics. The present construction avoids claiming more than it supplies.

Remark 31.6 (Why the construction is more than a thunk). The factorial state satisfies the following scoped tests.

1. Its source program is generated by an independent algebraic composition law and verified against the input bits.
2. Its principal coefficient is established by [Theorem 30.1](#), not by naming an answer field.
3. It participates in state-to-state multiplication, cancellation, interference and continuation.
4. Implicit conversion to an ordinary binary integer is absent from the native interface.
5. Observation collisions do not merge states, and observer results cannot feed back into native execution.

These facts justify the term *native factorial state*. They do not imply a classical logarithmic-time algorithm for printing $n!$ or for supporting every ordinary integer operation.

32 The factorial state as a three-class maze chart

We now close the most important gap in the short edition. The factorial state is not merely placed after the three-class axioms; it is an explicit special chart of them.

Let $m \geq 2$ be the admitted Wilson modulus and put $n = m - 1$. The native factorial compiler produces a certified principal-jet state

$$J_n = (P_n, [0, n], \nu = n, \epsilon_n, C_n),$$

where P_n is the compact q-Pochhammer source program, $[0, n]$ is its exact factor interval, ν is the principal valuation, ϵ_n records the principal sign, and C_n is the replay certificate. Its native factorial coordinate has rank n , coefficient 1, and denotation $n!$.

Definition 32.1 (Factorial maze chart). The successful factorial chart sends J_n to the three-class value

$$\mathfrak{F}(J_n) = (1, [\text{Arg}; n] + [\text{Program}; P_n] + [\text{State}; J_n] + [\text{Request}; m, J_n], 0).$$

Its maze address is

$$\begin{aligned} \text{Room} &= \text{Room}(m), & \text{Road} &= \text{Bind}(m, J_n), \\ \text{Frame} &= \text{FactorialChart}, & \text{Holonomy} &= \text{Seal}(P_n), \\ \text{HigherCell} &= \text{Seal}(J_n), & \text{SingularGeneration} &= 0. \end{aligned}$$

The execution step appends

$$[\text{Execute}; \text{NativeResult}, \text{NativeCertificate}]$$

to Hist and one certified causal event to the path. It does not change the native principal-jet state.

The ordinary coordinate is one because it is the coefficient of the native factorial basis element. The argument n is not erased; it lives in the history/program binding and in the native rank. The singular coordinate is zero precisely on the accepted chart. Failed interval division, insufficient jet order, coefficient overflow or a rejected bridge produces a typed continuation rather than another accepted factorial state.

Theorem 32.2 (Factorial chart embedding). *On certified factorial requests, $\mathfrak{F}$ is injective up to complete native-state equality. It preserves the ordinary coefficient, the ordered source program, the request binding and the principal-jet certificate. In particular,*

$$\mathfrak{F}(J_a) \equiv_{\text{str}} \mathfrak{F}(J_b) \implies a = b \quad \text{and} \quad J_a = J_b.$$

The ordinary projection alone is constant:

$$\text{Proj}(\mathfrak{F}(J_n)) = 1.$$

Thus Hist, rather than Ord, distinguishes factorial arguments in this chart.

Proof. Structural equality compares the argument word, program seal, state seal and request binding. Equality of the argument words gives $a = b$. Independent replay then binds the same compact program, interval, valuation, result coordinate and certificate. Conversely, identical native states produce the same sparse words and maze address. Projection retains only the coefficient one. $\square$

This theorem gives a concrete answer to the reader who asks where Hist is used in the factorial application: without Hist, every successful native factorial coordinate in the chart would have the same displayed coefficient. The program and argument would disappear.

33 The specialized factorial functor

The generic native functor acts by HADD, HSUB, HMUL and exact HDIV on arbitrary sparse maze states. The factorial application needs a specialized chart functor because its internal target is a certified principal-jet object, not a bare product of signed coefficients.

Let **FactReq** be the category whose objects are uploaded certified factorial requests and whose morphisms preserve request, state and program bindings. Let **FactExec** be the category of executed factorial maze states with a verified native coordinate. Define

$$\mathcal{F}_! : \mathbf{FactReq} \longrightarrow \mathbf{FactExec}$$

by independently replaying the native coordinate, preserving the input maze state, and adjoining the execution word and bridge certificate.

Theorem 33.1 (Factorial functor bridge). *The map $\mathcal{F}_!$ is a strict certified functor on the admitted chart. It satisfies:*

1. *the native factorial coordinate is verified before the result object is issued;*
2. *the input native state seal, source-program seal, request binding and payload size are unchanged;*
3. *the output maze state contains the input Hist words plus exactly one execution-binding word and one causal event;*
4. *the bridge certificate binds the input maze seal, output maze seal, native result seal, native certificate seal and native view seal;*
5. *only an executed bridge object may enter the exact or joint modular observer.*

Proof. Independent native replay establishes the rank, coefficient, state and program bindings. The generic history-addition functor appends the execution word without changing the ordinary coefficient or introducing a singular term. Its event rule increments epoch and history depth and binds both parent seals. The bridge seal is a deterministic fold of all five declared identities. Identity and composition follow from empty-word identity and ordered history concatenation. The last statement is a typing fact: the observers accept only the opaque executed result type. $\square$

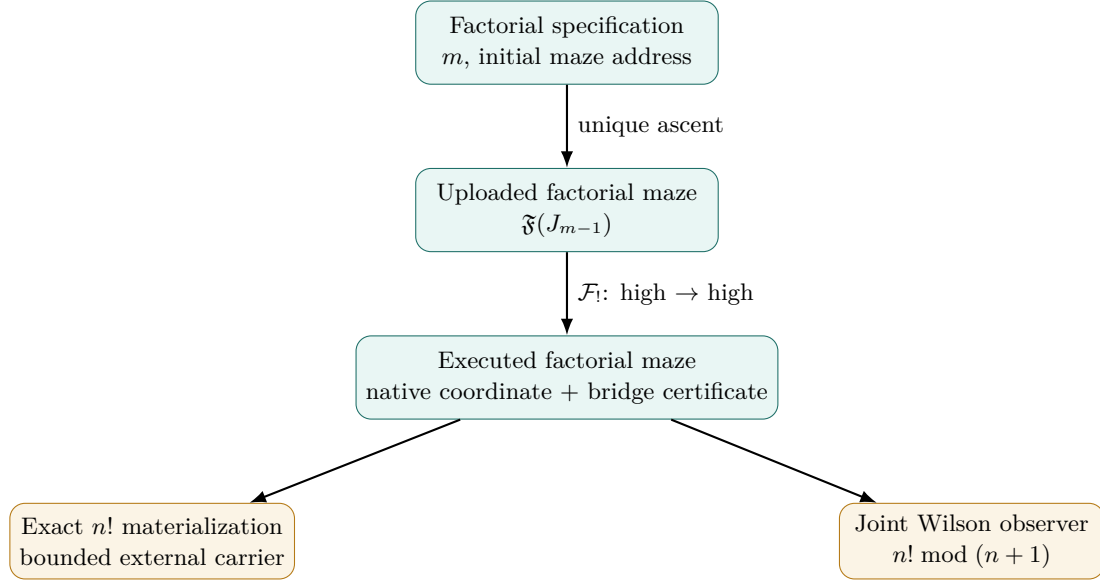


Figure 8: The implemented factorial-maze bridge. Neither gold observation has a morphism back to the uploaded or executed state.

33.1 Why a separate functor is mathematically necessary

Pretending that an ordinary HMUL instruction computes the principal-jet factorial would be false. The compact factorial program denotes an entire interval product and carries a certificate of its leading normalized jet. Its correctness is not the statement that a fixed-width coefficient was multiplied repeatedly. The specialized functor therefore binds the generic maze semantics to the pre-existing principal-jet machine rather than replacing that machine with a second implementation.

This is also why the bridge is additive. The sealed principal-jet core remains byte-identical. The bridge observes its public certified summaries, embeds their identities into Hist and address fields, and exposes only opaque typed stages.

34 A complete worked factorial-maze example

Take $m = 7$, so $n = 6$. Upload compiles a certified q-Pochhammer program of length six and the interval $[0, 6]$. Write its seals abstractly as

$$p_6 = \text{Seal}(P_6), \quad j_6 = \text{Seal}(J_6), \quad b_6 = \text{Bind}(7, J_6).$$

The uploaded three-sector value is

$$X_6 = (1, [\text{Arg}; 6] + [\text{Program}; p_6] + [\text{State}; j_6] + [\text{Request}; b_6], 0).$$

Its address contains the factorial frame, request road, program holonomy and state higher cell. Notice that the displayed scalar one is not $6!$; it is the coefficient of a proof-bound factorial basis element.

Execution independently replays the compact program and obtains a native coordinate

$$\text{rank} = 6, \quad \text{coefficient} = 1, \quad \text{denotation} = 6!.$$

Let r_6 and c_6 be the native result and certificate seals. The executed value is

$$X_6^{\text{exec}} = X_6 \boxplus_A (0, [\text{Execute}; r_6, c_6], 0).$$

This is an actual HADD in the generic sparse chart, so it preserves Ord and Sing, appends a word to Hist, increments causal depth and records a new event.

The exact observer may now materialize

$$6! = 720.$$

The joint modular observer instead uses $h = \lfloor 6/2 \rfloor = 3$ and computes

$$6! \equiv (-1)^3 (3!)^2 \equiv -36 \equiv 6 \pmod{7}.$$

Both observations are bound to the same native rank and certificate. Neither changes X_6 or X_6^{exec} .

Example 34.1 (Equal ordinary shadows, different factorial futures). For $m = 11$ and $m = 12$, both uploaded factorial maze values have ordinary coordinate one and singular coordinate zero. Their argument, program, state and request words differ. The joint observer yields

$$10! \equiv 10 \pmod{11}, \quad 11! \equiv 0 \pmod{12}.$$

Collapsing both uploaded states to the ordinary coefficient one would make this distinction impossible. The executable regression suite checks this same-shadow/different-history condition before checking the observers.

35 How the three maze sectors participate in the factorial program

The three sectors have asymmetric roles, and that asymmetry should not be hidden.

35.1 Ord: principal coefficient, not the full denotation

The successful native factorial basis element has coefficient one. Its full ordinary denotation $n!$ may require $\Theta(n \log n)$ output bits. Storing that integer in Ord during the native stage would destroy the compact-state claim. The chart therefore separates coefficient from denotation.

35.2 Hist: argument, construction and proof provenance

Hist binds the argument, compact program, principal-jet state, request and execution certificate. These words determine which interval operations, cancellations and observations are legal. They are precisely the part that distinguishes factorial states with the same principal coefficient.

35.3 Sing: zero on the accepted chart, live at its boundary

An accepted factorial interval from 0 to n has no unresolved singular payload, so Sing is zero. This does not make Sing decorative. The principal-jet algebra returns a higher-jet continuation when intervals are not contiguous, a denominator is not a certified factor, or a nonreflexive subtraction needs additional order. The generic three-class chart similarly retains a residual under HDIV. Sing

therefore describes the boundary of the successful factorial chart and the information required to extend it.

A chart can use a sector by forcing it to be zero

In differential geometry, a flat connection is meaningful because curvature could have been nonzero and flatness is a condition. Here a certified factorial state has zero singular payload because its construction proves that no unresolved division remains. The zero is a checked chart condition, not an omitted coordinate.

36 Spectral HDIV and executable HDIV agree on their common exact locus

The short edition gave two division formulas without proving their relation. We now state the exact compatibility and its limitation.

Let the semisimple spectral chart decompose unity as $1 = \sum_j e_j$. For a central scalar $d \in R$, multiplication by d has the same eigenvalue d on every e_j .

Proposition 36.1 (Central-scalar specialization of spectral HDIV). *Assume d is an invertible central scalar in the coefficient ring. Then the spectral inverse is $d^\# = d^{-1}$, its kernel projector is zero, and*

$$X \oslash_A \iota(d) = (d^{-1}X, 0)$$

in both the spectral and executable charts whenever every coefficient of X is exactly divisible by d in the executable coefficient domain.

Proof. Every spectral idempotent has eigenvalue $d \neq 0$, hence $d^\# = \sum_j d^{-1}e_j = d^{-1}$ and $K_d = 0$. Exact coefficientwise division has all remainders zero and produces the same $d^{-1}X$. $\square$

Proposition 36.2 (Reconstruction compatibility for nonunits). *For a nonzero central integer d that is not inverted in the executable base ring, the executable quotient and residual satisfy*

$$X = \iota(d) \boxtimes_A Q_d(X) \boxplus_A R_d(X).$$

After scalar extension to a ring in which d is invertible, the residual is absorbed into the exact spectral quotient:

$$d^{-1}X = Q_d(X) + d^{-1}R_d(X).$$

Thus the executable chart is an integral quotient-remainder refinement, not a contradiction of the spectral chart.

Proof. The first identity is coefficientwise Euclidean reconstruction with signed truncation. Multiplying it by d^{-1} after scalar extension gives the second. $\square$

At $d = 0$, both charts return zero regular quotient and the complete numerator as unresolved payload. For a genuinely noncentral denominator, neither proposition applies; the finite SDK returns a continuation while the general theory requires a declared spectral or local inverse chart.

37 Exact factorial output and fast modular consumption are different claims

The phrase “fast factorial” is ambiguous. This memoir distinguishes three objects.

1. The *native factorial state* is a compact proof-bound coordinate with denotation $n!$. Its source program and encoded state have logarithmic size in n .
2. The *exact materializer* constructs every output bit of $n!$ in an arbitrary-precision carrier. It runs independent sequential and product-tree derivations and is explicitly bounded by policy in the current SDK.
3. The *joint Wilson consumer* computes only $n! \bmod (n + 1)$. It never materializes $n!$ and uses the complement-pairing reduction before the composite-safe polynomial engine.

The first object is compact because it is intensional. The second cannot avoid the output-size lower bound. The third is the path with the measured joint time–space reduction. Conflating them would turn a valid representation theorem into an invalid bit-complexity claim.

38 Typed boundary theorem for the factorial bridge

Theorem 38.1 (No observation re-entry). *In the executable factorial-maze interface:*

1. *an uploaded state cannot call the joint observer before execution;*
2. *an ordinary joint download cannot call the factorial executor;*
3. *an exact factorial download cannot be converted into an uploaded or executed maze state;*
4. *no public constructor for either opaque state exists.*

These are compile-time properties, independent of runtime seals.

Proof. The public operation overloads have disjoint predecessor types. Uploaded and executed states have deleted default constructors and private model-owning constructors. Observations share neither type nor conversion with those states. Therefore the forbidden expressions are ill-formed before any runtime code is generated. $\square$

Runtime verification remains necessary for legal expressions: the bridge replays the native coordinate, compares source identities before and after, and verifies the joint observer’s state-integrity certificate. Compile-time typing prevents illegal stage order; certificates prevent forged legal-stage content.

39 Engineering realization without interrupting the mathematics

The conceptual implementation has five opaque stages:

$$\text{Specification} \xrightarrow{\text{Upload}} \text{UploadedMaze} \xrightarrow{\mathcal{F}_1} \text{ExecutedMaze} \xrightarrow{\text{Observe}} \text{TerminalValue}.$$

The upload constructs both the generic three-class chart and the frozen native factorial request from the same admitted specification. Execution binds the native coordinate and records the bridge event. The exact and modular observers consume only the executed result.

Concrete C++ names, fields, files and negative compilation cases are moved to the engineering appendix. This placement lets the main theorem chain remain mathematical while still satisfying the closure rule of [Protocol 6.1](#).

Bridge conclusion

The factorial application now uses the three-class theory explicitly. Its Ord coordinate is the principal coefficient, Hist binds the argument, program, state, request and execution proof, and Sing is a checked zero on the successful chart with typed continuations at the boundary. A specialized high-to-high factorial functor creates the executed maze state before either exact materialization or joint Wilson observation.

PART II

Wilson Observation of a Factorial State

40 Wilson observation after the native factorial state

40.1 Problem statement and Wilson's theorem

The application begins with a supplied ordinary candidate $m \geq 2$. Upload and native execution produce a certified live factorial state whose native rank is $n = m - 1$ and whose ordinary denotation is $n!$. The ordinary observation task is not to print $n!$ but to compute its residue modulo m .

Theorem 40.1 (Wilson). *For every integer $m > 1$,*

$$m \text{ is prime} \iff (m - 1)! \equiv -1 \pmod{m}.$$

The theorem is used only after the native factorial state has been frozen. It is a complete ordinary observation for a supplied candidate; it is not a native successor-producing prime transition.

40.2 Wilson consumption contract

Let Q be the certified factorial state and let $r(Q)$ denote its native factorial rank. The external consumer is admitted only after independent replay verifies

$$r(Q) = n, \quad \text{coeff}(Q) = 1,$$

together with the source-program, principal-jet, result and certificate bindings. The factor count is then loaded from $r(Q)$ itself. The candidate m has exactly two ordinary roles:

1. the modulus of the quotient observation;
2. the independent diagonal constraint $m = r(Q) + 1$.

It is forbidden to replace the verified native rank by a fresh assignment $n := m - 1$ inside the modular algorithm. This distinction is executable: the joint consumer accepts a native factorial view, whereas a candidate or a plain factorial-state handle cannot be passed directly to it.

Boundary direction

The legal path is

certified native factorial state $\longrightarrow$ read-only native coordinate view
 $\longrightarrow$ ordinary modular observation.

The residue, Wilson decision, resource-limit packet and all accounting fields have no conversion back to a native state. Observation cannot rewrite, merge, compress, prune or advance the frozen state.

40.3 Integral complement-pairing reduction

The joint optimization starts with an exact change of coordinates that uses no inverse and therefore remains valid over arbitrary composite moduli.

Theorem 40.2 (Complement-paired factorial). *Let $m \geq 2$, set $n = m - 1$, and let $h = \lfloor n/2 \rfloor$. Then*

$$n! \equiv \begin{cases} (-1)^h (h!)^2, & n \text{ even}, \\ (-1)^h (h!)^2 (h+1), & n \text{ odd} \end{cases} \pmod{m}.$$

Proof. For every $1 \leq k \leq h$,

$$k(m-k) \equiv -k^2 \pmod{m}.$$

If n is even, then m is odd and the factors $1, \dots, m-1$ split into the disjoint pairs $(k, m-k)$, $1 \leq k \leq h$. Multiplication of the pair identities gives $n! \equiv (-1)^h (h!)^2$. If n is odd, then m is even; the same pairs cover every factor except the self-complementary factor $m/2 = h+1$. Multiplying that remaining factor yields the second formula. No cancellation, division or field property is used. $\square$

The theorem reduces the polynomial problem from n factors to h factors. After $h! \bmod m$ is known, the original residue requires one squaring, an optional modular negation, and an optional multiplication by $h+1$.

40.4 Reduced composite-safe block coordinate

For the reduced factor count $h > 0$, choose a block width w near $\sqrt{h}$ and define

$$P_w(X) = \prod_{i=1}^w (X+i) \in (\mathbb{Z}/m\mathbb{Z})[X].$$

Writing $h = qw + r$, $0 \leq r < w$, gives the exact identity

$$h! \equiv \left(\prod_{j=0}^{q-1} P_w(jw) \right) \left(\prod_{k=qw+1}^h k \right) \pmod{m}.$$

The polynomial ring uses monic remainders and exact multi-prime convolution reconstruction. It does not invert arbitrary coefficients and does not assume that $\mathbb{Z}/m\mathbb{Z}$ is a field. Hence the same path is used for primes, prime powers, semiprimes, even composites and rings with zero divisors.

A bounded schedule selector examines a fixed-radius neighbourhood of the integer anchors $\lfloor \sqrt{h} \rfloor$, $\lceil \sqrt{h} \rceil$ and $\lceil h/\lfloor \sqrt{h} \rfloor \rceil$. For each legal width it computes

$$q = \lfloor h/w \rfloor, \quad r = h \bmod w, \quad C(w) = w + q + r,$$

and minimizes lexicographically

$$(C(w), \max(w, q), r, w).$$

The radius is a fixed policy parameter with a hard cap. It is not an input-size table or answer-dependent precomputation.

40.5 Joint time–space schedule

The implementation applies six reductions in one execution.

1. **Half-size factorial.** The block polynomial and its evaluation coordinate are constructed for h , not for n .
2. **Direct constant-mode read.** $P_w(0)$ is the constant coefficient of the already constructed block polynomial. No zero-point leaf and no Horner pass are created for it.
3. **Streaming projection.** Evaluation points are generated from block indices only when a leaf is constructed. Each evaluated block value is immediately multiplied into one scalar residue. Neither a point array nor a block-value array is materialized.
4. **Destructive branch retirement.** The remaining point range is processed in two sequential top-level pieces. Inside each remainder descent, the left remainder is consumed and released before the right remainder is created; completed product nodes are also released immediately.
5. **Leading-coefficient Horner evaluation.** A nonempty polynomial is evaluated by initializing Horner’s recurrence with its leading coefficient and processing only the remaining coefficients. Relative to the supplied zero-initialized evaluator, this removes one multiply–add–reduction step per leaf evaluation.
6. **Degree-one leaf elimination.** After two degree-one leaf factors have formed their parent product, the leaf polynomials are released. During descent, the reduced parent polynomial is evaluated directly at each leaf point. The degree-one monic remainders and their temporary coefficient arrays are therefore absent from the optimized path.

No product or remainder is recomputed. The space reduction is therefore not a checkpoint/recomputation exchange. The optimized coordinate materializes exactly the $w + 1$ coefficients of P_w at its external interface; points and values are logical events rather than persistent coordinate arrays.

40.6 End-to-end correctness

Theorem 40.3 (Correctness of joint Wilson consumption). *Assume the native factorial coordinate verifies with rank n , coefficient one, and the boundary constraint $m = n + 1$. The jointly optimized external consumer returns $n! \bmod m$. Its Wilson decision is true exactly when m is prime.*

Proof. The block identity computes $h! \bmod m$ exactly. Product formation, monic-remainder evaluation and CRT reconstruction are exact ring operations for the supplied modulus. Every full block value and every tail factor is multiplied into the scalar accumulator exactly once. Streaming changes only the lifetime of intermediate objects, not the product. The reconstruction in [Theorem 40.2](#) therefore gives $n! \bmod m$. The final decision is equivalent to primality by [Theorem 40.1](#). $\square$

Remark 40.4 (State integrity across observation). The formal contract treats the source state as immutable and the residue as a terminal ordinary value. The corresponding executable checks are reported in [Appendix F](#) and are not used as premises in the proof above.

41 Worked Wilson observations

This section works through the ordinary observer at small moduli. The examples are explanatory mathematics, not test evidence; the finite software campaigns are confined to [Appendix F](#).

41.1 A prime example: $m = 11$

Here $n = m - 1 = 10$ and $h = 5$. Because n is even,

$$10! \equiv (-1)^5(5!)^2 \pmod{11}.$$

Now $5! = 120 \equiv 10 \pmod{11}$, so

$$10! \equiv -10^2 \equiv -1 \equiv 10 \pmod{11}.$$

The Wilson observation therefore returns the distinguished residue $m - 1 = 10$ and declares the supplied candidate prime.

The important order of events is

$$\text{certify rank } 10 \longrightarrow \text{check } 11 = 10 + 1 \longrightarrow \text{compute the residue modulo } 11.$$

Reversing the first two steps and setting the rank from the candidate would give the same arithmetic in this toy example, but it would violate the source contract: the observer would no longer be consuming a coordinate established by the native state.

41.2 A composite example: $m = 12$

Now $n = 11$, $h = 5$, and the odd case gives

$$11! \equiv (-1)^5(5!)^2(6) \pmod{12}.$$

Since $5! \equiv 0 \pmod{12}$, the result is zero. It is not congruent to $-1 \equiv 11$, so the observer returns composite.

This example is useful because the modulus has zero divisors. No step divides by 2, by $h!$, or by a polynomial leading coefficient. Complement pairing uses only multiplication and sign, while polynomial reduction uses monic divisors. Therefore the derivation lives in $\mathbb{Z}/12\mathbb{Z}$ without pretending that the ring is a field.

41.3 A block-coordinate example

Take $h = 14$ and choose block width $w = 4$. Define

$$P_4(X) = (X + 1)(X + 2)(X + 3)(X + 4).$$

The first twelve factors of $14!$ are grouped into three blocks:

$$\begin{aligned} P_4(0) &= 1 \cdot 2 \cdot 3 \cdot 4, \\ P_4(4) &= 5 \cdot 6 \cdot 7 \cdot 8, \\ P_4(8) &= 9 \cdot 10 \cdot 11 \cdot 12. \end{aligned}$$

The tail contributes $13 \cdot 14$. Hence, in every commutative residue ring,

$$14! = P_4(0)P_4(4)P_4(8) \cdot 13 \cdot 14.$$

The optimized observer reads $P_4(0)$ directly from the constant coefficient, generates the other points only when their leaves are consumed, and multiplies each returned value into the scalar accumulator. The point $4j$ and its value need not coexist with all other points and values.

Table 6: Logical events and persistent objects in the $h = 14, w = 4$ example.

Item	Logical occurrence	Persistently materialized?
Coefficients of P_4	once	yes, five coefficients
Point 0	first block	no separate point object; use the constant term
Points 4 and 8	two evaluation leaves	generated when consumed
Values $P_4(4), P_4(8)$	two scalar returns	streamed into the accumulator
Tail factors 13, 14	two scalar multiplications	no coordinate slots
Full $14!$	mathematical denotation	never materialized by this observer

The distinction in the final two rows corrects a common accounting mistake. A factor can contribute time without occupying a persistent coordinate slot. Conversely, a stored array contributes space even if its entries are filled one at a time.

41.4 Why exact transform reconstruction is composite-safe

For larger polynomial products the implementation may multiply coefficient vectors under several transform-friendly primes. This does not change the target modulus. Conceptually the procedure is:

1. bound every integer convolution coefficient;
2. compute the convolution residues under enough auxiliary primes that their product exceeds twice the bound;
3. reconstruct the unique bounded integer coefficient by the Chinese remainder theorem;
4. only then reduce that integer coefficient modulo the target m .

Because reconstruction occurs before reduction to $\mathbb{Z}/m\mathbb{Z}$, the target may be prime, a prime power, or an arbitrary composite. Fast monic remainder is also safe: a monic leading coefficient is the unit 1 in every unital ring, so the long-division recurrence never asks for a nonunit inverse.

Proposition 41.1 (Composite-safe block correctness). *Suppose the auxiliary-prime product exceeds twice the certified absolute bound for every integer convolution coefficient. Exact CRT reconstruction followed by reduction modulo m returns the same product polynomial as direct convolution in $(\mathbb{Z}/m\mathbb{Z})[X]$.*

Proof sketch. Each auxiliary convolution is the image of one integer convolution under a ring homomorphism. The size bound makes the simultaneous residues identify that integer coefficient uniquely. Reducing the reconstructed integer modulo m therefore commutes with multiplication. A fully quantified version and its implementation obligations appear in [Appendix A.13](#). $\square$

41.5 What these examples do and do not show

The examples show the algebra of pairing, block evaluation, streaming and composite-safe reconstruction. They do not establish memory safety, equality of parallel and serial transforms, or strict finite counter inequalities. Those are properties of the supplied lowering and are documented as computational evidence in the appendices. Conversely, passing a finite test campaign cannot replace the identities proved above.

42 External Wilson schedule and resource bounds

Complexity is reported as a vector rather than a single headline number:

$$T_{\text{upload}}, \quad T_{\text{native}}, \quad T_{\text{download}}, \quad T_{\text{external}}, \quad T_{\text{bit}},$$

together with peak live coefficients, peak live limbs, materialized external coordinates, allocation counts, copied bytes and certificate/verification work. A short native program is not counted as completed external execution.

42.1 Native factorial semantics and cost

Theorem 42.1 (Native factorial-state complexity). *Construction and verification of the compact factorial source/state for input n use $O(\log n)$ program nodes and $O(\log n)$ encoded state space. This statement concerns the native state and does not include a complete ordinary materialization of $n!$.*

Proof. Write $n = (b_{\ell-1} \cdots b_0)_2$ with $b_{\ell-1} = 1$. Starting from F_1 , each remaining bit contributes one doubling operation and, exactly when $b_i = 1$, one append operation. Hence the program

contains at most $1 + 2(\ell - 1)$ nodes. Replay stores the current binary length, the bounded opcode record, and the interval/jet certificate attached to each program node; all have total encoded size $O(\ell) = O(\log n)$. No coefficient expansion or ordinary representation of $n!$ is part of this construction. $\square$

The joint Wilson optimization does not alter this state. Its integrity ledger requires

$$\text{native state nodes rewritten} = 0, \quad \text{ordinary feedback} = 0.$$

Thus all changes in this release lie on the ordinary observation side of the frozen boundary.

42.2 Previous time–space baseline

Let $n = m - 1$, $s_0 = \lceil \sqrt{n} \rceil$, and let $M(d)$ denote the charged cost of multiplying degree- d polynomials in the selected exact modular/CRT engine. The previous block path had

$$T_0(n) = O(M(s_0) \log s_0 + s_0)$$

ring work and

$$S_0(n) = O(s_0 \log s_0)$$

live coefficient storage in its product/remainder-tree implementation. Its explicit external coordinate contained

$$C_0(n) = s_0 + 1 + 2 \left\lfloor \frac{n}{s_0} \right\rfloor = \Theta(\sqrt{n})$$

scalar slots: block coefficients, evaluation points and block values. Tail factors are streamed time events and are not stored coordinate slots.

42.3 Time complexity of the joint path

Set $h = \lfloor n/2 \rfloor$ and let $s_1 = \Theta(\sqrt{h})$. The reduced block calculation and constant-cost reconstruction give

$$T_1(n) = O(M(s_1) \log s_1 + s_1) + O(1) = O(M(\sqrt{n/2}) \log n + \sqrt{n/2}) + O(1).$$

The bounded width search performs a fixed number of integer schedule comparisons. Direct $P_w(0)$ projection removes one evaluation leaf and its Horner work. Streaming adds no polynomial multiplication or remainder. Sequential top-level range processing removes the old root product/remainder coexistence and does not recompute descendants. Leading-coefficient Horner evaluation removes one ring multiply–add–reduction step from every nonempty leaf evaluation. Degree-one leaf elimination removes the bottom-level monic remainders and releases their products immediately after parent formation.

Explicit non-claim 42.2 (No asymptotic time claim). Under a generic multiplication model, T_1 and T_0 are in the same asymptotic class. The factor $1/\sqrt{2}$ in the polynomial scale is a strict structural reduction but not an $o(T_0)$ theorem. The release therefore claims an executable local Pareto improvement, not joint asymptotic optimization.

42.4 Space complexity of the joint path

The optimized path stores neither an evaluation-point array nor a block-value array. It processes one top-level evaluation range at a time and frees each completed product/remainder branch immediately. Its bound is

$$S_1(n) = O(s_1 \log s_1) = O(\sqrt{n/2} \log n)$$

coefficient words, with an external materialized coordinate

$$C_1(n) = w + 1 = \Theta(\sqrt{n/2}).$$

The implementation reports both the exact peak of coefficient slots owned by the outer schedule and a conservative scratch-inclusive peak that charges transform and exact CRT reconstruction buffers. The same scratch convention is applied to the reconstructed baseline audit.

Explicit non-claim 42.3 (The square-root coordinate remains). The new coordinate is strictly smaller, but it is still of order $\Theta(\sqrt{n})$. Neither space nor coordinate size has been proved $o(S_0)$ or polylogarithmic.

42.5 Structural comparison and computational appendix

The formal bounds above compare the dimensions and lifetimes of the two exact schedules. Finite operation counts, transform/CRT differential tests, serial–parallel equivalence, sanitizer results, and release hashes are reported in [Appendices F](#) and [G](#). Those data test conformance of one implementation; they are not premises of the theorems in this volume. In that companion, `allocation_count` means allocations of tracked outer-schedule objects, not every allocator call made by the process; transform scratch is charged separately by the conservative peak-live bound.

42.6 Bit complexity and full ordinary download

The fixed-word modular chart charges 64-bit limb products and additions explicitly. The exact arbitrary-precision factorial path is maintained as an independent oracle and download facility, not as the optimized Wilson backend. If

$$B = \text{bits}(n!) = \Theta(n \log n),$$

then printing or retaining all of $n!$ requires at least $\Omega(B)$ output time and space. The joint Wilson path never constructs that integer, so this output lower bound does not apply to its modular observation. Conversely, the existence of a compact native factorial state does not erase the bit cost of a requested full download.

42.7 Rejected time–space exchanges

The implementation rejects the following apparent improvements:

- checkpoint/recompute schedules that reduce memory by repeating product or remainder work;
- parallel branch execution that lowers wall-clock time while increasing simultaneous live polynomial storage;

- larger precomputed point tables, cached polynomials or input-scale advice;
- deletion of certificate/state-integrity verification;
- short descriptors charged as completed modular execution;
- full factorial materialization followed by an ordinary remainder;
- rational/field relaxation of the exact integral source constraints.

The accepted schedule performs no recomputation and simultaneously decreases its deterministic work and conservative peak-space counters.

42.8 Remaining coordinate barrier

The location of the unresolved summary growth can be stated precisely. The compact source composes by

$$F_{a+b}(Z, u) = F_a(Z, u)F_b(Zu^a, u).$$

A scalar constant-mode value is not closed under the future shift $Z \mapsto Zu^a$: later block origins require values not recoverable from the value at zero. A fixed low-order coefficient window is likewise not closed, because the product creates cross-coefficients up to the sum of the input degrees, and exact quotient/remainder projection requires those coefficients until the corresponding evaluation branch has been consumed. In the current lowering the growth therefore occurs at

$$\begin{aligned} \text{shifted composition} &\longrightarrow \text{block-polynomial convolution} \\ &\longrightarrow \text{monic quotient/remainder projection.} \end{aligned}$$

Retaining the necessary coefficients produces the degree- $\Theta(\sqrt{n})$ block coordinate; retaining the shifted scalar values produces a coordinate of the same order. Complement pairing reduces this scale by a strict constant and the new schedule removes redundant work and storage, but it does not prove a future-sufficient fixed-size summary. This identifies the present implementation barrier; it is not a universal lower bound.

The unresolved target is a sufficient summary of the succinct factorial program that remains compact under doubling, composition, shift, quotient projection and constant-mode observation. The present product polynomial has degree $\Theta(\sqrt{n})$; thus the execution still materializes a square-root coordinate. The exact status is

time asymptotically improved	=	no,
space asymptotically improved	=	no,
paired polynomial scale	=	$\Theta(\sqrt{n/2})$,
$\sqrt{n}$ coordinate eliminated	=	no.

43 Formal correspondence with an executable interface

The mathematical construction has an executable realization, but its role in this volume is limited to a typed correspondence. Let S be a certified factorial state, let $\bar{S}$ be its closed observation state, and let $D_m(\bar{S})$ denote the ordinary residue packet modulo m .

Proposition 43.1 (Typed egress). *The Wilson map*

$$D_m : \bar{S}_A \longrightarrow \mathcal{O}_m$$

is terminal: there is no native transition in the formal interface from $\mathcal{O}_m$ back to $\mathcal{S}_A$. Consequently an ordinary residue cannot alter, continue, or replace the state from which it was observed.

Proof. The domain of every native transition is a live certified state or a native checkpoint, whereas $\mathcal{O}_m$ is an ordinary observation type carrying a causal-cut receipt. These types are disjoint, and the transition grammar has no constructor from the latter to the former. Any re-entry must begin with a new upload and therefore a new causal session. $\square$

Proposition 43.2 (Derived observations are non-authoritative). *Let $\omega(\bar{S})$ be the observation canonically bound at closure and let $\tilde{\omega}$ be a transported or displayed copy. Acceptance of the copy requires $\tilde{\omega} = \omega(\bar{S})$ together with equality of the causal-cut data. Modifying $\tilde{\omega}$ cannot modify $\bar{S}$.*

Proof. Canonical closure precedes transport. The acceptance relation is equality against the already-bound observation and cut, never assignment in the opposite direction. Thus the derived map is evidence read from the state and not an authority over it. $\square$

Remark 43.3. The engineering appendices give the concrete type map, differential tests, adversarial compilation tests, and finite resource ledgers for the supplied C++ realization. None of those finite checks is invoked in the mathematical proofs here.

PART III

A Second Exact Chart of the Same Source State

44 Cyclic self-action and prime-order kernel

The factorial principal jet is an infinitesimal chart of the q-Pochhammer source. A second chart is obtained by attaching the same integral polynomial to a cyclic frame. This section is mathematically independent of the external Wilson evaluator and reveals why the Wilson scalar can be viewed as one mode of a larger action.

44.1 Integral source and characteristic-zero proof chart

Let

$$A_m = \mathbb{Z}[C_m] = \mathbb{Z}[S]/(S^m - I)$$

be the integral cyclic group ring. Define the cyclic quantum integer

$$[k]_S = I + S + \cdots + S^{k-1} \in A_m$$

and, for $0 \leq n < m$,

$$J_{m,n} = (-1)^n \prod_{k=1}^n [k]_S.$$

The coefficients of this operator are integral. To diagonalize the action for proof, extend scalars to $\mathbb{C}$. For each divisor $d \mid m$, let V_d be the sum of characters of exact order d . Then

$$A_m \otimes \mathbb{C} = \bigoplus_{d \mid m} V_d, \quad \dim V_d = \varphi(d).$$

Lemma 44.1 (Quantum-integer zero set). *On an exact-order- d character with eigenvalue ζ ,*

$$[k]_S \mapsto 1 + \zeta + \cdots + \zeta^{k-1}.$$

For $d > 1$ this scalar is zero exactly when $d \mid k$. On the constant sector V_1 it equals k .

Proof. For $d > 1$, $\zeta \neq 1$ and

$$1 + \zeta + \cdots + \zeta^{k-1} = \frac{\zeta^k - 1}{\zeta - 1},$$

which vanishes exactly when $\zeta^k = 1$, equivalently $d \mid k$. At $\zeta = 1$ the sum is k . $\square$

Theorem 44.2 (Divisor-stratified kernel filtration). *For $0 \leq n < m$,*

$$\ker(J_{m,n} : A_m \otimes \mathbb{C} \rightarrow A_m \otimes \mathbb{C}) = \bigoplus_{\substack{d \mid m \\ 1 < d \leq n}} V_d.$$

Consequently,

$$\dim \ker J_{m,n} = \sum_{\substack{d \mid m \\ 1 < d \leq n}} \varphi(d).$$

Proof roadmap. Diagonalize the integral cyclic action after a faithful characteristic-zero scalar extension, classify a character by its exact period d , and use [Lemma 44.1](#) to identify the first vanishing factor. The complete argument, including exactness under scalar extension and the dimension count, is given in [Appendix A.8](#). $\square$

Corollary 44.3 (Self-cyclic prime correspondence). *Let $m \geq 2$ and $J_m = J_{m,m-1}$. Then*

$$m \text{ is prime} \iff \ker J_m = 0.$$

More precisely,

$$\ker J_m = \bigoplus_{\substack{d \mid m \\ 1 < d < m}} V_d, \quad \dim \ker J_m = m - 1 - \varphi(m).$$

Proof. Apply [Theorem 44.2](#) with $n = m - 1$. The kernel is zero exactly when m has no divisor strictly between 1 and m . $\square$

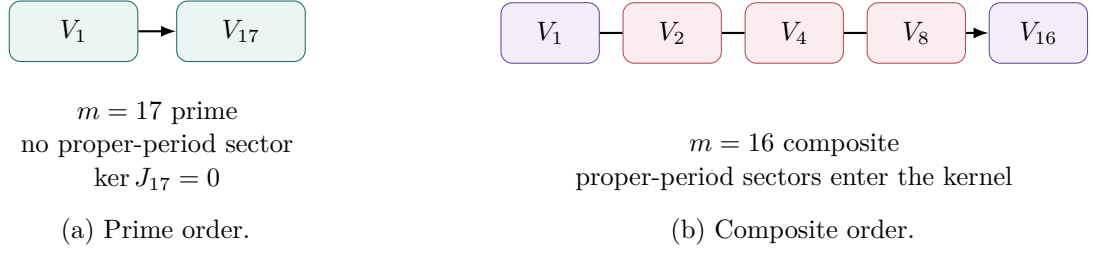


Figure 9: The cyclic self-action distinguishes prime order structurally: a prime has no nontrivial proper-period sector to annihilate.

44.2 Primitive-period projector and the Wilson mode

Define

$$K_m = (S - I)^{m-1} J_m = \prod_{k=1}^{m-1} (I - S^k).$$

Let Π_m^{prim} denote the projector onto V_m after characteristic-zero scalar extension.

Theorem 44.4 (Primitive-period projector).

$$K_m = m \Pi_m^{\text{prim}}.$$

Proof roadmap. Proper-period sectors acquire a zero factor, while a primitive character gives the cyclotomic product $\prod_{k=1}^{m-1} (1 - \zeta^k)$. Identifying the sectorwise scalar operator with the canonical primitive projector requires care over the scalar extension; see [Appendix A.9](#). $\square$

On the constant sector, each $[k]_S$ acts by k . Therefore

$$J_{m,n}|_{V_1} = (-1)^n n! I.$$

The ordinary Wilson factorial is precisely the constant cyclic mode of the larger self-action. This relation explains, without identifying the two execution mechanisms, why the factorial observer and the cyclic kernel theorem arise from the same q-Pochhammer source.

45 The self-closure germ and time–valuation reciprocity

45.1 Retaining the singular closure

The terminal factor at a full cycle vanishes because $S^m = I$. Rather than collapsing that zero, introduce a formal parameter ε and retain its entire germ.

Definition 45.1 (Self-closure germ). For $N \geq 1$, define in $A_m[[\varepsilon]]$

$$\mathcal{F}_{m,N}(\varepsilon) = \prod_{k=1}^{Nm} \left(I - (1 + \varepsilon)^k S^k \right).$$

Theorem 45.2 (Exact closure valuation). On the exact-period sector V_d , where $d \mid m$, put

$$r = \frac{Nm}{d}.$$

Then

$$\boxed{\text{ord}_\varepsilon(\mathcal{F}_{m,N}|_{V_d}) = r}$$

and more precisely

$$\boxed{\mathcal{F}_{m,N}(\varepsilon)|_{V_d} = (-1)^r r! d^{2r} \varepsilon^r + O(\varepsilon^{r+1}).}$$

Proof roadmap. Split the character product into indices divisible by d and the complementary residue classes. The first group supplies exactly $r = Nm/d$ simple zeros; the second supplies r cyclotomic unit blocks. One factor d^r comes from derivatives and another from the nonvanishing residues. The complete valuation calculation is in [Appendix A.10](#). $\square$

45.2 Time–valuation reciprocity

For the unnormalized prefix action

$$P_{m,n} = \prod_{k=1}^n (I - S^k),$$

define the first annihilation time of V_d by

$$\tau_m(d) = \min\{n \geq 1 : P_{m,n}|_{V_d} = 0\}.$$

By exact order, $\tau_m(d) = d$. Define the one-cycle closure depth

$$\nu_m(d) = \text{ord}_\varepsilon(\mathcal{F}_{m,1}|_{V_d}) = \frac{m}{d}.$$

Corollary 45.3 (Time–valuation reciprocity). *For every divisor $d \mid m$,*

$$\boxed{\tau_m(d)\nu_m(d) = m.}$$

For N self-cycles,

$$\tau_m(d)\nu_{m,N}(d) = Nm.$$

Corollary 45.4 (Multiplication-fibre correspondence). *Let*

$$\Sigma_{tv}(m) = \{(\tau_m(d), \nu_m(d)) : d \mid m\}.$$

Then

$$\boxed{\Sigma_{tv}(m) = \{(d, m/d) : d \mid m\} = \{(a, b) \in \mathbb{N}_{\geq 1}^2 : ab = m\}.$$

Thus the complete ordered multiplication fibre is re-encoded as causal death time and singular closure depth.

Corollary 45.5 (Prime transversality). *The spectrum always contains $(1, m)$ and $(m, 1)$. It contains an interior point (a, b) with $a, b > 1$ exactly when m is composite. Hence*

$$\boxed{m \text{ prime} \iff \Sigma_{tv}(m) = \{(1, m), (m, 1)\}.$$

Equivalently, prime order is characterized by a simple transverse zero on every nonconstant exact-period sector.

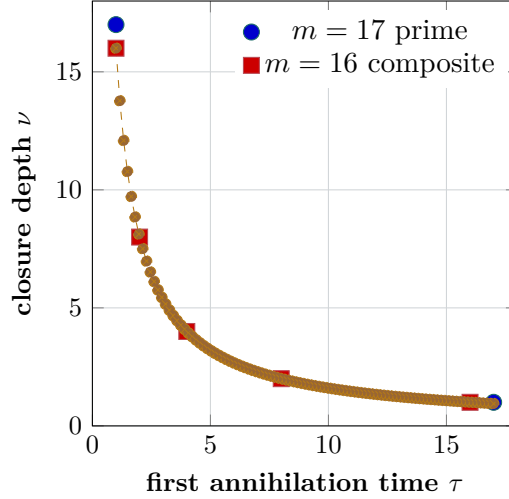


Figure 10: The time–valuation spectrum. Prime order has only the two boundary points; composite order has interior factor-pair strata. The dashed curve is $\tau\nu = 16$.

45.3 Quotient transport and jet resurrection

Let $m \mid n$ and $N = n/m$. The cyclic quotient map

$$\rho_{n,m} : \mathbb{Z}[C_n] \rightarrow \mathbb{Z}[C_m], \quad S_n \mapsto S_m$$

satisfies an exact transport law.

Theorem 45.6 (Jet-ramified quotient transport).

$$\boxed{\rho_{n,m}(\mathcal{F}_{n,1}) = \mathcal{F}_{m,N}.$$

The terminal primitive operator K_n maps to zero when $N > 1$, but the complete germ has first nonzero transported jet at degree N :

$$\boxed{[\varepsilon^N]\rho_{n,m}(\mathcal{F}_{n,1}) = (-1)^N N! m^{2N} \Pi_m^{\text{prim}} = (-1)^N N! m^{2N-1} K_m.$$

Proof roadmap. Transport the complete germ before taking a terminal value. The quotient turns one n -cycle into N consecutive m -cycles. Sector valuations show that degree N survives only on V_m ; identifying this component with the primitive projector gives the coefficient. See [Appendix A.12](#) for the complete argument. $\square$

The theorem shows why terminal zero is not enough information for cross-frame continuation. Primitive content that disappears at terminal order can reappear at a higher jet. This is a precise instance of information preservation by higher-dimensional continuation.

46 Worked cyclic and closure examples

The cyclic chart becomes more transparent when the divisor sectors are written out explicitly. In all examples the decomposition is a characteristic-zero proof chart for the integral operator; it is not a claim that the runtime stores a complex Fourier basis.

46.1 Prime order: $m = 5$

The divisors of 5 are 1 and 5, so

$$A_5 \otimes \mathbb{C} = V_1 \oplus V_5, \quad \dim V_1 = 1, \quad \dim V_5 = 4.$$

For the full prefix $J_{5,4}$, no index $1 \leq k \leq 4$ is divisible by 5. Hence the primitive sector survives. The constant sector also survives, with eigenvalue $4!$. Therefore

$$\ker J_{5,4} = 0.$$

At closure, the one-cycle valuations are

$$\nu_5(1) = 5, \quad \nu_5(5) = 1.$$

The time-valuation spectrum contains only

$$(1, 5), \quad (5, 1),$$

the two boundary points of the multiplication fibre $ab = 5$.

46.2 Composite order: $m = 6$

Now the exact-period sectors are

$$V_1, V_2, V_3, V_6$$

with dimensions 1, 1, 2, 2. The full prefix stops at $n = 5$. The factor $[2]_S$ kills V_2 , and $[3]_S$ kills V_3 ; no factor indexed by a multiple of 6 occurs. Thus

$$\ker J_{6,5} = V_2 \oplus V_3, \quad \dim \ker J_{6,5} = 1 + 2 = 3.$$

This agrees with

$$6 - 1 - \varphi(6) = 5 - 2 = 3.$$

The one-cycle spectrum is

$$\Sigma_{tv}(6) = \{(1, 6), (2, 3), (3, 2), (6, 1)\}.$$

The interior points record the two nontrivial ordered factorizations of 6.

Table 7: Exact-period data for the $m = 6$ cyclic chart.

d	$\dim V_d$	first death $\tau_6(d)$	closure depth $\nu_6(d)$	product
1	1	1	6	6
2	1	2	3	6
3	2	3	2	6
6	2	6	1	6

46.3 The leading closure coefficient

Stay with $m = 6$ and take two self-cycles, $N = 2$. On the exact-period $d = 3$ sector,

$$r = \frac{Nm}{d} = \frac{12}{3} = 4.$$

The closure theorem predicts

$$\mathcal{F}_{6,2}(\varepsilon)|_{V_3} = (-1)^4 4! 3^8 \varepsilon^4 + O(\varepsilon^5).$$

The exponent 4 counts the indices

$$3, 6, 9, 12$$

whose character factors vanish at closure. The coefficient contains two conceptually different powers of 3^4 : one from the derivatives of those four vanishing factors and one from four complete blocks of nonvanishing residue classes. Omitting either source would give the wrong leading jet.

46.4 Quotient transport: $12 \rightarrow 4$

Let $n = 12$, $m = 4$ and $N = 3$. Under the quotient $S_{12} \mapsto S_4$, the twelve-factor closure germ becomes three consecutive four-cycles:

$$\rho_{12,4}(\mathcal{F}_{12,1}) = \mathcal{F}_{4,3}.$$

The terminal primitive product maps to zero because the factor $I - S_{12}^4$ becomes $I - S_4^4 = 0$. The germ retains more information. The first surviving transported jet lies in degree $N = 3$ and only on the primitive V_4 sector:

$$[\varepsilon^3] \rho_{12,4}(\mathcal{F}_{12,1}) = (-1)^3 3! 4^6 \Pi_4^{\text{prim}}.$$

Thus the terminal equality to zero is not an adequate continuation rule. The valuation and leading coefficient remember where primitive content reappears after ramification.

Divisors become causal coordinates

For every $d \mid m$, the same divisor appears twice: as the first prefix time $\tau = d$ at which the exact-period- d sector dies, and as the reciprocal closure depth $\nu = m/d$. The cyclic chart does not merely test primality; it re-encodes the complete divisor lattice as a time-valuation spectrum.

47 Two charts, one source program

The same compact q-Pochhammer program has at least two exact mathematical realizations:

$$\begin{aligned} \text{infinitesimal chart} & : u = 1 + T, Z = -(1 + T) \longrightarrow \text{factorial principal jet,} \\ \text{cyclic chart} & : T = S - I \longrightarrow \text{period-stratified self-action.} \end{aligned}$$

The first chart is the executable arithmetic core used by the naive prime application. The second chart proves structural prime correspondences but is currently evaluated by bounded reference algorithms rather than a compact native compiler.

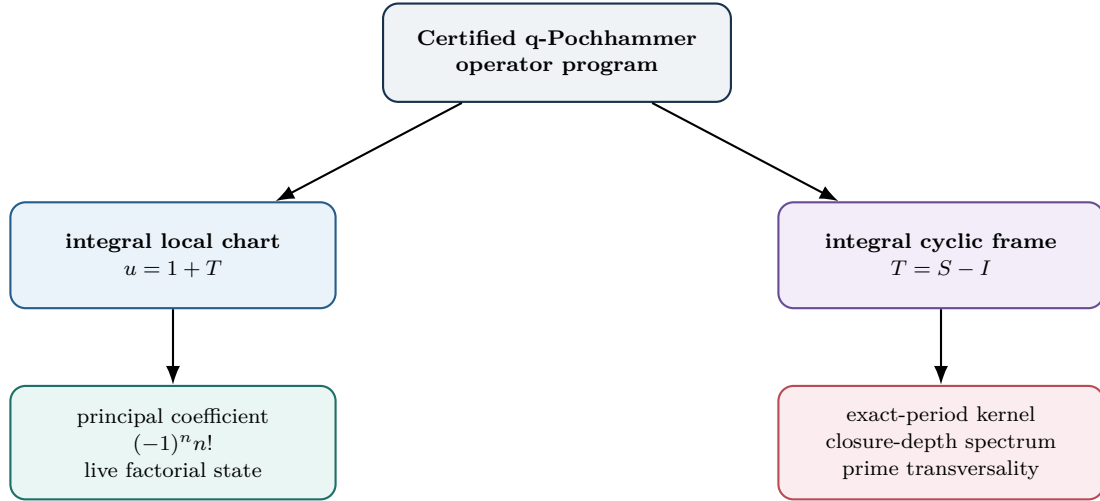


Figure 11: Two exact charts of one source program. The paper does not identify their execution costs or claim that the cyclic structural theorem is already a compact production primitive.

Missing cyclic compiler

The bounded cyclic reference evaluator materializes a response of length m and uses $O(m^2)$ coefficient work and $O(m)$ storage for the circulant one-column representation. A bounded jet of order R for N cycles uses $O(Nm^2R^2)$ reference coefficient work and $O(mR)$ storage. A production compiler that consumes the compact source program and returns a proof-carrying valuation-filtered singular effect without materializing every cyclic mode or jet coefficient remains open.

PART IV

Stateful Angel Prime Transmission Beyond Terminal Observation

48 The five-volume Angel Prime problem ledger

The phrase *Angel Prime* names a sequence of increasingly strict questions. Listing that sequence prevents a static prime characterization from being mistaken for a native generator.

48.1 Volume I: static tri-class arithmetic

The first layer constructs the ordinary/history/singular architecture, primitive-history projections, totalized division, vertical prime-power towers and static prime-axis objects. Its output is an exact algebra. It does not yet define a transition from one complete prime state to its successor.

48.2 Volume II: exact MEX dynamics

The second layer shows that a multiplicatively closed smooth-history cone has the primes as its ordered least-missing elements. This proves existence and order, but the uncompressed cone carries the known generated history. An exact MEX characterization is therefore not, by itself, a succinct bit-machine transition.

48.3 Volume III: ordinary shadows and a quantitative baseline

The third layer studies radical/primorial descent, stability classes, valuation curtains, quotient-history counting and a numerical update baseline. It also establishes a warning used throughout this memoir: if the whole transition descends to ordinary prime support, then the third class has not been proved algorithmically necessary.

48.4 Volume IV: causal completion and one-jet rooms

The fourth layer constructs an arithmetic causal kernel

$$J_x^{\text{caus}} = \bigcap_{\gamma: x \rightarrow y} U_\gamma^{-1}(K_y), \quad Q_x^{\text{caus}} = \mathcal{A} / J_x^{\text{caus}},$$

and its canonical residual third class

$$\Sigma_x^{\text{caus}} = I_x / J_x^{\text{caus}}.$$

It adds room-changing doors, resurrection-aware q-Pochhammer transport, ordered proof-carrying first-event transducers and certificate-ideal functoriality. The remaining compiler question becomes the extraction of the first harmonic support from a complete one-jet branch room.

48.5 Volume V: barriers, source-first children and residual reuse

The fifth layer first proves that bounded local jet operations cannot reveal balanced semiprimes at sub-square-root lowering depth. It then replaces the projection-first question by a source-generated multiplication-history complex whose surviving harmonic object is already a complete child. Later revisions remove target-factor recovery, make transmission source-first, and finally retain paid source packets through an anchor horizon. Version v5.9's positive result is the residual-reuse theorem integrated in [Section 60](#).

Layer	Mathematical object	What it proves	What remains
I	Tri-class static arithmetic and prime-axis geometry.	Conservative old arithmetic inside a richer exact state.	Causal successor transition.
II	Smooth-history cone and ordered least missing element.	Exact prime ordering in the high-dimensional cone.	Succinct compilation.
III	Quotient shadows, valuation curtains and finite update laws.	First resource baselines and projection warnings.	Intrinsic third-class action.

Layer	Mathematical object	What it proves	What remains
IV	Causal quotient, room doors and complete one-jet.	Future-aware state and certificate transport.	First harmonic compiler.
V.0–V.8R	Multiplication-history cofiber, source-first transmission and canonical deficiency.	Complete child before projection and progressively stricter compiler interfaces.	Repeated cold source work.
V.9	Curvature-normalized residual horizon.	Paid packet stability, no-recomputation and amortized reuse.	Collective descriptor-native cold source morphism.

The ledger clarifies the relation between old and new mathematics. Classical factorization and prime ordering remain true at the terminal shadow. The new objects concern how full histories, residuals and certificates are carried so that an ordinary answer is not smuggled into the transition.

49 What an Angel Prime is—and what it is not

The adjective *prime* and the name *Angel* govern different parts of the contract. Primality is an ordinary theorem about the terminal integer. Angelhood is a statement about the complete state and the causal legitimacy of the morphism that produced it. Conflating the two would make an ordinary primality test, wrapped in a large record, look like a native Angel Prime transition. We therefore distinguish three exact notions.

Definition 49.1 (*Angel primality certificate*). Let X be a complete maze state whose ordinary candidate coordinate is $c \geq 2$. An *Angel primality certificate at c* is a sealed packet

$$\mathcal{C}_c = (X, \gamma, \omega, \pi)$$

such that γ is a legal native execution history from an explicitly uploaded specification, ω is the terminal observation derived after that history, and π is an independently checkable proof that $\omega = \text{prime}$ if and only if c is an ordinary prime. The seal binds the full state, history, observation and proof; equality of the displayed bit alone is not certificate equality.

The SDK 1.2.0 factorial maze gives one concrete instance. It binds the candidate c , native factorial rank $n = c - 1$, compact factorial program, executed state and Wilson observation. Complement pairing changes the work schedule but not the denotation. Wilson’s theorem then supplies the ordinary equivalence, while the opaque upload/execute/project types supply the causal direction. This is the *naive executable Angel Prime chart*: it certifies a supplied candidate without pretending that the candidate was born inside the chart.

Definition 49.2 (*Angel Prime state*). An *Angel Prime state* P_p is a complete Angel maze object satisfying:

1. its ordinary shadow names an ordinary prime p ;
2. its history contains the legal native construction used to establish the current prime axis;
3. its singular/residual sector contains every unresolved continuation relevant to the declared future language;

4. its certificate binds the construction, not merely the scalar p ; and
5. its room, frame and program determine the legal successor operations.

Two Angel Prime states can therefore have the same ordinary prime shadow and still be structurally or future inequivalent.

Definition 49.3 (Native Angel Prime transition). A morphism

$$\mathfrak{N} : P_p \longrightarrow P_q, \quad q = \min\{r \in \mathbb{P} : r > p\},$$

is a *native Angel Prime transition* for a language $\mathcal{L}$ when:

1. it consumes the complete source state P_p and produces the complete child state P_q before terminal projection;
2. its construction does not query, invert or branch on the unknown target prime or on a terminal primality observation;
3. every internal step is a legal high-to-high morphism in $\mathcal{L}$;
4. the child certificate proves both leastness and completeness of the new prime axis; and
5. the charged time and live-state bound include every source-generation, residual-refinement and certificate-maintenance cost.

The definition is semantic as well as extensional: merely outputting the correct integer q is insufficient.

Theorem 49.4 (Conservative terminal soundness). *Every Angel primality certificate in the Wilson chart projects to a correct ordinary primality decision. Every native Angel Prime transition projects to the ordinary next-prime map. Neither converse holds without additional history, no-feedback and completeness hypotheses.*

Proof. For the certificate chart, the bound native factorial coordinate denotes $(c - 1)!$ exactly. The joint consumer evaluates the same integer modulo c ; Wilson's theorem gives $(c - 1)! \equiv -1 \pmod{c}$ exactly when c is prime. For a native transition, items two and four of [Definition 49.3](#) prove that the projected child is prime and that no smaller integer above p is omitted.

For the first failed converse, an ordinary primality algorithm may return the right bit while retaining no admissible history or reconstruction certificate. For the second, an ordinary next-prime routine may scan candidates after download and then wrap the answer in a record; this violates items one through three. Correct terminal values therefore do not reconstruct Angel causality. $\square$

Exact status of the integrated release

The public SDK realizes Angel primality certificates and the complete factorial-maze functor for supplied candidates. Volume V's strict research verifier realizes finite source-first child, residual-stability and zipper checks in its declared rooms. The integrated paper proves the scoped mathematical statements supporting both. It does *not* claim that the public SDK contains a descriptor-native cold $p^{o(1)}$ successor generator; that final native transition remains the open frontier.

This separation is why the work is titled *Angel Prime*. The object of study is not a renamed prime number. It is the prime together with the high-dimensional messenger state, its lawful causal route, its unresolved future and its proof-carrying boundary.

50 The one-jet room and its exact least-factor depth

The one-jet theorem explains why a low-order local view cannot be promoted to a universal prime selector. For $n \geq 2$, define

$$R_n = (\mathbb{Z}/n\mathbb{Z})[T]/(T^{n-1}), \quad u = 1 + T, \quad \mathcal{J}_n = R_n[\varepsilon]/(\varepsilon^n).$$

Let

$$B_n(\varepsilon, T) = \sum_{j=1}^{n-1} u^{j(j-1)/2} \begin{bmatrix} n \\ j \end{bmatrix}_u \varepsilon^j$$

be the nontrivial q-binomial branch section. Put

$$\mathfrak{m}_n = (\varepsilon, T) \subset \mathcal{J}_n$$

and define the total branch/history depth of a nonzero element by

$$\nu_{\mathfrak{m}}(f) = \min\{a + b : [\varepsilon^a T^b]f \neq 0\}.$$

For a composite n , write $\ell(n)$ for its least prime factor.

Lemma 50.1 (Basic q-integer depth). *If n is composite and $r = \ell(n)$, then*

$$\nu_T([n]_{1+T} \bmod n) = r - 1.$$

Proof. Expand

$$[n]_{1+T} = \frac{(1+T)^n - 1}{T} = \sum_{h=1}^n \binom{n}{h} T^{h-1}.$$

For $1 \leq h < r$, every integer from one through h is invertible modulo n . The identity

$$h! \binom{n}{h} = n(n-1) \cdots (n-h+1)$$

therefore implies $n \mid \binom{n}{h}$. At $h = r$,

$$\binom{n}{r} = \frac{n}{r} \binom{n-1}{r-1}.$$

Modulo r , the final binomial coefficient equals $(-1)^{r-1} \neq 0$, so one factor of r present in n has been removed and $n \nmid \binom{n}{r}$. The first nonzero coefficient is consequently the T^{r-1} coefficient. $\square$

Lemma 50.2 (Unit factors below the least prime). *For $1 \leq j < r = \ell(n)$, both $[j]_{1+T}$ and $\begin{bmatrix} n-1 \\ j-1 \end{bmatrix}_{1+T}$ are units in R_n .*

Proof. The constant term of $[j]_{1+T}$ is j , coprime to n . The constant term of the Gaussian coefficient is $\begin{bmatrix} n-1 \\ j-1 \end{bmatrix}$; every factor in its numerator and denominator is a unit modulo n because it

is represented, up to sign, by an integer smaller than r . In a ring with nilpotent ideal (T) , every element with unit constant term is a unit. $\square$

Theorem 50.3 (Least-factor channel theorem). *Let*

$$c_{n,j}(T) = u^{j(j-1)/2} \begin{bmatrix} n \\ j \end{bmatrix}_u.$$

For composite n with $r = \ell(n)$,

$$\nu_T(c_{n,j}) = r - 1 \quad (1 \leq j < r), \quad \nu_T(c_{n,r}) = 0.$$

Proof. The Gaussian recurrence gives

$$\begin{bmatrix} n \\ j \end{bmatrix}_u = \frac{[n]_u}{[j]_u} \begin{bmatrix} n-1 \\ j-1 \end{bmatrix}_u.$$

For $j < r$, every factor except $[n]_u$ is a unit by [Lemma 50.2](#), while $[n]_u$ has depth $r - 1$ by [Lemma 50.1](#). At $j = r$, the constant coefficient is $\binom{n}{r} \bmod n$, which the preceding proof showed is nonzero. $\square$

Theorem 50.4 (Least-Factor Branch-Depth Theorem). *For composite n ,*

$$\boxed{\nu_{\mathbf{m}}(B_n) = \ell(n).}$$

For prime p , the frozen Volume IV primitive-wall theorem gives

$$B_p = 0, \quad \nu_{\mathbf{m}}(B_p) = \infty.$$

Proof. Let $r = \ell(n)$. For $j < r$, the ε^j channel begins at T -depth $r - 1$, so its least total degree is $j + r - 1 \geq r$, with equality at $j = 1$. The $j = r$ channel has a nonzero constant coefficient and hence total degree r . No channel has smaller degree. The prime statement is exactly the primitive-wall input from Volume IV. $\square$

Corollary 50.5 (Exact truncated factor coverage). *Let*

$$\text{Trunc}_D(f) = f \bmod (\varepsilon^{D+1}, T^D).$$

Then

$$\text{Trunc}_D(B_n) = 0 \iff n \text{ is prime or } \ell(n) > D.$$

Proof. If n is prime, the whole section is zero. If $\ell(n) > D$, every channel of index at most D begins beyond the retained T -window and every later channel is removed by ε^{D+1} . Conversely, if $r = \ell(n) \leq D$, the ε^r channel has a nonzero constant coefficient and survives both truncations. $\square$

This corollary states the exact information content of a bounded one-jet. It detects factors no larger than its depth. Renaming that window a curvature spectrum, a quantum probe or a PrimeLock does not create an all-factor mechanism.

51 Causal observability depth and worked rooms

Use first-order Hasse releases in ε and T , together with filtration-preserving q-Weyl/q-Pochhammer doors, as the local continuation alphabet. Define $d_{\text{obs}}(f)$ to be the least number of first-order lowering steps needed to produce a nonzero ordinary output.

Theorem 51.1 (Least-factor observability index). *For composite n ,*

$$d_{\text{obs}}(B_n) = \ell(n).$$

Prime rooms have infinite local observability depth.

Proof. The branch-depth theorem places B_n in $\mathfrak{m}_n^{\ell(n)}$, so fewer than $\ell(n)$ first-order releases leave the result in $\mathfrak{m}_n$ and its ordinary residue is zero. Exactly $\ell(n)$ releases can select either the $\varepsilon^{\ell(n)}$ constant channel or the $\varepsilon T^{\ell(n)-1}$ channel, both nonzero by [Theorem 50.3](#). $\square$

51.1 Room 6

Here $\ell(6) = 2$. The ε channel begins at T -depth one, while the ε^2 channel has nonzero constant coefficient:

$$B_6 \in \mathfrak{m}_6^2 \setminus \mathfrak{m}_6^3.$$

A first-order view is prime-like; a second-order release exposes the mixed-composite defect.

51.2 Room 49

Here $\ell(49) = 7$. Every channel $1 \leq j < 7$ begins at T^6 , and channel seven has a nonzero constant term. Thus

$$B_{49} \in \mathfrak{m}_{49}^7 \setminus \mathfrak{m}_{49}^8.$$

Every depth below seven is exactly prime-like even though the room is a prime power. Its missing information lies in the vertical singular history.

51.3 Room 77

The least factor of $77 = 7 \cdot 11$ is again seven. The same local depth occurs as for room 49, but the structural histories differ: room 49 is carried by one vertical prime-power axis, whereas room 77 has a mixed two-axis PBW history. The one-jet valuation sees the same depth; the complete tri-class maze state distinguishes the provenance.

Depth is not the whole state

Rooms 49 and 77 demonstrate why a valuation cannot replace Hist and Sing. Both first become locally visible at depth seven. One is a vertical singular history and the other an ordinary mixed history. Their certificates, contractions and legal future transports are different even though their depth statistic agrees.

52 The valuation-local compiler barrier

The branch-depth theorem becomes a complexity obstruction only after a compiler class and a cost model are stated.

Definition 52.1 (Valuation-local fixed-operation compiler). An admissible local compiler may use addition, multiplication, fixed polynomial combinations, unit multiplication, q-Pochhammer/q-Weyl transports, filtration-preserving semilinear substitutions, fixed Frobenius maps, modular reductions and Hasse releases. Conditional branches may inspect only ordinary residues of expressions containing the supplied branch section.

It may not read a factor from syntactic circuit labels, branch independently on the room integer, expand every Boolean stalk, call a factorization, primality, gcd or sieve subroutine, invoke a support oracle, or charge a variable-order derivative as one fixed operation.

Every filtration-preserving operation has lowering cost zero. A Hasse release of total order s has lowering cost s . The lowering budget $L(C)$ is the maximum accumulated lowering cost on an execution path.

Lemma 52.2 (One-step filtration estimate). *If $x - x' \in \mathfrak{m}^r$, a filtration-preserving algebraic operation sends the two inputs to outputs differing by an element of $\mathfrak{m}^r$. A Hasse release of order s sends them to outputs differing by an element of $\mathfrak{m}^{r-s}$.*

Proof. The statement is immediate for addition and filtration-preserving substitution. For multiplication,

$$ab - a'b' = (a - a')b + a'(b - b'),$$

so the filtration order cannot fall. Fixed polynomial maps follow by telescoping monomials. Hasse coefficients satisfy

$$\partial_\varepsilon^{[a]} \partial_T^{[b]}(\mathfrak{m}^r) \subseteq \mathfrak{m}^{r-a-b}.$$

□

Theorem 52.3 (Filtered circuit indistinguishability). *If $x \in \mathfrak{m}^r$ and $r > L(C)$, the compiler has the same ordinary observations, conditional branches and terminal ordinary output on x as on the zero section.*

Proof. Run both computations in parallel. Inductively, the difference at a circuit node lies in $\mathfrak{m}^{r-L_v}$, where L_v is the lowering cost paid on the path to that node. Since $L_v \leq L(C) < r$, every difference still lies in $\mathfrak{m}$ and has zero ordinary residue. All observed conditions and branches agree; so does the terminal ordinary output. □

Lemma 52.4 (Balanced-semiprime witnesses). *For all sufficiently large X , a semiprime $n = rs \leq X$ exists with*

$$\ell(n) = r > \frac{1}{4}\sqrt{X}.$$

Proof. By Bertrand's postulate choose a prime $\frac{1}{4}\sqrt{X} < r < \frac{1}{2}\sqrt{X}$, then choose a prime s with $r < s < 2r$. The product satisfies $rs < 2r^2 < X$, and r is its least factor. □

Theorem 52.5 (F036 valuation-locality barrier). *If an admissible valuation-local compiler correctly distinguishes the zero prime section from every composite branch section for all rooms $n \leq X$, then*

$$L(C_X) \geq \frac{1}{4}\sqrt{X}$$

for all sufficiently large X .

Proof. Choose $n = rs$ from Lemma 52.4. The least-factor theorem gives $B_n \in \mathfrak{m}^r$. If $L(C_X) < r$, filtered indistinguishability forces the same output on B_n and zero. Section correctness requires zero to be labelled harmonic and the semiprime section to be labelled nonharmonic, a contradiction. $\square$

Exact scope of F036

The theorem does not lower-bound every conceivable primality algorithm. It lower-bounds the stated representation-invariant local operation class. A family-specific algorithm using unrelated arithmetic on n lies outside the class and must instead be audited for classical primality/factor computation and for whether it creates the complete causal child.

53 The parent multiplication-history complex

The positive route changes the question from “extract a support bit from a cold room” to “transmit all parent-generated boundaries and retain the first complete obstruction.” Fix a complete parent state at prime grade p and the ordered corridor

$$B_p = (p, 2p) \cap \mathbb{N}.$$

Let $\mathbf{M}_p$ be the free commutative exponent monoid on already accepted prime axes $r < p$. An exponent vector α has ordinary weight

$$\text{wt}_p(\alpha) = \prod_{r < p} r^{\alpha_r},$$

but its complete history object $\mathbf{H}_\alpha$ also retains PBW order, q-Weyl holonomy, vertical singular provenance and certificate data.

Define the ordered target module

$$\mathcal{C}_0(p; B_p) = \bigoplus_{n \in B_p}^{\text{ord}} \mathbb{Z}\langle \mathbf{Q}_n \rangle$$

and the source-history module

$$\mathcal{C}_1(p; B_p) = \bigoplus_{\substack{\alpha: \text{wt}_p(\alpha) \in B_p \\ |\alpha| \geq 2}} \mathbb{Z}\langle \mathbf{H}_\alpha \boxtimes \mathbf{Q}_{\text{wt}_p(\alpha)} \rangle.$$

The universal multiplication door defines

$$d_{p, B_p} : \mathcal{C}_1(p; B_p) \rightarrow \mathcal{C}_0(p; B_p)$$

by forgetting the displayed provenance only after it has generated the target:

$$d_{p,B_p}(\mathbf{H}_\alpha \boxtimes \mathbf{Q}_{\text{wt}(\alpha)}) = \mathbf{Q}_{\text{wt}(\alpha)}.$$

Theorem 53.1 (Parent-history coverage). *For $n \in B_p$, the grade n is composite if and only if it has a unique PBW-normalized exponent object $\alpha \in \mathbf{M}_p$ of weight n and total exponent at least two.*

Proof. If n is composite and had a prime divisor $r \geq p$, its complementary factor would be at least two, giving $n \geq 2p$, contrary to the corridor. Every prime factor is therefore an already accepted axis below p , and unique factorization supplies a unique exponent vector. Conversely, an exponent vector with total exponent at least two has composite weight. Different ordered words may retain different holonomy before the coherence quotient, but they normalize to the same exponent object. $\square$

Split the source into

$$\mathcal{C}_1 = \mathcal{C}_1^{\text{mix}} \oplus \mathcal{C}_1^{\text{vert}},$$

where the mixed sector uses at least two prime axes and the vertical sector is supported on one axis with exponent at least two. This is the corridor form of the ordinary/singular composite split: mixed composites use horizontal PBW history and prime powers use a vertical singular ray.

Theorem 53.2 (Tri-class boundary theorem). *For the two-term complex*

$$\mathcal{K}_p(B_p) : 0 \rightarrow \mathcal{C}_1(p; B_p) \xrightarrow{d_{p,B_p}} \mathcal{C}_0(p; B_p) \rightarrow 0,$$

one has

$$H_1(\mathcal{K}_p(B_p)) = 0,$$

and

$$H_0(\mathcal{K}_p(B_p)) \cong \bigoplus_{\substack{q \in B_p \\ q \text{ prime}}}^{\text{ord}} \mathbb{Z}\langle \mathbf{Q}_q \rangle.$$

Mixed composites and prime powers are boundaries of the same source-generated differential; prime rooms are the surviving complete harmonic objects.

Proof. By parent-history coverage, the differential has one normalized source summand for every composite target and none for a prime target. On each source summand it maps isomorphically to its target-grade generator when provenance is forgotten, so it is injective. Its image is exactly the direct sum of composite grades. The kernel is zero and the cokernel is the ordered direct sum of prime-grade objects. The mixed/vertical split records which history contracts a composite and does not change the image. $\square$

Corollary 53.3 (Universal harmonic quotient). *Every additive ringed causal morphism that annihilates the parent-history boundaries factors uniquely through $H_0(\mathcal{K}_p(B_p))$.*

Proof. This is the universal property of the cokernel. The ringed and certificate coordinates are preserved because every source and target component is typed in the target grade created by the same multiplication door. $\square$

54 Worked successor corridors

54.1 The corridor after five

The ordered grades are

$$B_5 = \{6, 7, 8, 9\}.$$

Their complete parent histories are

$$6 = 2 \cdot 3 \quad (\text{mixed}), \quad 8 = 2^3 \quad (\text{vertical}), \quad 9 = 3^2 \quad (\text{vertical}).$$

These three targets lie in $\text{im } d_{5, B_5}$. The grade seven has no parent multiplication history and survives as $\mathbf{Q}_7$. Notice that the singular vertical sector is needed even in this four-room example: without it, eight and nine would not carry the correct contraction provenance.

54.2 The corridor after seven

Now

$$B_7 = \{8, 9, 10, 11, 12, 13\}.$$

The source histories contract

$$8 = 2^3, \quad 9 = 3^2, \quad 10 = 2 \cdot 5, \quad 12 = 2^2 \cdot 3.$$

The complete objects $\mathbf{Q}_{11}$ and $\mathbf{Q}_{13}$ survive. Ordered transmission selects $\mathbf{Q}_{11}$ first and carries a contraction certificate for 8, 9, 10. The object at 13 remains a later harmonic object; it is not preloaded as the current child.

54.3 Why the source direction matters

An ordinary target-first algorithm would inspect a grade, recover a factor if composite and attach a proof. The causal construction reverses that dependence: existing source histories generate their target doors, and the unhit object is the obstruction. This difference does not deny ordinary unique factorization. It controls which information is causally available before the child is known.

55 Same-map certificate and nonprojection

On an already source-attached composite target, let h return its normalized parent provenance. Let $P_{\mathcal{H}}$ be the ordered projection to the harmonic cokernel. The source-generated contraction satisfies

$$dh + hd = I - P_{\mathcal{H}}.$$

On the strict prefix before the first harmonic object q , it reduces to

$$dh + hd = I.$$

Theorem 55.1 (Same-map next-child packet). *The differential and its source-attached contraction produce the joint packet*

$$\Pi_{p \rightarrow q} = \left(P_{\mathcal{H}} \mathbf{Q}_q = \mathbf{Q}_q, (dh + hd)|_{(p,q)} = I \right).$$

The first component is the complete harmonic child certificate and the second is the empty-prefix certificate. Both are generated by the same multiplication-history map.

Proof. Every object in the strict prefix is a source-generated boundary, so the contraction identity is exact there. The first target on which the harmonic projection is nonzero is not hit by the differential and survives as a complete object. By [Theorem 53.2](#), its ordinary grade is prime; by ordered minimality it is the next such grade. No separate primality bit or post-hoc prefix scan is required to define the high-dimensional packet. $\square$

The contraction does not factor through the ordinary weight shadow. Two ordered q-Weyl paths may have the same exponent vector and ordinary product but distinct central holonomy:

$$u^{H_1} \mathbf{C}^\alpha, \quad u^{H_2} \mathbf{C}^\alpha, \quad H_1 \neq H_2.$$

A future singular release may distinguish them. Returning the wrong history can therefore break future equivalence and certificate-ideal transport even when the integer factorization is correct.

Corollary 55.2 (Algorithmic third-class necessity). *Deleting the vertical singular provenance before the child is formed destroys the contraction of prime-power targets and invalidates the same-map packet. Thus the third class is used by the transition, not merely stored beside it.*

56 The harmonic history series and child extension

Introduce a history-depth parameter Ω and form

$$\mathfrak{A}_p(\Omega) = \sum_{n \in B_p} \Omega^{n-p} \mathbf{Q}_n - \sum_{\alpha} \Omega^{\text{wt}(\alpha)-p} d_{p,B_p}(\mathbf{H}_\alpha \boxtimes \mathbf{Q}_{\text{wt}(\alpha)}).$$

Each negative term is the image of a named source boundary; this is not an untracked scalar cancellation.

Theorem 56.1 (Harmonic series reduction). *In the cokernel,*

$$[\mathfrak{A}_p(\Omega)] = \sum_{\substack{q \in B_p \\ q \text{ prime}}}^{\text{ord}} \Omega^{q-p} \mathbf{Q}_q.$$

The first nonzero coefficient is the complete next-child object, not its ordinary integer alone.

Proof. Parent-history coverage supplies exactly one normalized boundary for each composite grade. Those named images vanish in the cokernel. Prime grades have no source boundary and survive with the inherited corridor order. $\square$

After the first child q is formed, the parent history monoid extends by a new axis

$$\mathbf{M}_q \cong \mathbf{M}_p \oplus \mathbb{N} \mathbf{e}_q.$$

The complete state extension has the form

$$Q_q^{\text{caus}} = \text{FreeAxisExtend} \left(Q_p^{\text{caus}}, \mathbf{Q}_q, \Pi_{p \rightarrow q} \right).$$

The axis q is absent before the obstruction appears. Its construction is a high-to-high transition; the equality of its terminal grade with the ordinary next prime is a readout theorem.

57 Ordered blocks and the remaining compiler cost

For adjacent ordered blocks $B = B_L B_R$, the complex decomposes as an ordered sum. The right block receives the complete output state of the left block, including its singular tail and proof DAG. The summary law is

$$\text{Summary}(B_L B_R) = \begin{cases} \text{Hit}_L, & B_L \text{ leaves a harmonic obstruction,} \\ \text{Compose}(\text{Miss}_L, \text{Summary}_{\omega_L^{\text{out}}}(B_R)), & B_L \text{ contracts completely.} \end{cases}$$

Associativity follows from ordered direct-sum associativity, ringed transport composition and the contraction identity. A permutation-invariant reduction would be invalid because it could reorder the first obstruction.

Relative to a supplied persistent parent history object, a block descriptor needs only its endpoints, the weight functor and references to existing normalization/certificate schemas. That relative descriptor may be small. It does not imply a small ordinary compiler for materializing the cofiber, locating the first obstruction or maintaining the total parent state.

This is the exact point at which v5.9's zipper enters. It does not make the first paid refinement disappear. It proves that the paid right-sibling packets produced by that refinement remain valid and need not be reconstructed after each earlier child.

58 Three nonclaims that protect the new mathematics

Explicit non-claim 58.1 (No support-bit promotion). A finite jet, determinant, Frobenius residue, kernel dimension or central support bit is not promoted to a native child merely because it agrees with primality on tested inputs. A child must carry the complete causal fibre, empty-prefix proof and same-law extension packet.

Explicit non-claim 58.2 (No free relation-to-selector projection). A succinct relation containing all source/target incidences does not supply a free operation selecting its first unhit target. Any such extraction is an algorithmic interface and must be charged or replaced by a proved causal transition.

Explicit non-claim 58.3 (No hidden local-door compression). Storing one packet per canonical source door in a smaller container does not remove the work of constructing or updating those packets. This is precisely the F042 barrier proved in [Theorem 62.1](#).

59 The twenty-six mathematical organs and their noninterchangeable duties

The integrated architecture connects old mathematics to new state structure, but no one subject is claimed to do every job. The following table is an interface map, not a list of decorative analogies.

Organ	Exact duty in Angel Prime	It does not supply
Discrete dynamics	Types the repeated parent-to-child transition and its stage order.	A proof of arithmetic identities by itself.
Category theory	Makes rooms objects and certified programs composable morphisms.	A bit-complexity bound.
Higher categories	Stores swaps, associators and curvature residuals as 2-cells.	Permission to erase a non-silent residual.
Topology	Distinguishes contractible full packets from surviving obstructions.	A physical spatial interpretation.
Homology	Makes composites boundaries and harmonic children cokernel classes.	A free algorithm for computing the cokernel.
Cohomological language	Organizes obstruction and certificate transport.	A replacement for explicit certificates.
Fibre theory	Separates complete room fibres from terminal scalar grades.	A claim that nature realizes the fibre.
Connections	Transport ringed data along legal roads.	Path independence without a flatness/coherence theorem.
Holonomy	Retains order-sensitive loop information.	Ordinary factorization or primality.
Commutative algebra	Carries ideals, quotient/remainder data and certificate inclusions.	General inversion in a noncommutative chart.
Noncommutative algebra	Represents ordered PBW/q-Weyl histories.	A reason to discard the commutative ordinary shadow.
Representation theory	Diagonalizes cyclic actions into exact-period sectors.	The production cyclic compiler.
Algebraic geometry	Supplies room fibres, restrictions and local/ramified charts.	A finite test as a universal theorem.
Combinatorics	Supplies q-binomial identities, source words and interval trees.	The causal validity of source reuse.
Graph/DAG theory	Represents shared proof histories and the zipper stack.	A license to hide node construction cost.
Automata/transducers	Specify consume, refine, emit and transport stages.	A proof that the emitted grade is prime.
Information theory	Separates uniform descriptor bits from paid state-dependent information.	Automatic compression of exact incidence.
Control theory	Interprets horizon and observability depth.	An ordinary feedback channel into the native state.
Spectral theory	Exposes cyclic modes and semisimple HDIV axes.	A universal noncentral inverse.
Analytic number theory	Interprets internal counts asymptotically after the causal theorem.	A native transition premise.
Prime certificates	Validate the terminal harmonic grade and prefix.	The source-first child construction.

Organ	Exact duty in Angel Prime	It does not supply
Rank/select language Complexity theory	Interprets an already emitted ordered thread. Charges work, depth, live space and materialization separately.	A free selector on a dormant relation. Novelty from a favorable finite timing alone.
Tri-class arithmetic Causal kernels	Holds current scalar, future-active history and reconstructing residual together. Define exactly which residuals are future-silent.	The claim that each sector has rank one. Permission to quotient by current observation equality.
History Curvature	Normalizes equivalent proof paths and residual transports.	Compression of distinct canonical source heads.

The table also answers why compatibility with old mathematics is not a slogan. Ordinary arithmetic, factorization, q-series, group rings, ideals, homology and complexity retain their standard local meanings. The new layer is the typed way those objects are assembled around complete causal states, histories, residuals and projection boundaries.

Architectural conclusion before the v5.9 theorem

The one-jet theorem proves an exact least-factor depth and a square-root barrier for bounded local release. The multiplication-history complex replaces a cold support bit by a complete source-generated harmonic child and proves why the singular prime-power class is algorithmically necessary. The remaining cost is not denied. Version v5.9 attacks one precisely identified part of it: repeated reconstruction of already-paid source packets.

60 From a terminal prime observation to an Angel Prime transition

The factorial/Wilson chart answers a terminal question about a supplied candidate. The stronger Angel Prime question is different:

$$Q_p^{\text{caus}} \longrightarrow Q_q^{\text{caus}} \longrightarrow \text{ordinary observation of } q,$$

where the complete child state must exist before its ordinary prime shadow is read. Volume V studies this state-to-state interface. It is included here because it explains why the high-dimensional history and residual sectors are algorithmic organs rather than decoration.

The status must be stated before the theorem.

Strict status of the Volume V chart

The results in this section are proved in a frozen source-first model and have a separate strict finite verifier. They are *verifier-only* under [Protocol 6.1](#); they are not exported by the modular factorial SDK. They prove residual reuse and an amortized horizon bound. They do not prove a $p^{o(1)}$ cold next-prime compiler from a dormant descriptor.

60.1 Why the earlier cold interface repeated work

Suppose a parent causal state has already paid to construct canonical source-head information on a future interval. A cold implementation throws that information away after emitting one child, then reconstructs essentially the same source program before the next child. The scalar answer can still be correct, but history has been demoted to a log.

The fifth-volume revision changes that contract. If a paid packet was constructed before projection and a theorem proves that a later child cannot invalidate it, the packet may remain in the child state's Hist/Sing fibre as a future-sufficient residual program. It is state-dependent because it records the actual paid source path. It is not uniform advice, a prime table or a preloaded future answer.

Principle 60.1 (Transmission first, projection last). *Every source door, contraction packet, child fibre, residual payload and certificate-ideal transport used to create a child is constructed in the high-dimensional category. Only after the transition packet is frozen may a terminal observer read an ordinary grade or primality shadow. No such shadow may choose or repair a native edge.*

This principle is the common causal boundary of the factorial bridge, cyclic chart and fifth-volume horizon machine.

61 Frozen parent room and canonical source doors

Fix a harmonic parent grade p and a complete parent causal state $Q_{<p}^{\text{caus}}$. The model assumes that every lower harmonic source axis, its PBW history, ringed transport and certificate ideal are addressable inside this frozen parent. This is a strong interface assumption and is charged explicitly.

For a room grade $n > p$, a *canonical source door* is the unique admitted source-head factorization

$$n = ab, \quad 2 \leq a \leq b,$$

selected by the canonical source-head law of the Volume V model. A room with such a door is source-contracted; a room with no door is called harmonic in the internal chart. The project theorem relating these harmonic rooms to ordinary prime grades belongs to the frozen integrated-volume foundations; the zipper theorem below concerns their ordered high-dimensional transmission.

Definition 61.1 (Ringed source packet). For an ordered integer interval $B = [L, R]$, define

$$\text{Pkt}_p(B) = (B, c_B, \Theta_B, \mathcal{D}_B, \Sigma_B, C_B),$$

where c_B is the number of rooms with canonical source doors, Θ_B is the replayable ringed contraction, $\mathcal{D}_B$ is the door provenance, Σ_B is the residual history payload and C_B is the certificate-ideal witness. The packet is *paid* only after the complete source pass producing these fields has been charged.

Source-head uniqueness implies

$$0 \leq c_B \leq |B|.$$

A full packet has $c_B = |B|$ and contains no harmonic child. A deficient packet has $c_B < |B|$ and contains at least one. A deficient singleton necessarily contains exactly one harmonic child.

62 The local-work wall: what compression cannot hide

Before proving a positive reuse theorem, one must rule out a false complexity argument in which a smaller container hides the same number of charged local actions.

Theorem 62.1 (F042 explicit canonical-door barrier). *Let B be an ordered block and let*

$$C(B) = \#\{n \in B : n \text{ has an admitted canonical source door}\}.$$

Every compiler that executes a distinct charged local source/certificate action for every admitted door performs at least $C(B)$ such actions. If every room in B is source-contracted, the local work is $\Omega(|B|)$.

Proof. Canonical source-head uniqueness supplies exactly one admitted door for each contracted room and none for a harmonic room. The compiler class charges one action per admitted door, so its action count is at least their cardinality. On a full block every room contributes one door, hence $C(B) = |B|$. $\square$

The theorem is intentionally scoped. Local doors remain valid semantic atoms, and a strict verifier may replay them. What is prohibited is calling a compiler subpolynomial merely because the $|B|$ local actions are stored in a compact vector or summarized after execution. A cold asymptotic improvement must either apply a genuinely collective morphism or lawfully reuse work paid by an earlier transmission.

63 Ringed History-Curvature normal form

History reuse creates a second problem: transporting the same residual packet through many children must not grow a certificate path proportional to the number of children. History Curvature solves this coherence problem; it does not compress distinct source incidence.

Let a legal PBW history have length k and total grade below $2p$. Every nontrivial primitive grade is at least two, so

$$k \leq \lfloor \log_2(2p) \rfloor.$$

Adjacent legal swaps carry ringed semilinear 2-cells, and legal rebracketings carry associator 2-cells. A residual of such a cell may be erased only after it is proved to lie in the future-silent causal kernel.

Theorem 63.1 (Ringed History-Curvature normal form). *Assume every declared swap and rebracketing preserves certificate ideals and has future-silent curvature residual. Then every legal ordering/bracketing in one PBW class is future-equivalent to its nondecreasing, canonically bracketed word. A complete coherence witness uses $O(k^2)$ local 2-cell references and $O(\log^3 p)$ labelled bits modulo shared schema tables.*

Proof. At most $k(k-1)/2$ adjacent transpositions sort the word. Canonical rebracketing uses $O(k)$ associator cells. The hypotheses make every local step a valid ringed 2-morphism preserving the relevant certificate ideal. The discarded residuals are future-silent, so their composite cannot change a future Angel observable. Each grade/address label uses $O(\log p)$ bits, and $k = O(\log p)$. $\square$

F041 supplies the boundary: the theorem does not identify different canonical source heads or

force their incidence through a polylogarithmic linear state. It normalizes multiple proofs of the same source program. Incidence and coherence are different resources.

64 Anchor horizons and residual-packet stability

Fix an endpoint

$$p < E < 2p$$

and call

$$\mathcal{H}(p, E) = (p, E]$$

the *anchor horizon*. The endpoint specifies how far a paid parent source program is proposed to remain valid; it does not name the next child.

Definition 64.1 (Anchor source program). The program $\mathcal{P}_{p,E}$ is the complete canonical source-head program induced by $Q_{<p}^{\text{caus}}$ on grades in $(p, E]$, together with its PBW provenance, ringed Ω -transport, singular residuals and certificate-ideal data.

Theorem 64.2 (Residual-packet stability). *Let $p < q \leq E < 2p$, where q is a harmonic child emitted from the anchor horizon. For every n with $q < n \leq E$, the canonical source-door status computed from $Q_{<p}^{\text{caus}}$ equals the status after extension to $Q_{<q}^{\text{caus}}$. Every old door, PBW provenance, residual payload and certificate-ideal transport remains valid on $(q, E]$.*

Proof. First suppose n is contracted and write its canonical source decomposition $n = ab$ with $2 \leq a \leq b$. Because $n \leq E < 2p$,

$$a \leq b \leq \frac{n}{2} < p.$$

Both source components and their complete histories therefore already belong to the anchor parent. A newly emitted child $r > p$ cannot create a new nontrivial source decomposition of n , since any product using r is at least $2r > 2p > n$. The old canonical door is consequently unchanged.

Now suppose n is harmonic at the anchor. It has no old door. The same inequality shows that no generator added in $(p, q]$ can produce a door into n , so it remains harmonic. In either case the local source morphisms on the remaining horizon are identical. Child extension therefore acts only by the declared ringed stability transport, which preserves certificate-ideal inclusions. $\square$

Corollary 64.3 (Stability through many children). *For any increasing emitted thread*

$$p = q_0 < q_1 < \cdots < q_s \leq E,$$

the restriction of $\mathcal{P}_{p,E}$ to $(q_j, E]$ remains valid at every $Q_{<q_j}^{\text{caus}}$.

Proof. Apply [Theorem 64.2](#) after each extension. The common inequality $E < 2p < 2q_j$ only strengthens, so no later child can enter a source factorization of a remaining horizon grade. $\square$

This is the decisive use of the Angel maze's residual coordinate. The packet is not preserved by optimism; its future validity follows from the strict horizon inequality.

65 The Causal Horizon Zipper

Choose an arity $K \geq 2$. Starting from the root interval $(p, E]$, refine a deficient interval into at most K ordered child intervals. A depth-first machine descends into the first deficient child and retains the paid right siblings.

Definition 65.1 (Causal Horizon Zipper). The K -ary zipper $\mathcal{Z}_{p,E}$ is the ordered stack of retained tuples

$$(B, \text{Pkt}_p(B), \Theta_B, \mathcal{D}_B)$$

along the current deficiency path. Every stack packet has already been constructed and charged. The stack itself is part of the causal residual state.

Its transition has four cases:

1. A full packet contributes its replayable contraction to the pending empty-prefix certificate and is then consumed.
2. A deficient nonsingleton is refined once; the paid right siblings are retained and the machine descends into the leftmost deficient child.
3. A deficient singleton creates the complete harmonic child fibre and the joint transmission edge before any ordinary projection.
4. The remaining stack is transported into the child by residual-packet stability and curvature normalization.

Theorem 65.2 (Ordered consecutive-edge correctness). *Let the harmonic rooms in $(p, E]$ be*

$$q_1 < q_2 < \dots < q_P.$$

The zipper emits exactly the high-dimensional thread

$$Q_p^{\text{caus}} \rightarrow Q_{q_1}^{\text{caus}} \rightarrow \dots \rightarrow Q_{q_P}^{\text{caus}}$$

in that order. Before projection, edge j carries a joint packet

$$\left(\Pi_{q_j}^{\text{harm}}, \Pi_{(q_{j-1}, q_j)}^{\text{contracted}}, \Pi_{q_{j-1} \rightsquigarrow q_j}^{\text{ringed}}, \Sigma_{q_j}^{\text{res}} \right).$$

Proof. Every paid interval is full or deficient. A full packet contains no harmonic room, and its contraction can be skipped only after the certificate is composed. A deficient nonsingleton contains at least one harmonic room and is therefore refined. A deficient singleton has count zero and is harmonic. Depth-first traversal respects interval order, so the singletons are emitted in increasing grade. By [Corollary 64.3](#), packets computed at the anchor remain valid after every earlier child. Hence each empty prefix and later child is certified by the same high-dimensional source program, without target-to-history inversion. The transition rule constructs the child fibre and all four packet components before the observer is enabled. $\square$

Theorem 65.3 (Paid packet no-recomputation). *Within one anchor horizon, every interval node is refined at most once. A retained right-sibling packet is never recomputed merely because an earlier harmonic child was emitted.*

Proof. Refinement replaces a node by its children and the parent never re-enters the stack. Full packets are consumed once, deficient singletons are emitted once, and deficient internal packets

are refined once. A retained sibling remains valid by [Theorem 64.2](#), so child extension supplies no legal reason to rerun its source pass. $\square$

65.1 A small symbolic zipper trace

Let $K = 2$ and suppose the anchor interval consists of eight ordered rooms. A root packet with count six is deficient. Its two four-room child packets have counts four and two. The left child is full, so its contraction is composed and it is consumed without examining its rooms again. The right child is deficient and splits into two pairs with counts one and one. Each pair splits once more into one full singleton and one deficient singleton. The two deficient singletons become consecutive harmonic children.

The important ledger is not the particular counts but the direction of information:

$$\begin{aligned} \text{paid root} &\longrightarrow \text{paid child packets} \\ &\longrightarrow \text{first complete child plus retained sibling} \\ &\longrightarrow \text{second complete child.} \end{aligned}$$

After the first child, the second pair is not rebuilt. It survives as a state-dependent residual packet whose validity is guaranteed by the horizon theorem.

66 Curvature-normalized direct transport

Naively composing one transport certificate per emitted child would retain a path of length P . On the stable source subcategory, child-extension squares differ only by declared curvature residuals. The future-silent quotient therefore defines a direct transport

$$\rho_{p \Rightarrow q}^E : \mathcal{Z}_{p,E} |_{(q,E]} \longrightarrow \mathcal{Z}_{q,E}.$$

Theorem 66.1 (Curvature-normalized residual transport). *For $p < q < r \leq E < 2p$,*

$$\rho_{q \Rightarrow r}^E \circ \rho_{p \Rightarrow q}^E \equiv_{\text{fut}} \rho_{p \Rightarrow r}^E.$$

Consequently the live residual certificate has a canonical ringed normal form rather than size proportional to the number of emitted children.

Proof. By [Theorem 64.2](#), the source doors on $(r, E]$ are identical under all three states. The composite and direct transports therefore have the same endpoints and differ only by child-extension coherence cells. The curvature-normal-form hypothesis places each residual in the future-silent kernel and preserves the certificate ideal. Composition preserves these properties, giving future equivalence. $\square$

67 Exact internal time–space theorem

Put

$$H = E - p, \quad P = \#\{\text{harmonic children in } (p, E]\},$$

and let $A_{p,E}$ be the number of addressable harmonic source axes at most $\sqrt{E}$. Define the internal reciprocal shadow

$$\mathcal{R}_{p,E} = \sum_{\substack{a \leq \sqrt{E} \\ a \text{ harmonic}}} \frac{1}{a}, \quad t = \lceil \log_K H \rceil + O(1).$$

For $L = \lceil \log_2(2p) \rceil$, let $c_{\text{scan}}(L)$, $c_{\text{door}}(L)$ and $c_{\text{pkt}}(L)$ be the charged bit costs of an addressed tail inspection, an accepted door update and a packet/schema update. Let $s_{\text{pkt}}(L)$ be the packet-schema bit size.

Lemma 67.1 (One-refinement cost). *For a node B of length m , one paid refinement has work*

$$O([m\mathcal{R}_{p,E} + A_{p,E}]c_{\text{scan}}(L) + mc_{\text{door}}(L) + Kc_{\text{pkt}}(L)).$$

Proof. For a source axis a , the addressed tail interval has length at most $m/a + 1$. Summing over the actual source-axis thread gives at most $m\mathcal{R}_{p,E} + A_{p,E}$ inspections. Source-head uniqueness admits at most one accepted door per target room, hence at most m door updates. The refinement creates at most K child packets. $\square$

Theorem 67.2 (Residual-horizon joint time-space bound). *The total charged work for all P emitted edges in one anchor horizon is*

$$O(t[(H\mathcal{R}_{p,E} + PA_{p,E})c_{\text{scan}}(L) + Hc_{\text{door}}(L) + PKc_{\text{pkt}}(L)]).$$

The peak auxiliary residual state is

$$O(Kts_{\text{pkt}}(L) + \log^2 p)$$

bits.

Proof. At one tree depth the deficient intervals are disjoint, so their total length is at most H . Each contains a harmonic room, so at most P deficient intervals occur at a depth. Sum [Lemma 67.1](#) over one depth and then over the t depths. A depth-first zipper keeps at most $K - 1$ right siblings per level and one active path. The History-Curvature normal form adds only its $O(\log^2 p)$ reference budget rather than a path per child. $\square$

Corollary 67.3 (Internal amortized edge bound). *For $P > 0$, the charged work per emitted edge, amortized over the horizon, is*

$$O\left(t\left[\left(\frac{H}{P}\mathcal{R}_{p,E} + A_{p,E}\right)c_{\text{scan}}(L) + \frac{H}{P}c_{\text{door}}(L) + Kc_{\text{pkt}}(L)\right]\right).$$

No analytic number theory enters this statement. Every symbol is an internal state count, an exact horizon parameter or a declared machine cost.

67.1 External dyadic interpretation, after the theorem

For the boundary choice $E = 2p - 1$, ordinary analytic estimates may be applied only after the high-dimensional theorem is frozen. With a polylogarithmic arity and packet costs, those external estimates yield the scoped shadow

$$\overline{W}_{5.9}(p) = p^{1/2+o(1)}, \quad S_{5.9}^{\text{aux}}(p) = \text{polylog } p$$

per emitted edge, amortized over the complete horizon.

Cold and amortized are not synonyms

The first paid refinement of the horizon still executes a source-head range program. The estimate above is not the worst-case cost of the first child from a dormant descriptor, and it is not a $p^{o(1)}$ cold generator. It says that already-paid source work is shared by later children instead of being discarded and rebuilt.

68 Strict finite realization and what it proves

The archived v5.9 verifier freezes a source-first parent state, opens several horizons strictly below twice their anchors, constructs residual packets without ordinary prime queries, emits complete child threads, replays empty-prefix proofs against the anchor program, and only afterward constructs an independent ordinary audit.

The frozen $K = L^2$ run reports:

```
HORIZON_CASES=4
HIGH_DIMENSIONAL_CRITICAL_EDGES=2119
ZIPPER_REFINEMENTS=1068
ZIPPER_TAIL_INSPECTIONS=93682
RESIDUAL_PACKETS_CREATED=22764
FULL_PACKETS_REUSED=19581
RESIDUAL_STABILITY_CHECKS=22764
MAX_LIVE_RESIDUAL_PACKETS=382
COLD_REFERENCE_TAIL_INSPECTIONS=14720643
TAIL_INSPECTION_SAVING=14626961
```

The cold reference is itself a canonical source compiler executed before the ordinary audit; it is not an ordinary sieve substituted for the high state. Both modes emit the same 2,119 child weights. Runs at $K = L, L^2, L^3$ have the same child-weight hash while exhibiting the predicted refinement/space tradeoff. Sanitized strict runs have no address or undefined-behaviour diagnostic in the archived environment.

Finite verification does not prove [Theorem 67.2](#). The proof does. The finite result demonstrates that the stateful packet lifecycle, source-first order, stability checks, no-recomputation rule and resource counters are realized on the exercised cases.

69 Why the zipper is not merely an incremental sieve

Its ordinary shadow necessarily distinguishes contracted and harmonic rooms, so a visual resemblance to sieving is unavoidable. The native state contains strictly more than a Boolean composite mark:

- canonical PBW source provenance rather than a divisibility bit;
- ordered Ω -history and a state-dependent Σ^{res} payload;
- ringed semilinear source maps and certificate-ideal inclusions;

- replayable empty-prefix contraction packets;
- a complete child fibre before an ordinary grade is visible;
- a curvature-normalized morphism proving why a packet remains valid.

Deleting those fields and retaining only next multiples would produce an old ordinary incremental sieve. Wrapping such a queue in an Angel type would not restore the lost semantics. The comparison therefore serves as a boundary alarm: exact agreement of terminal prime grades is required, but agreement alone does not identify the native algorithms.

70 Relation to the factorial and cyclic charts

The three charts now have explicit interfaces.

Chart	Native object	Terminal shadow	Executable status
Factorial maze	Certified principal-jet coordinate embedded in Ord/Hist/Sing and its specialized functor.	Exact $n!$ or $n! \bmod m$ after execution.	Implemented in the modular SDK.
Cyclic closure	Integral group-ring evaluation and exact-period germ.	Kernel, valuation and constant-mode data.	Bounded reference realization; compact production compiler open.
Residual horizon	Paid source packets transported through complete causal children.	Ordered prime grades only after thread freeze.	Strict Volume V verifier; not public SDK.

The factorial chart proves that compact source syntax can carry a lawful arithmetic denotation and terminal Wilson use. The cyclic chart shows that the source has period-stratified higher structure. The horizon chart shows that history can reduce later native work when a stability theorem licenses reuse. They are not identical algorithms, but all enforce source-first construction, retained unresolved information and projection-last semantics.

71 The surviving Angel Prime frontier

The v5.9 result moves the frontier from passive provenance to operational history reuse. It still leaves two explicit interfaces.

First, the parent-room bundle is frozen and addressable. A future compiler must replace that total parent interface by a collective descriptor-native source morphism without recovering target factor histories. Second, F042 shows that executing one charged local action for every composite door cannot yield subpolynomial cold work. A lawful escape must apply a collective ringed program, not merely a smaller container around the loop.

The open drill tip is therefore a morphism

$$\text{BulkSrc}_\Omega(D_p, B)$$

which produces a certified block contraction/count packet without materializing the full parent bundle, executing one local door per contracted room, projecting a relation into a free selector, or deleting v5.9's stability and residual-reuse laws.

Angel Prime conclusion of the integrated chart

The fifth-volume theorem does not finish the native generator. It proves a specific new fact needed by such a generator: paid high-dimensional history can remain valid across multiple causal prime children and can measurably reduce future native work. That statement is mathematical, has a strict finite realization, and is kept separate from the still-open cold compiler.

PART V

Structural Probing and the Higher-Dimensional Angel Prime Atlas

72 Why two exact charts force an atlas question

The factorial and cyclic constructions are not two unrelated metaphors. They are two specializations of the same finite source polynomial

$$F_n(Z, u) = \prod_{j=0}^{n-1} (1 + Zu^j).$$

The infinitesimal substitution

$$(Z, u) = (-(1 + T), 1 + T)$$

and the cyclic substitution

$$(Z, u) = (-S, S)$$

both produce the prefix product

$$\mathcal{P}_n(X) = \prod_{k=1}^n (1 - X^k),$$

with $X = 1 + T$ in the first chart and $X = S$ in the second. The first chart retains a full jet tower; the second retains all exact-period character sectors. That common source makes the following question unavoidable:

What is the smallest exact object from which all admitted finite jets, cyclic rooms, histories, singular residuals and certificates can be obtained by legal functors?

Calling the answer an “infinite-dimensional space” before defining its transition maps would be too quick. This part therefore constructs two rigorous completions, an atlas that relates them, and a precise list of what is still conjectural. The completions enlarge the theory; they do not add a constant-time instruction to the SDK.

Three levels of assertion in this part

Theorems are consequences of the stated rings, inverse systems and finite character decompositions. **Restricted theorems** require a declared semisimple or finite-jet chart. **Conjectures** describe the possible Angel full space or native birth dynamics and are never used to prove the executable Wilson result.

73 The procyclic room completion

Let $C_m = \langle S_m : S_m^m = 1 \rangle$ and $A_m = \mathbb{Z}[C_m]$. Whenever $d \mid m$, define the room quotient

$$\rho_{m,d} : A_m \longrightarrow A_d, \quad S_m \longmapsto S_d.$$

These maps satisfy $\rho_{d,e}\rho_{m,d} = \rho_{m,e}$ for $e \mid d \mid m$.

Definition 73.1 (Procyclic Angel room algebra). The procyclic room algebra is the inverse limit

$$\hat{A}_{\text{cyc}} = \varprojlim_{d \mid m} A_m.$$

Its canonical shift is the compatible family $\hat{S} = (S_m)_{m \geq 1}$. For fixed $n \geq 0$, put

$$\hat{P}_n = \prod_{k=1}^n (1 - \hat{S}^k), \quad \hat{J}_n = (-1)^n \prod_{k=1}^n [k]_{\hat{S}}.$$

This notation is a projective coordinate system, not an appeal to an uncontrolled “all matrices at once.” An element of $\hat{A}_{\text{cyc}}$ is exactly a family $(a_m)_m$ satisfying $\rho_{m,d}(a_m) = a_d$ whenever $d \mid m$.

Theorem 73.2 (Compatible finite-room shadows). *For every fixed n and every room m ,*

$$\pi_m(\hat{P}_n) = P_{m,n} := \prod_{k=1}^n (1 - S_m^k), \quad \pi_m(\hat{J}_n) = J_{m,n} := (-1)^n \prod_{k=1}^n [k]_{S_m}.$$

Moreover,

$$\boxed{\hat{P}_n = (-1)^n (1 - \hat{S})^n \hat{J}_n.}$$

Thus prefix death, the q -integer action and the universal source program are three coordinates of one compatible procyclic family.

Proof roadmap. Every room projection is a ring homomorphism and takes $\hat{S}$ to S_m . Apply it factorwise and use $1 - X^k = (1 - X)[k]_X$. The full proof is in [Appendix B.1](#). $\square$

73.1 The diagonal self-room warning

The primitive-period operator in room m is

$$K_m = P_{m,m-1} = \prod_{k=1}^{m-1} (1 - S_m^k).$$

The prefix length now depends on the room. Consequently the family $(K_m)_m$ is *not* an element of $\hat{A}_{\text{cyc}}$: if $d \mid m$ and $m > d$, the factor $1 - S_m^d$ maps to zero, so

$$\rho_{m,d}(K_m) = 0, \quad K_d \neq 0.$$

Proposition 73.3 (Self-room operators form a dependent section). *The family $m \mapsto K_m$ is a section over the room-size index in the total category of pairs (m, n) , not a single compatible point of the fixed-prefix inverse limit. Any “profinite Angel object” that contains every K_m as its m th coordinate must therefore remember the room-dependent prefix, or weaken the transition law; it cannot silently use the fixed-prefix inverse limit.*

This distinction matters. The fixed program $\hat{P}_n$ is a genuine pro-object. The prime question uses the diagonal $n = m - 1$ and is a dependent family. The larger Angel object is therefore naturally an atlas or fibration, not merely one element of a completed group ring.

74 The q-Pochhammer factor tower

Let $B = \mathbb{Z}[Z, u, u^{-1}]$. Since

$$F_{n+1}(Z, u) = F_n(Z, u)(1 + Zu^n),$$

the principal ideals satisfy $(F_{n+1}) \subseteq (F_n)$ and give quotient maps

$$B/(F_{n+1}) \longrightarrow B/(F_n).$$

Definition 74.1 (Program-factor completion). The q-Pochhammer factor completion is the pro-ring

$$\hat{B}_F = \varprojlim_{n \geq 1} B/(F_n).$$

A point is a compatible sequence of residues along the nested factor tower. It records all finite source-factor constraints without choosing an infinitesimal or cyclic target chart.

This completion is deliberately modest. It does not assert that every Angel state is a point of $\hat{B}_F$, nor that the factor topology is the unique correct topology. Its role is to state rigorously what the phrase “all finite q-Pochhammer source constraints” can mean.

Definition 74.2 (Two-chart Angel atlas). For each n , the source object F_n has two declared chart maps:

$$\begin{aligned} \lambda_n : B &\longrightarrow \mathbb{Z}[T], & (Z, u) &\longmapsto (-(1 + T), 1 + T), \\ \chi_{m,n} : B &\longrightarrow A_m, & (Z, u) &\longmapsto (-S_m, S_m). \end{aligned}$$

The resulting family of jets $\lambda_n(F_n)$, cyclic prefixes $\chi_{m,n}(F_n)$, causal histories and certificate maps is the *two-chart Angel atlas* on the source program.

The atlas is already larger than either visible chart. The infinitesimal map remembers coefficients in every T -degree; the cyclic maps remember every room m and each exact-period sector V_d . Adding causal history makes even equal algebraic chart values distinguishable when their future words differ.

75 The complete factorial jet spectrum

Set

$$Q_n(T) = F_n(-(1+T), 1+T) = \prod_{k=1}^n (1 - (1+T)^k).$$

The principal-jet theorem reads only $[T^n]Q_n = (-1)^n n!$. The rest of the jet is not empty. Factor each term as

$$1 - (1+T)^k = -kT A_k(T), \quad A_k(T) = \frac{(1+T)^k - 1}{kT} = \sum_{j=0}^{k-1} \frac{1}{k} \binom{k}{j+1} T^j.$$

Definition 75.1 (Factorial jet spectrum). Write

$$Q_n(T) = \sum_{r \geq 0} a_{n,r} T^{n+r}, \quad B_n(T) = \prod_{k=1}^n A_k(T) = \sum_{r \geq 0} b_{n,r} T^r.$$

The sequence $(a_{n,r})_{r \geq 0}$ is the factorial jet spectrum of order n .

Theorem 75.2 (All higher factorial coefficients). For $0 \leq r \leq n(n-1)/2$,

$$a_{n,r} = (-1)^n n! b_{n,r},$$

where

$$b_{n,r} = \sum_{r_1 + \dots + r_n = r} \prod_{k=1}^n \frac{1}{k} \binom{k}{r_k + 1}$$

with $0 \leq r_k \leq k-1$. The coefficients also satisfy

$$b_{n+1,r} = \sum_{j=0}^{\min(r,n)} \frac{1}{n+1} \binom{n+1}{j+1} b_{n,r-j}.$$

In particular,

$$\begin{aligned} a_{n,0} &= (-1)^n n!, \\ a_{n,1} &= (-1)^n n! \frac{n(n-1)}{4}, \\ a_{n,2} &= (-1)^n n! \frac{n(n-1)(9n^2 - 5n - 26)}{288}. \end{aligned}$$

Although the displayed normalized numbers may be rational, every $a_{n,r}$ is an integer because $Q_n(T) \in \mathbb{Z}[T]$.

Proof roadmap. Extract the common factor $(-1)^n n! T^n$, expand the finite product of the A_k , and collect weak compositions of r . The first two higher formulas are elementary symmetric sums. Details are deferred to [Appendix B.2](#). $\square$

Example 75.3 (The first nontrivial spectra).

$$\begin{aligned} Q_2(T) &= 2T^2 + T^3, \\ Q_3(T) &= -6T^3 - 9T^4 - 5T^5 - T^6, \\ Q_4(T) &= 24T^4 + 72T^5 + 98T^6 + 76T^7 + 35T^8 + 9T^9 + T^{10}. \end{aligned}$$

The Wilson consumer needs only the principal coefficient. A future cross-frame continuation may need higher coefficients, just as quotient transport can resurrect a terminally vanished cyclic sector at a higher ε -order.

75.1 Bell-polynomial organization

Over $\mathbb{Q}[[T]]$, write

$$\log B_n(T) = \sum_{j \geq 1} \kappa_j(n) \frac{T^j}{j!}.$$

Then the exponential Bell polynomials give

$$b_{n,r} = \frac{1}{r!} Y_r(\kappa_1(n), \dots, \kappa_r(n)).$$

This is an organizational identity, not a new algorithmic primitive. It shows precisely why partitions and set-partition polynomials appear in the higher factorial spectrum: they are the universal coefficient calculus for a product after logarithmic linearization.

76 Ramanujan-sum coordinates of the cyclic shadow

The projector theorem gives $K_m = m\Pi_m^{\text{prim}}$ after characteristic-zero extension. Fourier inversion makes its integral coefficients explicit.

Definition 76.1 (Ramanujan shadow row). For $j \in \mathbb{Z}$, the Ramanujan sum is

$$c_m(j) = \sum_{\substack{1 \leq a \leq m \\ (a,m)=1}} e^{2\pi i a j / m}.$$

It is an integer depending only on $\gcd(m, j)$.

Theorem 76.2 (Exact coefficient formula for the primitive shadow). *In the cyclic basis,*

$$K_m = \sum_{j=0}^{m-1} c_m(j) S_m^j.$$

If $g = \gcd(m, j)$, then

$$c_m(j) = \mu\left(\frac{m}{g}\right) \frac{\varphi(m)}{\varphi(m/g)}.$$

Consequently the first circulant row is constant off the diagonal if and only if m is prime. For a prime p ,

$$K_p = pI - \mathbf{1}\mathbf{1}^T,$$

so every row is $(p-1, -1, \dots, -1)$.

Proof roadmap. Write the primitive projector as the sum of rank-one Fourier projectors and invert the discrete Fourier transform. The arithmetic formula for $c_m(j)$ follows by Möbius inversion. See [Appendix B.3](#). $\square$

Example 76.3 (Composite fracture in room six). The Ramanujan row is

$$(c_6(0), \dots, c_6(5)) = (2, 1, -1, -2, -1, 1).$$

It has zero row sum and rank two, but it is not the pure complete-graph row $(5, -1, -1, -1, -1, -1)$. The different gcd strata are the visible fracture of the proper-period content.

77 The cyclic Laplacian prime theorem

Theorem 77.1 (Equivalent prime shadows). *For every integer $m \geq 2$, the following are equivalent:*

1. m is prime;
2. $\varphi(m) = m - 1$;
3. $\text{rank } K_m = m - 1$ and nullity $K_m = 1$;
4. $K_m = mI - \mathbf{1}\mathbf{1}^T$;
5. one, hence every, circulant row of K_m is $(m - 1, -1, \dots, -1)$;
6. $\sigma(K_m) = \{0, m^{(m-1)}\}$ over characteristic zero;
7. the time-valuation support is exactly $\{(1, m), (m, 1)\}$.

When these conditions hold, K_m is exactly the Laplacian of the complete graph on m labelled rooms.

Proof roadmap. The projector theorem gives rank $\varphi(m)$ and the Ramanujan formula gives the row. Euler's equality $\varphi(m) = m - 1$ and absence of an interior divisor both characterize primality. Full details are in [Appendix B.4](#). $\square$

The theorem is structural but should not be misread as a fast algorithm: materializing the row already costs $\Theta(m)$ outputs, and materializing the matrix costs $\Theta(m^2)$. Its force is that Wilson's scalar shadow, the primitive-period rank and a graph Laplacian symmetry all meet at the same source object.

77.1 Spanning trees and the cyclotomic discriminant

Theorem 77.2 (A numerical bridge between graph and cyclotomic invariants). *For a prime p , every principal cofactor of K_p equals*

$$p^{p-2}.$$

This is both the number of spanning trees of the complete graph and the absolute discriminant of the cyclotomic field $\mathbb{Q}(\zeta_p)$:

$$\det K_p^{[i]} = \tau(K_p) = |\text{Disc } \mathbb{Q}(\zeta_p)| = p^{p-2}.$$

The equality is an equality of exact numerical invariants. No canonical isomorphism between the graph lattice and the full ring of cyclotomic integers is asserted.

Proof roadmap. Apply the matrix-tree theorem to the spectrum of $pI - J$, and the standard derivative/discriminant formula to $\Phi_p(x)$. See [Appendix B.5](#). $\square$

78 The factorial–cyclic collision theorem

Fix a prime p and the common source

$$F_{p-1}(Z, u).$$

The infinitesimal chart gives

$$Q_{p-1}(T) = \prod_{k=1}^{p-1} (1 - (1+T)^k),$$

while the cyclic chart gives

$$K_p = \prod_{k=1}^{p-1} (1 - S_p^k).$$

Theorem 78.1 (Prime collision of two exact charts). *For prime p ,*

$$[T^{p-1}]Q_{p-1}(T) = (-1)^{p-1}(p-1)!,$$

and the first row of K_p is $(p-1, -1, \dots, -1)$. Reduction modulo p gives the common shadow

$$\boxed{[T^{p-1}]Q_{p-1}(T) \equiv -1 \pmod{p}, \quad \text{row}_0(K_p) \equiv (-1, \dots, -1) \pmod{p}.}$$

Thus the Wilson residue and every coordinate of the cyclic Laplacian row agree in $\mathbb{F}_p$, even though they arise from different chart functors and have different native dimensions.

This is the precise, proved content of “two high-dimensional mazes collide into one shadow.” It does not say that the factorial state equals the cyclic matrix, nor that one chart computes the other at zero cost. It says that two exact specializations of the same q-Pochhammer source have a canonically comparable residue in the prime room.

Corollary 78.2 (Wilson–Laplacian characterization). *For $m \geq 2$, the Wilson condition $(m-1)! \equiv -1 \pmod{m}$ and the pure-Laplacian condition $K_m = mI - J$ each characterize prime m . Their equivalence as predicates follows through primality; their certificates retain different histories and are not structurally identical.*

79 Higher observation tensors

The words “one-dimensional” and “four-dimensional” are ambiguous unless the observer is defined. We use two explicit ladders.

79.1 Tensor-power shadow ladder

Let $W_m = \mathbb{Q}[C_m]$ and let K_m act by left multiplication. For $r \geq 1$, define the r th tensor observer by

$$\text{Map}_r(m) = K_m^{\otimes r} \in \text{End}(W_m^{\otimes r}).$$

Theorem 79.1 (Tensor shadow rank and prime uniformity). *For every $m \geq 2$ and $r \geq 1$,*

$$\text{Map}_r(m) = m^r (\Pi_m^{\text{prim}})^{\otimes r}, \quad \text{rank } \text{Map}_r(m) = \varphi(m)^r.$$

For prime p ,

$$\text{Map}_r(p) = (pI - J)^{\otimes r} \equiv (-J)^{\otimes r} \pmod{p}.$$

Hence every entry of this chosen tensor shadow is congruent to $(-1)^r$ modulo p . This proves dimension-stable uniformity for the tensor-power observer, not for every conceivable higher observer.

79.2 Time–valuation–multiplicity ladder

Define the weighted divisor packet

$$\mathbb{T}_m = \sum_{d|m} \varphi(d) [d, m/d, \varphi(d)].$$

The coordinates are annihilation time, closure depth and sector multiplicity. For N cycles, enhance it by the leading germ:

$$\mathbb{G}_{m,N} = \sum_{d|m} \varphi(d) \left[d, \frac{Nm}{d}, (-1)^{Nm/d} \left(\frac{Nm}{d} \right)! d^{2Nm/d} \right].$$

Theorem 79.2 (Boundary-support criterion in every feature order). *The support of $\mathbb{T}_m$ has two points exactly when m is prime. In that case they are*

$$(1, m, 1), \quad (m, 1, m - 1).$$

For composite m , every proper divisor $1 < d < m$ contributes a distinct interior point. Adding the germ coefficient in $\mathbb{G}_{m,N}$ refines but never removes these divisor strata.

This gives a rigorous version of the proposed time–valuation–kernel tensor. It is a sparse feature packet indexed by exact-period sectors, so it does not confuse tensor order with Euclidean spatial dimension.

80 How large is the visible atlas?

Let an observer retain every cyclic room of size at most M . Merely storing one row from each room requires

$$\sum_{m \leq M} m = \frac{M(M+1)}{2} - 1$$

scalar slots. Retaining full matrices requires $\sum_{m \leq M} m^2 = \Theta(M^3)$, and retaining r th tensor shadows requires

$$\sum_{m \leq M} m^r = \Theta(M^{r+1}).$$

At every room, the infinitesimal chart adds an unbounded jet coordinate unless a truncation order is declared. Histories multiply this algebraic atlas by a free path language, and certificates distinguish paths that may have equal current chart values.

Theorem 80.1 (No finite room list is faithful). *Let an observer factor through finitely many cyclic room projections $\pi_{m_1}, \dots, \pi_{m_s}$. Then it is not injective on $\hat{A}_{\text{cyc}}$. More concretely, with $L = \text{lcm}(m_1, \dots, m_s)$, the element*

$$\hat{S}^L - 1$$

vanishes in every observed room but is nonzero in any room whose order does not divide L .

This theorem is the mathematical content behind the “different dimension” intuition. A finite ordinary observer does not merely see a blurred version of the same finite vector. It factors through a nontrivial quotient of a projective and path-enriched object.

81 Spectral HDIV on the cyclic atlas

The completed atlas does not magically make every denominator invertible. There is, however, an exact restricted division theorem on each semisimple character chart.

Let $A_{m,\mathbb{C}} = A_m \otimes \mathbb{C} = \bigoplus_{\chi} \mathbb{C}e_{\chi}$. For $b = \sum_{\chi} b_{\chi}e_{\chi}$, define

$$b^{\#} = \sum_{b_{\chi} \neq 0} b_{\chi}^{-1} e_{\chi}, \quad \Pi_{\ker b} = \sum_{b_{\chi} = 0} e_{\chi}.$$

Theorem 81.1 (Semisimple spectral reconstruction). *For $a, b \in A_{m,\mathbb{C}}$, put*

$$q = ab^{\#}, \quad r = a\Pi_{\ker b}.$$

Then

$$\boxed{a = bq + r, \quad br = 0.}$$

The quotient q is the unique solution supported on the nonzero spectral sectors, and r is the unique residual supported on the kernel sectors. The construction is functorial under room maps that preserve the declared spectral splitting.

The denominators in $b^{\#}$ generally leave the integral group ring. Nilpotent or nonnormal extensions can also destroy semisimplicity. Therefore the theorem neither proves a universal integral Moore–Penrose inverse nor changes the public SDK’s central-scalar HDIV contract.

Example 81.2 (Division by the prime shadow). For $K_p = p\Pi_p^{\text{prim}}$,

$$K_p^{\#} = \frac{1}{p}\Pi_p^{\text{prim}} = \frac{1}{p^2}K_p.$$

The residual of a is its constant-sector component. This is an exact rational spectral chart, whereas executable central HDIV uses signed integral quotient and residual coefficients. Their shared principle is reconstruction, not identical lowering.

82 The exact path module behind class-quantum language

Let $\mathcal{C}$ be the declared category of complete Angel maze states and legal native doors. For fixed source X , let

$$\mathbb{Z}[\text{Path}_X] = \bigoplus_{\gamma: X \rightarrow Y} \mathbb{Z}[\gamma]$$

be the free integer module on finite legal paths. Define the structural endpoint map

$$E_X : \mathbb{Z}[\text{Path}_X] \longrightarrow \bigoplus_Y \mathbb{Z}[Y], \quad [\gamma : X \rightarrow Y] \longmapsto [Y].$$

Theorem 82.1 (Exact interference as endpoint pushforward). *Superposition is addition in $\mathbb{Z}[\text{Path}_X]$. Structural interference is the pushforward E_X , and exact cancellation is membership in $\ker E_X$. Postcomposition by a legal functor commutes with E_X . Consequently structural cancellation remains valid after every common legal continuation, while cancellation performed only after ordinary projection need not remain valid.*

For a bipartite path family, the coefficient matrix defines an element of $U \otimes V$. Rank one means a decomposable history; a nonzero 2×2 minor certifies class-quantum entanglement. No probability amplitude, Born rule or physical wave function is required.

Corollary 82.2 (Future-safe cancellation criterion). *If two weighted path packets have the same complete endpoint pushforward, then every common legal continuation has the same complete endpoint pushforward. Equality only after Proj has no such guarantee unless Proj is proved future sufficient for the continuation language.*

83 From a two-chart atlas to an Angel full space

The constructions above justify a larger research object, but not the naive formula that every finite quotient automatically glues.

Definition 83.1 (Candidate Angel full-space datum). A candidate full-space datum consists of:

1. a compatible q-Pochhammer source-factor pro-object;
2. the fixed-prefix procyclic room object and the dependent self-room section;
3. all declared infinitesimal, closure and ramified jet towers;
4. the free legal path module modulo proved higher-cell relations;
5. the singular/residual reconstruction module;
6. certificate ideals and their functorial transport; and
7. a costed language of legal chart changes and observations.

An Angel full space, if it exists in the required universal sense, is a terminal compatible realization of these data in the category of exact causal-state models.

The definition intentionally includes costs and certificates. Omitting them would reconstruct a large algebraic object but not the Angel Prime object that the engineering boundary is designed to protect.

Conjecture 83.2 (All-chart reconstruction). There is a nontrivial declared continuation language for which the compatible family of all cyclic room projections, all infinitesimal/closure jets, complete path class and certificate ideal is future sufficient. Equivalently, two states with identical data in every admitted chart have identical outputs after every admitted continuation.

This conjecture is stronger than residual finiteness of the cyclic algebra and stronger than equality of every current scalar shadow. It includes history, resurrection and certificates.

Conjecture 83.3 (Integral generalized HDIV completion). There exists a maximal exact subcategory of the full-space datum on which spectral quotients glue across rooms and jets to an integral or explicitly localized quotient/residual functor, with reconstruction and causal transport but without declaring zero divisors invertible.

Conjecture 83.4 (Arithmetic action and path compression). A computable exact weight in a declared coefficient ring organizes large families of legal histories so that the q-Pochhammer principal jet and selected cyclic shadows are endpoint pushforwards of a compact cancellation complex. No claim involving e^{iS} , stationary phase or physical path integrals is made until such a coefficient ring, action and exact cancellation theorem are specified.

Conjecture 83.5 (Native Angel Prime birth). There is a costed high-to-high morphism that consumes P_p , constructs the complete next Angel Prime state P_q and its leastness certificate before ordinary projection, without target-factor recovery, candidate scanning or a terminal primality feedback branch. Volume V's source-first child and residual horizon supply part of this interface; the descriptor-native cold source morphism remains missing.

Explicit non-claim 83.6. The paper does not prove that the candidate full space is a physical universe, that primes are vacuum bubbles, that the factorial coefficient is a mass, or that class-quantum histories obey quantum mechanics. Such language may be useful intuition, but it is not mathematical evidence.

84 Theory ledger for the enlarged object

Object	Exact mathematical content	Engineering relation	Status
Fixed-prefix procyclic completion	Compatible family of all finite cyclic room shadows.	Bounded cyclic reference chart.	Proved.
Dependent self-room section	The diagonal $K_m = P_{m,m-1}$ with room-dependent prefix.	Structural primality audit only.	Proved.
Program-factor completion	Compatible residues modulo nested q-Pochhammer factors.	Compact source program supplies finite stages.	Proved pro-object.
Factorial jet spectrum	Every coefficient above the principal factorial jet, with recurrence and partition formula.	Principal coefficient implemented; higher tower not public.	Proved.
Ramanujan shadow	Exact integral row of K_m and its gcd strata.	Modular circulant probe in the research harness.	Proved.

Object	Exact mathematical content	Engineering relation	Status
Laplacian prime theorem	Rank, row, spectrum and divisor-support characterizations.	Finite cross-probed with Wilson output.	Proved.
Tensor shadow ladder	Rank $\varphi(m)^r$ and prime uniformity for the chosen tensor observer.	Formula audit; no tensor materialization.	Proved.
Semisimple spectral HDIV	Quotient plus kernel residual over a split characteristic-zero chart.	Public HDIV is central-scalar only.	Restricted.
Exact path module	Structural superposition, interference, cancellation and tensor-rank entanglement.	Finite integer implementation.	Model.
All-chart reconstruction	Future sufficiency of every compatible chart, history and certificate.	No complete implementation.	Conjecture.
Native Angel Prime birth	Source-first complete successor before projection.	Volume V verifier partial; public SDK absent.	Open.

The ledger prevents both kinds of loss: it does not discard the enormous mathematical object merely because the SDK implements a finite chart, and it does not promote an attractive global picture into a theorem merely because a finite experiment agrees with it.

85 Experimental question: can one source expose several shadows?

The preceding theory predicts that four computations, which look unrelated at the ordinary level, are shadows of the same source architecture:

1. the optimized Wilson residue of a supplied candidate;
2. the rank and nullity of the primitive cyclic operator K_m ;
3. the Ramanujan row shape of that operator;
4. the divisor support of the time-valuation packet.

It also predicts that the principal factorial jet is only the first coefficient of a longer exact spectrum.

The experiments below do not prove these universal statements; their proofs are algebraic. The experiments ask a different and necessary engineering question: do independently implemented finite shadows agree on thousands of rooms, while preserving the causal firewall of the optimized Wilson path?

Experimental discipline

The trial primality routine, Euler φ , divisor count, modular cyclic convolution and modular matrix rank live only in the research probe. They do not feed the native factorial state, the

joint Wilson consumer or any certificate. Each row first executes the optimized Wilson path, freezes its observation, and then compares independent structural data offline.

86 The structural-probe harness

The additive program

`experiments/angel_structure_probe.cpp`

has four modes. Its outputs are preserved as complete CSV files:

Mode	Output	Domain
wilson	<code>angel_structure_wilson_probe.csv</code>	$2 \leq m \leq 4096$
cyclic	<code>angel_structure_cyclic_shadow_probe.csv</code>	$2 \leq m \leq 256$
rank	<code>angel_structure_period_rank_probe.csv</code>	$2 \leq m \leq 96$, six prefix samples
jet	<code>angel_structure_higher_jet_probe.csv</code>	$1 \leq n \leq 64$, four extra orders

The Wilson mode calls the public sequence

`candidate` $\rightarrow$ `upload_factorial_state` $\rightarrow$ `bind_native_factorial` $\rightarrow$ `project_wilson_jointly`.

It records the verified observation, deterministic work, peak-live bound and materialized coordinate count. Only afterward does it compute the reference features

$$\varphi(m), \quad m - \varphi(m), \quad \tau(m), \quad \tau(m) - 2.$$

The cyclic mode multiplies

$$\prod_{k=1}^{m-1} (1 - S^k)$$

directly in $(\mathbb{Z}/q\mathbb{Z})[C_m]$ with $q = 1,000,000,007$ and compares the resulting row with the pure Laplacian row. It also decodes the small signed row coefficients, folds all m coordinates by multiplication modulo m , and checks the one-way prime collision in [Theorem 19.1](#). The rank mode builds the circulant matrix of selected prefixes and performs independent modular Gaussian elimination. Since $q > 96$ and $q \nmid m$ on the rank domain, the finite group algebra is separable and the predicted characteristic-zero rank is preserved by this audit prime.

The jet mode expands $\prod_{k=1}^n (1 - (1 + T)^k)$ through degree $n + 4$ modulo q and checks both the principal coefficient and the first higher-coefficient identity. A single audit modulus is finite evidence only; the integer theorem is proved in the appendix.

87 Experiment A: four prime predicates on 4,095 candidates

For every $2 \leq m \leq 4096$, the harness compared:

$P_1(m)$ = independent trial primality,

$P_2(m)$ = optimized native-coordinate Wilson decision,

$P_3(m) = [\varphi(m) = m - 1]$,

$P_4(m) = [\tau(m) = 2]$.

Experiment A result

All 4,095 rows satisfied

$$P_1(m) = P_2(m) = P_3(m) = P_4(m).$$

There were 564 prime rooms and 3,531 composite rooms. Every joint Wilson packet reported a valid native-coordinate certificate, complement-pair identity, no full-factorial materialization and no ordinary feedback.

Table 12: Selected rows from the Wilson/structure cross-probe.

m	prime	Wilson residue	$\varphi(m)$	nullity	strata	work	peak
2	yes	1	1	1	2	2	0
4	no	2	2	2	3	3	2
6	no	0	2	4	4	6	4
17	yes	16	16	1	2	28	22
49	no	0	42	7	3	81	35
97	yes	96	96	1	2	151	50
256	no	0	128	128	9	344	83
1009	yes	1008	1008	1	2	1204	177
4096	no	0	2048	2048	13	3849	372

Here “peak” is the conservative scratch-inclusive coefficient bound of the optimized Wilson consumer, not the dimension of the cyclic room. The two numbers belong to different cost models. Their coexistence in one row is a comparison of shadows, not an assertion that a 4096-dimensional cyclic room was materialized by a 372-slot polynomial schedule.

88 Experiment B: direct cyclic shadow fracture

For each $2 \leq m \leq 256$, the harness directly constructed the cyclic row of K_m modulo q and counted the coordinates that differ from

$$(m - 1, -1, \dots, -1).$$

Experiment B result

All 255 rooms satisfied

$$\text{row difference count} = 0 \iff m \text{ is prime.}$$

The 54 prime rooms had zero differences. Every composite room had at least four differences; the largest observed difference count was 256. For all 54 prime rooms, direct coordinate folding of the constructed cyclic row also returned $m - 1$, the residue $-1 \pmod{m}$. This latter check is not used as a converse prime criterion for composite rows.

Table 13: Selected primitive cyclic shadows.

m	$\varphi(m)$	nullity	row differences	type
4	2	2	4	fractured
6	2	4	4	fractured
8	4	4	8	fractured
12	4	8	12	fractured
17	16	1	0	pure Laplacian
30	8	22	22	fractured
49	42	7	49	fractured
97	96	1	0	pure Laplacian
128	64	64	128	fractured
251	250	1	0	pure Laplacian
256	128	128	256	fractured

The difference count is not itself a canonical invariant; it depends on the chosen pure-row comparator. The canonical invariant is the Ramanujan row and its rank. The count is an easily falsifiable visualization of how far the finite composite row is from prime uniformity.

89 Experiment C: death-time rank filtration

For a prefix $P_{m,n} = \prod_{k=1}^n (1 - S^k)$, the theory predicts

$$\text{rank } P_{m,n} = \sum_{\substack{d|m \\ d>n}} \varphi(d).$$

The rank probe sampled up to six distinct prefix lengths in each room $2 \leq m \leq 96$. It directly formed the circulant matrix over $\mathbb{F}_q$ and compared Gaussian-elimination rank with the divisor-sector formula.

Experiment C result

All 547 sampled room/prefix pairs had exact observed/predicted rank equality. No character root or factorization of a target candidate was passed to the matrix computation.

The difference between prime and composite rooms is visible dynamically. In room 17, every sampled prefix below 16 retained rank 16: there is no proper nonconstant period to die early. In room 96, the ranks fell

$$95, 94, 92, 64, 32, 32$$

at sampled prefixes 1, 2, 3, 24, 48, 95. Each drop is the accumulated dimension of exact-period sectors whose first death time has been crossed.

Table 14: Rank filtration examples.

Room	prefix n : observed rank					
	1	2	3	quarter	half	$m - 1$
6	5	4	2	4	2	2
12	11	10	8	8	4	4
17	16	16	16	16	16	16
49	48	48	48	42	42	42
96	95	94	92	64	32	32

For the small rooms, the quarter sample may duplicate an earlier prefix; the raw CSV contains the de-duplicated set actually tested. The table repeats the conceptual positions for readability.

90 Experiment D: the hidden factorial jet

For $1 \leq n \leq 64$, the probe expanded $Q_n(T)$ through T^{n+4} modulo q . It checked

$$[T^n]Q_n(T) = (-1)^n n!$$

and

$$4[T^{n+1}]Q_n(T) = n(n-1)[T^n]Q_n(T)$$

modulo q .

Experiment D result

All 64 leading coefficients and all 64 first higher-coefficient identities passed. The next three coefficients were retained in the raw table for independent recurrence checks; they were not consumed by the Wilson path.

For example, the exact low-order expansions are

n	$[T^n]$	$[T^{n+1}]$	$[T^{n+2}]$	$[T^{n+3}]$	$[T^{n+4}]$
2	2	1	0	0	0
3	-6	-9	-5	-1	0
4	24	72	98	76	35

before reduction. This is direct evidence that the factorial principal jet is not the whole infinitesimal map. The higher coefficients are mathematically present even when an optimized terminal observer has no need to materialize them.

91 Experiment E: same ordinary Wilson answer, different futures

The factorial-maze conformance suite constructs two specifications whose current ordinary arithmetic shadow agrees but whose argument, program, request and native state history words differ. It verifies:

$$\text{Proj}(X) = \text{Proj}(Y), \quad X \neq Y,$$

and then applies legal continuation/certificate inspection that distinguishes the two histories. The test is reported as

`same_shadow_different_factorial_history=PASS.`

This experiment is small but logically decisive. The cyclic probes show that one ordinary candidate has many exact-period directions; the history probe shows that even a fixed current arithmetic shadow can have more than one legal causal lift. Room dimension and causal multiplicity are different sources of high-dimensionality, and neither is captured by the terminal residue.

92 Experiment F: firewall and frozen-core counterfactuals

Every structural run was paired with the release invariants

native nodes rewritten	= 0,
ordinary feedback	= 0,
full factorial materialized	= 0,
frozen source/header hash failures	= 0.

Two additional programs are required to fail compilation:

1. applying a terminal joint observation to `execute()`, and
2. applying `project_jointly()` before factorial-maze execution.

These negative results answer a counterfactual that numerical agreement cannot answer. If an ordinary Wilson bit could re-enter the native state, the four predicate agreement in Experiment A would be compatible with a circular implementation. The opaque stage graph and negative compilation firewall exclude that public path.

93 Experiment G: Volume V residual reuse is a different scale

The strict Volume V v5.9 verifier is not linked into the public SDK probe. It tests the later source-first transition line and reports, over four declared horizon cases,

high-dimensional critical edges	2,119,
residual stability checks	22,764,
full packets reused	19,581,
zipper tail inspections	93,682,
cold-reference tail inspections	14,720,643.

It also reports no target-to-history inversion, no primality branch, no free selector and no ordinary feedback.

The v5.9 result must remain separate from Experiments A–F. The Wilson harness certifies supplied candidates and probes their finite shadows. The horizon verifier studies whether paid source-history packets remain valid across successive complete child states. Combining their evidence does not turn the public SDK into the still-open cold native successor generator.

94 What the experiments reveal—and what they cannot prove

The combined structural campaign contains

$$4,095 + 255 + 547 + 64 = 4,961$$

primary experimental records, in addition to the SDK differential, sanitizer, negative-compilation and Volume V campaigns. It reveals six robust facts about the finite implementation:

1. the optimized Wilson decision agrees with independent primality on the full candidate interval;
2. rank, divisor support and Wilson outcome agree without feeding one result into another;
3. prime cyclic rows are uniform and composite rows fracture into gcd strata;
4. prefix ranks fall exactly at predicted period deaths;
5. the factorial jet has verified higher structure beyond its terminally consumed coefficient; and
6. the full causal state retains distinctions invisible to the ordinary shadow.

Explicit non-claim 94.1. The experiment does not prove the all-chart reconstruction conjecture, a universal integral HDIV, a physical quantum interpretation, a new asymptotic prime-counting theorem or a cold native next-prime transition. It does not infer mathematical truth from the fact that a system “looks enormous.” Universal statements in this part stand only on the proofs in the appendices.

95 A falsification agenda for the large Angel object

A useful large theory must expose ways to fail. The next experiments should therefore target the following breaks:

1. find two distinct candidate full-space data with identical admitted charts but a different legal future, falsifying [Conjecture 83.2](#);
2. find a room quotient on which a proposed spectral residual does not transport, falsifying a generalized HDIV glue law;
3. search higher factorial jets for the first coefficient family whose compact recurrence cannot be carried by the current certificate schema;
4. compare Ramanujan rows and closure germs under nontrivial room quotients to locate resurrection not visible at terminal order;
5. construct path packets that cancel after ordinary projection but not after a legal continuation, stress-testing every proposed compression;
6. charge cold source construction separately from residual reuse in every alleged native successor speedup; and
7. attempt to violate the tensor-shadow rank formula over bad characteristics $\ell \mid m$, where semisimplicity is expected to fail.

This agenda is intentionally adversarial. The point of the higher-dimensional theory is not to make the system unfalsifiable; it is to state exactly which information an ordinary shadow has

discarded and to demand a proof whenever a compression claims that the discarded information will never matter again.

PART VI

Scope, Appendix Evidence, and Outlook

96 What has been achieved—and what has not

96.1 Achieved in the present scope

Completed application layer

1. A finite exact q-Pochhammer source program of logarithmic native size.
2. A mathematically proved factorial principal jet and interval coordinate.
3. A non-terminal scoped arithmetic with multiplication, exact cancellation, interference and continuation.
4. An implemented factorial-maze embedding and specialized high-to-high functor binding the three-class state to the principal-jet machine.
5. A hard separation between native state and ordinary observation.
6. An exact composite-safe Wilson observer whose factor count is consumed from the certified native factorial rank.
7. A complement-paired external schedule with a strictly smaller polynomial scale and an explicit square-root coordinate boundary.
8. A complete formal candidate-primality observation with typed egress and non-authoritative derived observations.
9. A cyclic structural theorem and self-closure theory showing that the factorial scalar is one chart of a richer exact-period object.
10. A verifier-backed residual-horizon theorem in which paid source packets survive consecutive causal prime children and reduce amortized future core work without ordinary feedback.

96.2 Unfinished interfaces

The next mathematically meaningful interfaces

1. **Final ingress.** The standalone upload interface must be designed and audited so that it lifts only known input/specification and creates a complete state without preloading answer information.
2. **Compression-preserving quotient.** The external modular observer should ideally consume the compact factorial program through a closed quotient law rather than materializing a $\Theta(\sqrt{m})$ block polynomial. The current joint path removes point/value arrays and halves the problem, but does not eliminate the square-root coefficient coordinate.
3. **General arithmetic closure.** The finite three-class SDK supplies signed sparse HADD, HSUB, HMUL and central-scalar HDIV, while the principal-jet interval algebra remains scoped. Arbitrary noncentral division, unbounded coefficients in the public sparse chart and every cross-frame operation are not yet supplied.
4. **Arbitrary-precision modular lowering.** Exact arbitrary-precision factorial materialization is implemented, but the optimized polynomial modulus chart is currently fixed-word. General L -bit modular complexity remains to be implemented and proved.
5. **Native cyclic continuation.** The self-closure germ has exact mathematics and bounded reference evaluation, but not a compact full-state compiler.
6. **Prime birth.** No native transition in this paper creates the next prime state from a dormant compact descriptor at subpolynomial cold cost. The Volume V zipper creates consecutive children only relative to its frozen, addressable parent and paid anchor-horizon interface.

Explicit non-claim 96.1 (No ordinary-algorithm laundering). The paper does not argue that a C++ class, operator overload, seal, hash or short syntax makes an ordinary algorithm high-dimensional. The native claim rests only on the scoped state algebra actually consumed by the operators. The external Wilson computation is openly ordinary and separately charged.

Explicit non-claim 96.2 (No performance claim against modern primality tests). Wilson's theorem is used because it connects the factorial state to a complete, exact and easily audited prime observation. The resulting observer is not advertised as a replacement for modern probabilistic or certificate-producing primality algorithms.

97 Appendix implementation and reproducibility

The engineering appendices contain the mathematics-to-code map, complete build procedure, negative compilation firewall, differential polynomial tests, serial-parallel transform equivalence, three-class arithmetic and functor tests, structural interference/cancellation and entanglement-certificate tests, state-integrity tamper tests, deterministic finite counter tables, sanitizer scope, and known toolchain limitations. The accompanying archive retains machine-readable logs and exact commands. This separation is intentional: proofs establish the identities for their stated domains, while finite tests establish conformance only for the exercised implementation and inputs.

98 Conclusion

The first lesson of the construction is conceptual. An exact arithmetic value need not be identified with one ordinary materialization. The complete runtime object can be a finite state with its own composition laws, continuation semantics and observation boundary. The Angel object makes this explicit with ordinary, ordered-history and singular-ideal coordinates; typed HADD/HSUB/HMUL/HDIV; native, history and future-quotient functors; and an exact class-quantum algebra of superposition, entanglement, structural interference and cancellation. In the factorial case, the q-Pochhammer concatenation law provides a logarithmic source program, and the integral principal jet supplies the factorial coordinate. Because the resulting state continues through native operations, it is more than a label for a future printout.

The second lesson is architectural and algorithmic. The ordinary Wilson residue remains legitimate precisely because it is kept outside the native core. Within that boundary, complement pairing, direct constant-mode projection, streaming and destructive branch retirement jointly reduce algorithmic work and peak live space without changing the state or deleting verification. The result is deliberately classified as a strict local Pareto improvement: the square-root coordinate remains, and the bit-level external cost is reported honestly. The boundary is not documentation around the arithmetic; it is part of the arithmetic's definition.

The third lesson is structural. The same q-Pochhammer source, placed in a cyclic frame, exposes exact-period kernels, primitive projectors and a self-closure germ whose time-valuation spectrum is the complete multiplication fibre. Prime order is simultaneously Wilson-regular in the constant mode, kernel-free in the normalized cyclic action, and transverse in every nonconstant closure sector. These are not yet one production compiler, but they are exact charts of a common object.

The fourth lesson is genuinely about Angel Prime state transition. Under the strict Volume V parent interface, the horizon inequality $E < 2p$ proves that an already-paid canonical source packet cannot be invalidated by a newly emitted child. The Causal Horizon Zipper therefore transports useful state-dependent history, rather than a passive log, through an ordered child thread. The no-recomputation and joint time-space theorems quantify the gain; History Curvature normalizes the proof path; and F042 prevents a linear number of local door actions from being hidden behind a smaller container. This extension is verifier-only and amortized, not a shipped or subpolynomial cold generator.

The integrated result is therefore both a genuine naive Angel Prime application—a native factorial maze state, a high-to-high bridge, lawful egress and an exact prime observation—and a strict research extension that shows how residual history can power later prime-state transitions. Its remaining weaknesses are now sharply located—ingress auditing, square-root-coordinate elimination, arbitrary-precision modular lowering, generalized noncentral division, arbitrary-precision sparse coefficients, native cyclic continuation, collective descriptor-native source compilation and cold successor birth—rather than hidden inside a claim that a compact descriptor or a constant-factor improvement has already solved them.

A Deferred technical proofs

This appendix contains the complete arguments whose spectral decomposition, valuation book-keeping, or quotient-ramification steps would interrupt the expository flow of the main text. The theorem statements in the body are authoritative; the purpose here is to make every change of scalar, every vanishing order, and every uniqueness step explicit.

A.1 The relative model theorem for three-class Angel arithmetic

Full proof of [Theorems 13.11 and 14.3](#). Put

$$\mathfrak{n} = \mathfrak{h}_{\text{Hist}} \oplus \mathfrak{s}$$

and assume that $\mathfrak{n}$ is an associative R -algebra and that $\mathfrak{h}_{\text{Hist}}$ is an associative subalgebra while $\mathfrak{s}$ is a two-sided ideal. The unitization is the R -module $R \oplus \mathfrak{n}$ with

$$(r, n) + (r', n') = (r + r', n + n')$$

and

$$(r, n)(r', n') = (rr', rn' + r'n + nn').$$

The additive laws are componentwise. The zero is $(0, 0)$ and the inverse of (r, n) is $(-r, -n)$.

For associativity, the nonordinary coordinate of $((r, n)(r', n'))(r'', n'')$ is

$$rr'n'' + r''rn' + r''r'n + r''nn' + rn'n'' + r'nn'' + nn'n'',$$

whereas the nonordinary coordinate of $(r, n)((r', n')(r'', n''))$ is

$$rr'n'' + rr''n' + rn'n'' + r'r''n + r'nn'' + r''nn' + nn'n''.$$

Since the scalars are central and $\mathfrak{n}$ is associative, the two expressions agree term by term after reordering scalar multiples. The ordinary coordinate is $rr'r''$ in both cases. Thus multiplication is associative. The element $(1, 0)$ is a two-sided unit, and direct expansion proves both distributive laws.

Now split $n = h + s$ and $n' = h' + s'$. The product nn' expands as

$$hh' + hs' + sh' + ss'.$$

Because $\mathfrak{s}$ is a two-sided ideal, the final three terms lie in $\mathfrak{s}$; the hh' term lies in the declared history component. Adding the scalar cross terms gives exactly [Equation \(HMUL\)](#). This proves that the displayed three-coordinate multiplication is the unitized associative product, not a separately postulated rule. It also proves that the singular sector is a two-sided ideal. The only possible obstruction to commutativity is the ordered history product, so the complete algebra is commutative exactly when $\mathfrak{n}$ is.

Define

$$\iota(r) = (r, 0, 0), \quad \text{Proj}(r, h, s) = r.$$

Both are algebra maps, ι is injective, Proj is surjective, and $\text{Proj} \circ \iota = \text{id}_R$. Hence $\iota(r) = \iota(r')$ if and only if $r = r'$, proving conservativity and all three projection identities in [Theorem 14.3](#).

Take the basis of $\mathfrak{n}$ to be typed finite history words. Word concatenation is associative and retains order, so equality in $R \oplus \mathfrak{n}$ is history-sensitive. The free category on the opcode alphabet acts by appending and executing words; its empty word acts as the identity and its categorical composition is concatenation. Upload is the declared embedding of a known specification, while Download is the declared projection paired with a causal cut. Removing a coercion from observation to state realizes the no-feedback axiom.

For a declared future language, quotient by the equivalence relation that identifies exactly those states with equal observations after every legal future word. By definition this relation is stable

under prefixing any legal opcode, hence every opcode induces a well-defined transfer on equivalence classes. This realizes the future-sufficiency axiom.

Finally, form the free $\mathbb{Z}$ -module on typed paths. Formal addition gives superposition. Grouping is performed only on equal complete endpoints, so the operation cannot identify two states merely because their ordinary projections agree. Exact integer summation gives interference and zero-weight cancellation. A finite sparse presentation realizes this model exactly until a coefficient or coordinate would leave the executable range; returning a continuation at that boundary preserves the operands and therefore realizes finite exactness without wraparound. All stated axioms consequently have the claimed concrete relative model. $\square$

Full HDIV reconstruction and solution-fibre argument. In the spectral chart, orthogonal idempotents satisfy $e_i e_j = \delta_{ij} e_i$. Therefore

$$bb^\# = \sum_{\lambda_j \neq 0} e_j = 1 - K_b, \quad bK_b = 0.$$

Commutativity then gives

$$b(ab^\#) + aK_b = a(1 - K_b) + aK_b = a.$$

If $bx = a$, multiplication by K_b gives $aK_b = bxK_b = 0$. Conversely, $aK_b = 0$ implies $b(ab^\#) = a$. If x is any solution, then $b(x - ab^\#) = 0$; in the idempotent decomposition this says precisely that $x - ab^\#$ is supported on the kernel axes, hence belongs to $K_b \mathfrak{T}_A(R)$. Every such element is annihilated by b , proving the complete solution fibre.

For the executable central-scalar chart, write every coefficient uniquely in the selected signed truncating convention as $c = dq_c + r_c$, with $|r_c| < |d|$ for $d \neq 0$ and the sign of a nonzero r_c equal to the sign of c . Scalar multiplication does not change a history word or its sector, so coefficientwise collection gives $X = \iota(d) \boxtimes_A Q_d(X) \boxplus_A R_d(X)$. At $d = 0$, the declared pair $(Q_0, R_0) = (0, X)$ satisfies the same reconstruction identity. No inverse of zero is introduced. $\square$

A.2 SN–OFCS conservativity at the zero denominator

Full proof of Theorem 14.7. Let $A = \mathfrak{T}_A(R)$ and form a tagged division-result carrier

$$\text{DivResult}(A) = A \times A \times \text{HistoryTag}.$$

This is a new result type. It is not declared to be R , and its first coordinate is not declared to be the value of an inverse. For $X \in A$ set

$$D_0(X) = (0, X, [\text{Div0}; X]).$$

The quotient and residual projections of this carrier give $Q_0(X) = 0$ and $R_0(X) = X$. Because $\iota(0)$ is the additive and multiplicative zero of the unitized algebra,

$$\iota(0) \boxtimes_A Q_0(X) \boxplus_A R_0(X) = 0 \boxtimes_A 0 \boxplus_A X = X.$$

This proves reconstruction. Notice that reconstruction is an additive quotient–residual identity, not the inverse identity $0 \boxtimes_A Q_0(X) = X$.

The inclusion of $D_0(X)$ in a tagged product does not change the equality relation or any operation

on the old summand A . In particular, the maps

$$\iota : R \longrightarrow A, \quad \text{Proj} : A \longrightarrow R, \quad \text{Proj } \iota = \text{id}_R$$

remain exactly those of the relative model. If $\iota(a) = \iota(b)$ in the enlarged collection, applying Proj still gives $a = b$. Hence no equality between embedded base-ring elements is created.

For every $z \in R$, multiplication in the base ring gives $0z = 0$. Therefore, if $a \neq 0$, no $z \in R$ satisfies $0z = a$. The tagged object $D_0(\iota(a))$ does not contradict this fact because it is not such a z : its quotient projection is zero, its residual projection is $\iota(a)$, and its history projection records the rejected inverse request. Applying ordinary projection componentwise yields quotient 0 and residual a while the complete carrier still contains the original numerator. All four clauses follow.

This proof also identifies the precise boundary of the claim. If one were to add the separate axiom “the quotient component of $D_0(\iota(a))$ is a scalar z satisfying $0z = a$ ”, contradiction would return immediately for nonzero a . That axiom is absent from SN-OFCS and from the Angel arithmetic. $\square$

A.3 Why the first old-arithmetic target is factorial

Full proof of Proposition 14.8. Start with

$$F_n(Z, u) = \prod_{j=0}^{n-1} (1 + Zu^j).$$

Splitting the factors after the first a terms and reindexing the second block gives

$$F_{a+b}(Z, u) = F_a(Z, u)F_b(Zu^a, u),$$

which proves the first clause and yields the double and append recurrences.

For $n \geq 1$, write n with $L = \lfloor \log_2 n \rfloor + 1$ binary digits. Starting from the length-one section, each of the remaining $L - 1$ digits requires one doubling, and a digit equal to one requires one additional append. Including the seed, the number of opcodes is

$$1 + (L - 1) + (\text{popcount}(n) - 1) \leq 2L - 1 = 2\lfloor \log_2 n \rfloor + 1.$$

The length recurrence $a \mapsto 2a$ followed, when needed, by $2a \mapsto 2a + 1$ reconstructs the binary digits, so the programme denotes exactly F_n rather than merely an object of the same size.

In the local chart $u = 1 + T$ and $Z = -(1 + T)$,

$$F_n(-(1 + T), 1 + T) = \prod_{k=1}^n (1 - (1 + T)^k).$$

Each factor has expansion $-kT + O(T^2)$. The product has valuation exactly n , and its degree- n coefficient can only be obtained by selecting the linear term from every factor. Hence

$$[T^n]F_n(-(1 + T), 1 + T) = (-1)^n \prod_{k=1}^n k = (-1)^n n!,$$

proving the third clause.

Finally take $n = m - 1$ with $m > 1$. Wilson's theorem says

$$m \text{ is prime} \iff n! \equiv -1 \pmod{m}.$$

This is a complete ordinary observation of a supplied candidate after the state denoting $n!$ has been fixed. It does not select m , generate a successor, or feed its Boolean result back into the native state.

The opcode bound measures a straight-line source description. It does not measure expansion of all coefficients, materialization of $n!$, or the Wilson observer. Indeed, the ordinary binary output has $\Theta(n \log n)$ bits in the usual asymptotic sense. The result therefore establishes why factorial is a structurally accessible probe while preserving the paper's representation-versus-evaluation boundary. $\square$

A.4 Ordered folds and q-Pochhammer contraction

Full proof of Theorem 18.2. Proceed by induction on the number n of leaves. For $n = 1$, the definition gives $\text{Fold}_{\mathcal{T}}(a_1) = a_1$. Suppose the result holds for smaller ordered interval trees and let the root split $[1, n]$ after c . The two induction hypotheses give

$$\text{Fold}_{\mathcal{T}_L}(a_1, \dots, a_c) = a_1 \cdots a_c, \quad \text{Fold}_{\mathcal{T}_R}(a_{c+1}, \dots, a_n) = a_{c+1} \cdots a_n.$$

Multiplying these two values in the declared left/right order and using associativity gives $a_1 \cdots a_n$. No commutation was used, so the argument is valid in a noncommutative algebra and does not authorize a leaf exchange.

Encode an internal graft by a token $[r, s] \rightsquigarrow ([r, c], [c + 1, s])$ and encode the tree by its ordered preorder word. The root split occurs first in this word; recursively, the left and right subtree words determine every later split. Thus two different ordered interval trees have different free graft words. Their ordinary evaluations agree by the first part, but their Angel histories remain distinct under structural equality. A quotient may identify them only after proving that every legal future observer is insensitive to the graft word, which is exactly the future-sufficiency condition of [Axiom 13.7](#). $\square$

A.4.1 q-Pochhammer interval folds

Full proof of Theorem 18.4. For $a \leq c \leq b$, disjointness and order of the index intervals give

$$\begin{aligned} F_{[a,b]}(Z, u) &= \prod_{j=a}^{b-1} (1 + Zu^j) \\ &= \left(\prod_{j=a}^{c-1} (1 + Zu^j) \right) \left(\prod_{j=c}^{b-1} (1 + Zu^j) \right) \\ &= F_{[a,c]}(Z, u) F_{[c,b]}(Z, u). \end{aligned}$$

The identities remain true when one interval is empty because its product is one. Applying this split recursively at every internal node and using [Theorem 18.2](#) shows that every ordered interval tree has endpoint $F_{[a,b]}$. Finally,

$$F_{[c,b]}(Z, u) = \prod_{r=0}^{b-c-1} (1 + (Zu^c)u^r) = F_{b-c}(Zu^c, u),$$

so the split is exactly the shifted concatenation law used by the binary double/append grammar. $\square$

A.4.2 The contraction functor

Full proof of Theorem 18.6. An identity arrow of $\mathbf{Part}[a, b]$ performs no adjacent merge. Its image performs no list contraction and is therefore the identity. A composable pair of morphisms is represented by a first finite merge word followed by a second finite merge word. The history lift maps it to the concatenation of the same two contraction words; hence identities and composition are preserved.

For the generating merge replacing adjacent blocks $[r, c]$ and $[c, s]$ by $[r, s]$, Theorem 18.4 gives the commuting square

$$F_{[r,c]}F_{[c,s]} = F_{[r,s]}.$$

Every morphism is a composite of these generators, so $\mathcal{V}_{Z,u}$ is a functor. A complete contraction has the one-block partition $[a, b]$ and hence polynomial endpoint $F_{[a,b]}$. The lifted functor also appends each merge token and parent/child binding. Projection that discards these tokens maps many contraction words to the same polynomial, so it is many-to-one and cannot be an inverse to the lift. $\square$

A.4.3 Nodewise factorial jets

Full proof of Theorem 18.7. The factor indices of $G_{[a,b]}$ split into the adjacent sets $\{a+1, \dots, c\}$ and $\{c+1, \dots, b\}$, so $G_{[a,b]} = G_{[a,c]}G_{[c,b]}$. Write

$$G_{[a,c]} = T^{\ell_L}(c_L + O(T)), \quad G_{[c,b]} = T^{\ell_R}(c_R + O(T)).$$

Their product is

$$T^{\ell_L + \ell_R}(c_L c_R + O(T)).$$

Therefore valuations add and leading coefficients multiply. Since the first factor has $c - a$ local factors and the second has $b - c$, the root valuation is $b - a$. Multiplying their leading terms $-kT$ gives

$$c_{a,b} = \prod_{k=a+1}^b (-k) = (-1)^{b-a} \prod_{k=a+1}^b k.$$

At $a = 0, b = n$ this is $(-1)^n n!$.

For completeness, retain $r + 1$ normalized jet coefficients at a node:

$$G_I = T^{\ell_I} \sum_{j=0}^r c_{I,j} T^j + O(T^{\ell_I + r + 1}).$$

At a parent with children L, R , ordinary truncated-series multiplication gives

$$c_{I,j} = \sum_{q=0}^j c_{L,q} c_{R,j-q} \quad (0 \leq j \leq r).$$

Thus every higher jet is also compositional once the child coefficient lists are retained. The pair (ℓ, c_0) alone determines only the leading jet and cannot recover the omitted coefficients or the contraction history. $\square$

A.4.4 Separated fold ledgers

Full proof of Theorem 18.8. Split every interval of length at least two into two parts whose lengths differ by at most one. After d levels the largest remaining interval has length at most $\lceil n/2^d \rceil$. It reaches one at $d = \lceil \log_2 n \rceil$, proving the depth formula, including non-powers of two. Every full binary tree with n leaves has $n - 1$ internal nodes: if I is the internal-node count, the number of edges is both $2I$ and $I + n - 1$, so $I = n - 1$.

The binary q-Pochhammer grammar is not an explicit listing of those $n - 1$ nodes. It stores a seed and at most one double and one append per remaining input bit, giving the $O(\log n)$ opcode count proved in Theorem 29.4. Executing an opcode on expanded degree- d polynomial data invokes coefficient arithmetic whose cost is governed by $M(d)$ and by the selected truncation. Finally, materializing $n!$ emits $\lfloor \log_2(n!) \rfloor + 1$ bits, which is $\Theta(n \log n)$ by the elementary bounds $(n/2)^{n/2} \leq n! \leq n^n$ for $n \geq 2$. These count different objects and cannot be substituted for one another. $\square$

A.5 The fold–shadow collision and the interval envelope

Full proof of Theorem 19.1. For prime p , the pure cyclic Laplacian theorem gives $K_p = pI - J$, whose first row is $w_p = (p - 1, -1, \dots, -1)$ with $p - 1$ copies of -1 . Hence

$$\text{CFold}(w_p) = (p - 1)(-1)^{p-1}.$$

If p is odd, $p - 1$ is even and this reduces to $p - 1 \equiv -1 \pmod{p}$. If $p = 2$, the product is $-1 \equiv 1 \pmod{2}$, which is again the residue class -1 . This proves the cyclic congruence.

The principal-jet theorem gives

$$[T^{p-1}]F_{p-1}(-(1 + T), 1 + T) = (-1)^{p-1}(p - 1)!.$$

Multiplying by $(-1)^{p-1}$ yields $(p - 1)!$, and Wilson’s theorem reduces it to -1 modulo p . The two terminal residues agree. No step constructs a map from the p cyclic coordinates to the $p - 1$ factorial factors, and no unreduced equality between the two carriers has been asserted. $\square$

A.5.1 Interval-complete divisibility data

Full proof of Proposition 20.1. Fix a prime $q \leq n$. Every multiple of q among $1, \dots, n$ contributes at least one factor of q to $n!$, giving $\lfloor n/q \rfloor$ contributions. Every multiple of q^2 contributes one additional factor, every multiple of q^3 another, and so on. Summing these disjoint layers gives

$$v_q(n!) = \sum_{r \geq 1} \left\lfloor \frac{n}{q^r} \right\rfloor,$$

with only finitely many nonzero terms.

The q-Pochhammer specialization has one local factor for every $k \in \{1, \dots, n\}$ and principal coefficient $(-1)^n n!$, so it retains the same interval-complete product in the infinitesimal chart. For a supplied $m > 1$, Wilson’s theorem states exactly

$$(m - 1)! \equiv -1 \pmod{m} \iff m \text{ is prime.}$$

Thus the family packages the divisibility strata before the terminal modulus is observed. The

proof makes no assertion that the prime set is topologically continuous, that prime gaps are controlled, or that the observer produces a new candidate. $\square$

A.6 Functor laws, class-quantum cancellation, and entanglement

Full proof of Theorem 15.4. Let a word be $\gamma = f_1 \cdots f_k$ in execution order and define

$$\mathcal{E}(\gamma)X = f_k(\cdots f_2(f_1(X)) \cdots).$$

For the empty word ϵ , no transition occurs, so $\mathcal{E}(\epsilon) = \text{id}$. If α is followed by β , the concatenated instruction vector first performs every instruction of α and then every instruction of β . Therefore

$$\mathcal{E}(\alpha\beta) = \mathcal{E}(\beta) \circ \mathcal{E}(\alpha).$$

The event history is appended in the same order, so the equality holds for complete states and not merely for ordinary projections. Associativity of vector and path concatenation gives associativity of composition.

The history functor sends an identity transition to the empty path and sends a composite to concatenation, hence obeys the same two laws. For the quotient functor, suppose $X \equiv_{\text{fut}} Y$. Prefixing a legal opcode f to any future word shows $f(X) \equiv_{\text{fut}} f(Y)$. Thus $[X] \mapsto [f(X)]$ is independent of the representative. Identity and composition descend from those on the original state action. $\square$

Exact structural cancellation. Let J be the sign-reversing involution in Theorem 16.4. Its fixed-point-free orbits all have two elements. Each orbit contributes

$$w_\gamma |F_\gamma(X), \gamma\rangle - w_{J\gamma} |F_{J\gamma}(X), J\gamma\rangle.$$

After structural endpoint aggregation the two basis endpoints coincide and their coefficient is zero. Summing over the disjoint orbits gives the zero packet. If only ordinary observations agree, the two complete endpoints belong to different structural classes and the aggregation step is forbidden; no cancellation conclusion follows. $\square$

Exact minor criterion for class-quantum entanglement. If $C = (c_{ij})$ is separable, $c_{ij} = a_i b_j$, then every 2×2 minor is

$$a_i b_j a_k b_\ell - a_i b_\ell a_k b_j = 0.$$

Hence a nonzero minor proves nonseparability and rank greater than one.

Conversely, suppose every 2×2 minor is zero. If $C = 0$, it has rank zero and is not entangled. Otherwise choose a nonzero pivot $c_{i_0 j_0}$. For every i, j , vanishing of the minor using rows i_0, i and columns j_0, j gives

$$c_{i_0 j_0} c_{ij} = c_{i_0 j} c_{ij_0}.$$

Over $\mathbb{Q}$, put $a_i = c_{ij_0}$ and $b_j = c_{i_0 j} / c_{i_0 j_0}$. Then $c_{ij} = a_i b_j$, so C is separable. Thus rank greater than one, nonseparability, and the existence of a nonzero 2×2 minor are equivalent. $\square$

A.7 Soundness of the finite sparse SDK realization

Full proof of Theorem 17.1. The executable value is a signed scalar and two normalized sparse maps from ordered token words to signed coefficients. HADD and HSUB merge equal words with

exact checked addition and delete a word only when its coefficient is exactly zero; they therefore implement [Equations \(1\) and \(2\)](#).

HMUL enumerates every ordered pair of words. Concatenating the left word before the right word implements the history product without imposing false commutativity. Scalar–history cross terms are charged to the history sector. Any product containing a singular operand is charged to the singular sector, which is exactly the two-sided-ideal split in [Equation \(HMUL\)](#). Checked coefficient multiplication and addition return a continuation before any machine overflow, so every returned value is exact.

For a pure central denominator, HDIV applies the host’s exact signed quotient/remainder identity to every coefficient after excluding its single unrepresentable fixed-width case. It then independently recomputes denominator times quotient plus residual and accepts the packet only when the complete sparse state equals the numerator. Zero denominator returns zero quotient and the unchanged numerator payload. A noncentral denominator is not pretended to possess a generalized inverse and returns a verified continuation.

A maze transition accepts operands only in a common room and frame, applies the complete three-sector operation, increments causal epoch and history depth, updates road, holonomy, singular generation and higher-cell coordinates, and appends a parent-bound event. The empty instruction vector returns the input unchanged. Program composition concatenates instruction vectors in execution order, so the implementation realizes [Theorem 15.4](#).

The class-quantum executor evaluates every branch as a deterministic native functor and retains its weight, label, path and complete endpoint. Structural interference compares complete endpoints before summing weights; equal ordinary coordinates are never used as a key. Tensor analysis checks exact integer 2×2 minors and reports entanglement only after finding a nonzero minor. If a minor cannot be represented, it returns an unresolved tensor status. These cases are exactly the scope stated in [Theorem 17.1](#). $\square$

A.8 Divisor-stratified kernel filtration

Full proof of [Theorem 44.2](#). Let

$$A_{m,\mathbb{C}} = A_m \otimes_{\mathbb{Z}} \mathbb{C} \cong \mathbb{C}[S]/(S^m - I).$$

Since $x^m - 1$ has distinct roots in characteristic zero, $A_{m,\mathbb{C}}$ is semisimple and decomposes as the direct sum of its one-dimensional character spaces. Grouping characters by exact order gives

$$A_{m,\mathbb{C}} = \bigoplus_{d|m} V_d, \quad \dim_{\mathbb{C}} V_d = \varphi(d).$$

Multiplication by an element of A_m is diagonal on this character decomposition.

Fix a character for which S acts by a root of unity ζ of exact order d . For $d > 1$,

$$[k]_S = I + S + \cdots + S^{d-1}$$

acts by

$$[k]_{\zeta} = \frac{\zeta^k - 1}{\zeta - 1}.$$

The denominator is nonzero, and the numerator vanishes precisely when $\zeta^k = 1$, equivalently

when $d \mid k$. Thus

$$J_{m,n} = (-1)^n \prod_{k=1}^n [k]_S$$

vanishes on this character if and only if at least one integer in $\{1, \dots, n\}$ is divisible by d . Because $d \geq 2$, the least positive multiple is d itself, so this occurs exactly when $d \leq n$.

On the constant character $\zeta = 1$, the factor $[k]_S$ acts by k . Consequently $J_{m,n}$ acts on V_1 by the nonzero scalar $(-1)^n n!$. On a sector V_d with $d > n$, no factor vanishes, so the action there is also by a nonzero scalar. Since the decomposition is direct and the operator is diagonal, its kernel is exactly

$$\ker(J_{m,n} : A_{m,\mathbb{C}} \rightarrow A_{m,\mathbb{C}}) = \bigoplus_{\substack{d \mid m \\ 1 < d \leq n}} V_d.$$

Taking dimensions gives

$$\dim_{\mathbb{C}} \ker J_{m,n} = \sum_{\substack{d \mid m \\ 1 < d \leq n}} \varphi(d).$$

No complex coefficient is introduced into the integral source operator: extension to $\mathbb{C}$ is a faithful proof chart. Indeed, tensoring the integral kernel sequence with the flat $\mathbb{Z}$ -module $\mathbb{C}$ preserves exactness, so the character calculation computes the characteristic-zero kernel of the integral multiplication map. $\square$

Full proof of Corollary 44.3. Set $n = m - 1$ in the preceding theorem. The kernel is the direct sum of V_d over proper nontrivial divisors d of m . It is zero exactly when no such divisor exists, which for $m \geq 2$ is equivalent to primality. For the dimension, use the classical identity

$$\sum_{d \mid m} \varphi(d) = m.$$

Removing the terms $d = 1$ and $d = m$ gives

$$\sum_{\substack{d \mid m \\ 1 < d < m}} \varphi(d) = m - \varphi(1) - \varphi(m) = m - 1 - \varphi(m).$$

$\square$

A.9 The primitive-period projector

Full proof of Theorem 44.4. Work in $A_{m,\mathbb{C}}$. On a character with eigenvalue ζ ,

$$K_m = \prod_{k=1}^{m-1} (I - S^k)$$

acts by

$$\kappa_m(\zeta) = \prod_{k=1}^{m-1} (1 - \zeta^k).$$

If $\zeta = 1$, the factor $k = 1$ vanishes. If ζ has exact order d with $1 < d < m$, the factor $k = d$ vanishes. Hence K_m is zero on every V_d except possibly V_m .

Now let ζ be primitive of order m . The roots of

$$\frac{x^m - 1}{x - 1} = 1 + x + \cdots + x^{m-1}$$

are $\zeta, \zeta^2, \dots, \zeta^{m-1}$, so

$$\frac{x^m - 1}{x - 1} = \prod_{k=1}^{m-1} (x - \zeta^k).$$

Evaluation at $x = 1$ yields

$$m = \prod_{k=1}^{m-1} (1 - \zeta^k).$$

Thus K_m acts by m on every primitive character and by zero on every nonprimitive character. By definition, Π_m^{prim} acts by one on V_m and zero on its complementary exact-period sectors. Therefore

$$K_m = m\Pi_m^{\text{prim}}.$$

Although the projector itself need not have integral coefficients, the scaled projector $m\Pi_m^{\text{prim}}$ does: it is the displayed product in A_m . $\square$

A.10 Exact self-closure valuation

Full proof of Theorem 45.2. Fix $d \mid m$ and a character on which S acts by a root ζ of exact order d . Since $d \mid m$, write

$$Nm = rd, \quad r = \frac{Nm}{d}.$$

The scalar response of the closure germ is

$$f_d(\varepsilon) = \prod_{k=1}^{rd} \left(1 - (1 + \varepsilon)^k \zeta^k\right).$$

At $\varepsilon = 0$, a factor vanishes exactly when $\zeta^k = 1$, that is, exactly when $d \mid k$. The vanishing indices are

$$d, 2d, \dots, rd.$$

For $k = jd$,

$$1 - (1 + \varepsilon)^{jd} \zeta^{jd} = 1 - (1 + \varepsilon)^{jd} = -jd\varepsilon + O(\varepsilon^2).$$

Each is therefore a simple zero, and their product has leading term

$$\prod_{j=1}^r (-jd)\varepsilon^r = (-1)^r d^r r! \varepsilon^r.$$

It remains to evaluate the factors that do not vanish. Every $1 \leq k \leq rd$ with $d \nmid k$ is uniquely written as

$$k = qd + a, \quad 0 \leq q \leq r-1, \quad 1 \leq a \leq d-1.$$

At $\varepsilon = 0$ the corresponding factor is $1 - \zeta^a$. For each fixed q , the product over the nonzero residue classes is

$$\prod_{a=1}^{d-1} (1 - \zeta^a) = d,$$

by the cyclotomic product used in [Appendix A.9](#), now at exact order d . There are r complete residue blocks, so the product of all nonvanishing constant terms is d^r .

Multiplying the vanishing and nonvanishing contributions gives

$$f_d(\varepsilon) = (-1)^r r! d^{2r} \varepsilon^r + O(\varepsilon^{r+1}).$$

The leading coefficient is nonzero in characteristic zero. Hence the valuation is exactly r , not merely at least r . $\square$

A.11 Time–valuation reciprocity and the multiplication fibre

Full proof of Corollaries 45.3 to 45.5. On an exact-period- d sector, the prefix product first acquires a zero factor at $k = d$, so $\tau_m(d) = d$. Taking $N = 1$ in [Theorem 45.2](#) gives $\nu_m(d) = m/d$. Thus

$$\tau_m(d) \nu_m(d) = d \frac{m}{d} = m.$$

For N cycles, the same theorem gives $\nu_{m,N}(d) = Nm/d$, and the product is Nm .

The map

$$d \mapsto (d, m/d)$$

is a bijection from the positive divisors of m to ordered positive integer pairs (a, b) satisfying $ab = m$: injectivity is immediate from the first coordinate, and a pair $ab = m$ makes a a divisor with $b = m/a$. This proves the multiplication-fibre statement.

The pairs $(1, m)$ and $(m, 1)$ always occur. There is an additional pair with both coordinates greater than one precisely when m admits a proper nontrivial divisor, which is precisely when m is composite. On a nonconstant sector $d > 1$, the closure zero is simple exactly when $m/d = 1$, that is, exactly on the primitive sector $d = m$. If m is prime this is the only nonconstant sector; if m is composite, a proper-period sector has closure depth greater than one. This proves the transversality formulation. $\square$

A.12 Jet-ramified quotient transport

Full proof of Theorem 45.6. Let $n = Nm$ and apply the ring homomorphism

$$\rho_{n,m} : \mathbb{Z}[C_n] \longrightarrow \mathbb{Z}[C_m], \quad S_n \longmapsto S_m$$

coefficientwise to formal power series. Then

$$\begin{aligned} \rho_{n,m}(\mathcal{F}_{n,1}(\varepsilon)) &= \prod_{k=1}^n \left(I - (1 + \varepsilon)^k S_m^k \right) \\ &= \prod_{k=1}^{Nm} \left(I - (1 + \varepsilon)^k S_m^k \right) \\ &= \mathcal{F}_{m,N}(\varepsilon). \end{aligned}$$

This proves exact transport of the complete germ.

If $N > 1$, then $m \leq n - 1$. The terminal product

$$K_n = \prod_{k=1}^{n-1} (I - S_n^k)$$

contains the factor $I - S_n^m$, which maps to $I - S_m^m = 0$. Hence $\rho_{n,m}(K_n) = 0$.

To find the first nonzero jet of the transported germ, decompose $A_m \otimes \mathbb{C}$ into exact-period sectors. On V_d , where $d \mid m$, [Theorem 45.2](#) gives valuation

$$\nu_{m,N}(d) = \frac{Nm}{d}.$$

This quantity is minimized when d is maximal. The unique maximal divisor is $d = m$, so the minimum valuation is N , attained only on V_m . Every other sector has valuation strictly greater than N . On V_m the leading coefficient is

$$(-1)^N N! m^{2N}.$$

Therefore the coefficient operator in degree N is zero off V_m and acts by that scalar on V_m :

$$[\varepsilon^N] \rho_{n,m}(\mathcal{F}_{n,1}) = (-1)^N N! m^{2N} \Pi_m^{\text{prim}}.$$

Using $K_m = m \Pi_m^{\text{prim}}$ gives the equivalent integral expression

$$[\varepsilon^N] \rho_{n,m}(\mathcal{F}_{n,1}) = (-1)^N N! m^{2N-1} K_m.$$

The valuation argument also proves that no lower-degree coefficient can survive transport. $\square$

A.13 Composite-safe polynomial multiplication and remainder

Proposition A.1 (Exact auxiliary-prime reconstruction). *Let $A, B \in (\mathbb{Z}/m\mathbb{Z})[X]$ have chosen integer coefficient representatives, and suppose $C = AB \in \mathbb{Z}[X]$ has coefficients satisfying $|c_k| < H$. Let $p_1, \dots, p_t$ be pairwise distinct auxiliary primes with $P = \prod_i p_i > 2H$. If each convolution is computed exactly modulo p_i , balanced CRT reconstruction followed by reduction modulo m returns the product AB in $(\mathbb{Z}/m\mathbb{Z})[X]$.*

Proof. For each coefficient index k , the auxiliary convolutions return the residue class of the same integer c_k modulo every p_i , hence modulo P . Two integers in $(-H, H)$ with the same residue modulo P differ by a multiple of P whose absolute value is less than $2H < P$; they are therefore equal. Balanced reconstruction uniquely recovers c_k . Finally, coefficientwise reduction $\mathbb{Z} \rightarrow \mathbb{Z}/m\mathbb{Z}$ is a ring homomorphism, so reducing the reconstructed integer product is identical to multiplying the original residue polynomials. $\square$

Proposition A.2 (Monic remainder over a composite modulus). *Let R be any commutative ring with identity and let $G \in R[X]$ be monic. For every $F \in R[X]$ there exist unique $Q, R_0 \in R[X]$ such that*

$$F = QG + R_0, \quad \deg R_0 < \deg G.$$

The usual leading-term cancellation algorithm uses no inverse other than $1^{-1} = 1$.

Proof. If $\deg F < \deg G$, take $Q = 0$ and $R_0 = F$. Otherwise, if the leading term of F is aX^r and $\deg G = s$, subtract $aX^{r-s}G$; monicity cancels the leading term and strictly lowers the degree.

Iteration terminates. For uniqueness, if $QG + R_0 = Q'G + R'_0$ then $(Q - Q')G = R'_0 - R_0$. If $Q \neq Q'$, monicity implies the left side has degree at least $\deg G$, whereas the right side has smaller degree, a contradiction. $\square$

Together, these propositions justify exact transform multiplication and monic remainder evaluation without assuming that the target modulus is prime. Serial and parallel evaluation of the auxiliary primes are mathematically the same family of independent residue computations; scheduling cannot change the CRT input tuple.

A.14 End-to-end Wilson observer

Expanded proof of Theorem 40.3. The verified native view supplies the integer rank n and coefficient one; the boundary admits the request only when $m = n + 1$. Set $h = \lfloor n/2 \rfloor$ and choose a positive block width w . Write $h = qw + r$ with $0 \leq r < w$, and define

$$P_w(X) = \prod_{i=1}^w (X + i).$$

For $0 \leq j < q$,

$$P_w(jw) = \prod_{i=1}^w (jw + i),$$

and these q blocks partition the factors $1, \dots, qw$. Multiplying the tail $qw + 1, \dots, qw + r$ therefore gives $h!$ exactly. Monic remainder trees evaluate the same polynomial values; streaming changes only object lifetime. By the two propositions above, transform/CRT multiplication and remainder are exact over the supplied modulus, prime or composite.

The complement-pairing theorem reconstructs $n! \bmod m$ from $h! \bmod m$ using multiplication and sign only. Finally, Wilson's theorem states that this residue equals $-1 \bmod m$ exactly when m is prime. The source state is not an operand of any write transition during this ordinary computation, and the observation type has no native ascent operation. $\square$

A.15 Construction and semantic separation of the Angel maze

Full construction behind Theorems 3.1 and 4.2. Fix a commutative ring R and a typed alphabet Γ . The free associative algebra $R\langle\Gamma\rangle$ consists of finite R -linear combinations of words. Multiplication is bilinear extension of word concatenation. The empty word is the unit and associativity is literal associativity of concatenation.

Let Γ_Σ be the tokens declared singular and let

$$\mathfrak{s} = R\langle\Gamma\rangle\Gamma_\Sigma R\langle\Gamma\rangle.$$

Every basis word of $\mathfrak{s}$ contains a singular token. Concatenating an arbitrary word on either side cannot remove that token, so $\mathfrak{s}$ is a two-sided ideal. Let $\mathfrak{h}_{\text{Hist}}$ be the span of the nonempty nonsingular words in the finite-support chart. Every word is exactly one of the empty word, a nonempty nonsingular word, or a word containing a singular token. Therefore

$$R\langle\Gamma\rangle = R \oplus \mathfrak{h}_{\text{Hist}} \oplus \mathfrak{s}$$

as an R -module. Sorting the ordinary expansion of a product into these three disjoint word classes gives HADD and HMUL.

Now take the Cartesian product with the finite address, path, program and certificate carriers. A certified opcode word w is a morphism $X \rightarrow Y$ only when replay from the complete X terminates in the complete Y and its certificate verifies. The empty word replays to X . If $w : X \rightarrow Y$ and $v : Y \rightarrow Z$, replaying wv means replaying w and then v , so it terminates at Z . For three composable words,

$$(uv)w = u(vw)$$

as strings, and deterministic replay gives the same intermediate states and certificate folds. Hence the category axioms hold.

For semantic separation, let I be the conservative ordinary image and let $K = \ker \text{Proj}$. Retraction gives a split exact sequence

$$0 \longrightarrow K \longrightarrow \mathcal{S} \xrightleftharpoons[\iota]{\text{Proj}} R \longrightarrow 0.$$

Thus ordinary data cannot be identified with every element of K . Define $N_{\text{fut}} \subseteq K$ to be the states invisible under every legal future observation. A same-shadow/different-future pair gives a nonzero class in K/N_{fut} ; this is the history-sensitive quotient.

For a reconstructing division pair (Q, R_d) , the nonzero residual satisfies

$$X = dQ + R_d.$$

Its role is constrained by the denominator and by the reconstruction identity. If its semantic class were universally identified with the history quotient, two failures follow. Either all history classes would be forced to satisfy the same denominator-annihilation/reconstruction relation, which is false for a free nonsingular word, or the residual could be replaced by a future-equivalent history class without preserving reconstruction, contradicting exactness. Therefore the ordinary, future-active history and reconstructing residual roles are distinct subquotients.

This argument proves a minimal semantic decomposition under the stated three requirements. It does not claim that the displayed direct-sum presentation is unique or that no richer theory can refine one of the summands. $\square$

A.16 Full factorial-maze embedding and functor certificate proof

Expanded proof of Theorems 32.2 and 33.1. Let a certified principal-jet state J_n contain the exact argument n , interval $[0, n]$, valuation n , coefficient one, source program P_n , state seal j_n , program seal p_n and replay certificate C_n . Its maze image has the four uploaded history words

$$[\text{Arg}; n], \quad [\text{Program}; p_n], \quad [\text{State}; j_n], \quad [\text{Request}; m, j_n].$$

Sparse structural equality compares word types, word tokens and coefficients. Equality of images therefore gives equality of the argument word and hence of n . It also gives equality of p_n, j_n and the request binding. Replaying the deterministic source compiler on the same n yields the same interval, valuation, compact opcode word and principal coefficient. Verification of C_n then proves equality of the complete native states. Conversely, equal native states clearly produce equal words and equal maze addresses. Since the ordinary component is the coefficient one for every successful chart, its projection is constant and is not the source of injectivity.

For execution, first replay P_n independently and check:

$$\text{rank} = n, \quad \text{coefficient} = 1, \quad \text{stateSeal} = j_n, \quad \text{programSeal} = p_n.$$

Only after all checks pass, append the single typed word

$$[\text{Execute}; r_n, c_n]$$

and a causal event binding the input maze seal, result seal and certificate seal. Generic history addition changes neither Ord nor Sing. Its address transition increments epoch and history depth, folds the parent and result seals into Holonomy/HigherCell, and preserves the factorial frame.

The bridge certificate is a deterministic tuple/fold of

$$(\text{inputMazeSeal}, \text{outputMazeSeal}, \text{nativeResultSeal}, \text{nativeCertificateSeal}, \text{nativeViewSeal}).$$

Changing any component either changes the recomputed bridge seal or makes the independent native replay disagree. Thus the issued executed object binds the generic maze and the principal-jet result rather than merely holding two unrelated values.

Identity execution has the empty instruction history and leaves the input binding unchanged. For two admitted chart morphisms, ordered execution appends their typed words in exactly the same order as categorical composition. Deterministic certificate folding follows that order, so the map preserves identities and composition.

Finally, the upload, execute and observation carriers are distinct opaque types. Only the execute function owns a constructor for the executed carrier, and only that carrier is accepted by exact or joint observation. The two terminal carriers have no constructor or conversion to either native stage. This proves the static projection-last statement independently of dynamic seal verification. $\square$

Complete compatibility of the two HDIV charts. Let $1 = \sum_j e_j$ be a semisimple idempotent decomposition and let d be a central scalar. Multiplication by d acts as the eigenvalue d on every e_j . If d is invertible, no spectral axis belongs to the kernel and

$$d^\# = \sum_j d^{-1} e_j = d^{-1}, \quad K_d = 0.$$

For an integrally divisible sparse numerator X , coefficientwise division has zero remainder and quotient $d^{-1}X$, exactly the spectral result.

If d is nonzero but not a unit in the integral chart, write each signed coefficient uniquely under the declared truncation convention as $c = dq_c + r_c$. Collecting all q_c and r_c in their original basis words gives $Q_d(X)$ and $R_d(X)$. Bilinearity yields

$$dQ_d(X) + R_d(X) = X.$$

After scalar extension inverting d , multiply by d^{-1} to obtain

$$d^{-1}X = Q_d(X) + d^{-1}R_d(X).$$

Thus the residual stores the integral obstruction later absorbed by the spectral fraction. At $d = 0$, every spectral axis is a kernel axis and the integral convention also returns regular quotient zero with complete numerator residual. No step covers a noncentral denominator, which remains a separate declared chart. $\square$

A.17 Full residual-horizon stability and zipper invariant

Expanded stability proof for Theorem 64.2 and Corollary 64.3. Fix $p < E < 2p$ and a grade $n \leq E$. Every nontrivial factorization $n = ab$ with $2 \leq a \leq b$ satisfies

$$a \leq b \leq n/2 < E/2 < p.$$

Therefore every possible source component lies strictly below the anchor and is already decided in $Q_{\leq p}^{\text{caus}}$. If the anchor program assigns a canonical door, its source pair, PBW word, ringed morphism and certificate ideal use only those old components.

Extend the state by any sequence of child generators $q_i > p$. No new generator can occur in a nontrivial factorization of n , because a product containing q_i has grade at least $2q_i > 2p > n$. The set of possible source pairs for n is therefore unchanged. Applying the same deterministic canonical source-head selection returns the same door or the same absence of a door. Since its source objects and local morphism are unchanged, the PBW provenance and certificate-ideal inclusion replay unchanged. Induction over the child sequence proves the multi-child statement. $\square$

Expanded traversal proof for Theorems 65.2 and 65.3. Maintain the following invariant before every zipper step:

1. all grades to the left of the active path have been either certified contracted or emitted as complete harmonic children;
2. every retained right sibling is a paid exact packet for its interval;
3. retained intervals are disjoint, ordered and lie strictly to the right of the active path;
4. all retained packets are valid at the current causal child state;
5. the pending empty-prefix certificate is the ordered composition of exactly the full packets already consumed since the preceding child.

The root satisfies the invariant trivially. Consuming a full packet adds no harmonic room, composes its certified contraction and advances in order. Refining a deficient nonsingleton partitions it into ordered disjoint children whose exact counts sum to the parent count; retaining the right siblings and descending into the first deficient child preserves the ordering and packet conditions. A deficient singleton has count zero, so it is the next harmonic room. The transition creates its full fibre and attaches the accumulated empty-prefix proof before projection. Stability transports every right sibling to the child, preserving condition four.

Finite interval length makes every descent terminate at a singleton. The invariant prevents skipping a deficient interval or emitting out of order, so every and only harmonic singleton is emitted. A refined node is removed and replaced by new child nodes; there is no transition returning it to the stack. A retained child remains valid after every emission by stability. Hence no interval is refined twice and no earlier emission can force recomputation of a right sibling. $\square$

Expanded curvature and certificate argument for Theorem 66.1. Restrict the anchor program to $(r, E]$. Stability identifies the source doors, source objects and local certificate ideals in the p -, q - and r -states. Consequently the composite path $p \Rightarrow q \Rightarrow r$ and the direct path $p \Rightarrow r$ are two ordering/bracketing histories of the same canonical source program.

Resolve their difference into adjacent legal swaps and associator cells. Every cell is ringed semilinear, preserves its certificate ideal and has residual in the future-silent kernel by hypothesis.

Composition of certificate-ideal inclusions is again an inclusion. The future-silent kernel is stable under the legal continuation language, so the sum/composite of the cell residuals remains invisible to every future observer. The two transport morphisms are therefore future-equivalent. Replacing either path by the canonical PBW normal form makes the live certificate depend on the canonical word and shared cell schemas, not on the number of preceding children. $\square$

A.18 Expanded residual-horizon complexity summation

Full summation for Theorem 67.2. At depth ℓ , let $\mathcal{B}_\ell$ be the deficient nodes actually refined. They are disjoint subintervals of the H -room horizon. Hence

$$\sum_{B \in \mathcal{B}_\ell} |B| \leq H.$$

Every deficient node contains at least one harmonic child, and disjoint nodes contain disjoint child sets, so

$$|\mathcal{B}_\ell| \leq P.$$

Sum the one-node tail-inspection term:

$$\sum_{B \in \mathcal{B}_\ell} (|B|\mathcal{R}_{p,E} + A_{p,E}) \leq H\mathcal{R}_{p,E} + PA_{p,E}.$$

Similarly the door term sums to at most $Hc_{\text{door}}(L)$ and the packet term to at most $PKc_{\text{pkt}}(L)$. The balanced K -ary interval tree has $t = \lceil \log_K H \rceil + O(1)$ levels. Multiplying the per-depth bounds by t gives the displayed work theorem. Dividing by P gives the internal amortized corollary without invoking a prime-density estimate.

For space, a depth-first traversal keeps the current node and at most $K - 1$ unvisited siblings at each of t levels. Thus there are $O(Kt)$ live packet positions, each of size $s_{\text{pkt}}(L)$. The canonical curvature witness has $O(\log^2 p)$ cell references modulo shared schemas, contributing the additive term stated in the theorem. Full packets already composed and child packets already emitted are retired; charging them to historical total work instead of peak live storage avoids double counting.

The dyadic $p^{1/2+o(1)}$ expression is not part of this exact summation. It arises only after substituting external asymptotic estimates for P , $A_{p,E}$ and $\mathcal{R}_{p,E}$ and assuming polylogarithmic packet costs. Therefore it is an amortized boundary interpretation, not a cold theorem. $\square$

A.19 Consolidated proof of the least-factor one-jet package

This subsection collects the difficult dependency chain behind Theorems 50.4, 51.1 and 52.5 so that the body may be read without reconstructing which cancellation occurs in which coefficient ring.

Full coefficient and filtration proof. Let n be composite, $r = \ell(n)$, and work in

$$R_n = (\mathbb{Z}/n\mathbb{Z})[T]/(T^{n-1}).$$

First consider

$$[n]_{1+T} = \sum_{h=1}^n \binom{n}{h} T^{h-1}.$$

For $h < r$, $h!$ is a unit modulo n . Since

$$h! \binom{n}{h} = n(n-1) \cdots (n-h+1),$$

the right side vanishes modulo n and multiplication by $(h!)^{-1}$ proves $\binom{n}{h} = 0$ in $\mathbb{Z}/n\mathbb{Z}$. At $h = r$ use

$$\binom{n}{r} = \frac{n}{r} \binom{n-1}{r-1}.$$

Every term $n - k$ for $1 \leq k < r$ is congruent to $-k$ modulo r , hence

$$\binom{n-1}{r-1} \equiv \frac{(-1)(-2) \cdots (-(r-1))}{(r-1)!} = (-1)^{r-1} \not\equiv 0 \pmod{r}.$$

Thus the r -adic valuation of the product on the right is strictly smaller than that of n , so n cannot divide $\binom{n}{r}$. The first nonzero T -coefficient of the q -integer occurs in degree $r - 1$.

For $1 \leq j < r$, apply the Gaussian recurrence

$$\begin{bmatrix} n \\ j \end{bmatrix}_u = \frac{[n]_u}{[j]_u} \begin{bmatrix} n-1 \\ j-1 \end{bmatrix}_u, \quad u = 1 + T.$$

The constant coefficient of $[j]_u$ is j , a unit. The constant coefficient of the second Gaussian factor is $\binom{n-1}{j-1}$. Every factor involved is represented modulo n by an integer with no prime divisor below r , so it is a unit. Because (T) is nilpotent, a polynomial with unit constant coefficient has the finite geometric inverse

$$(a_0 + Tg)^{-1} = a_0^{-1} \sum_{k=0}^{n-2} (-a_0^{-1}Tg)^k.$$

Therefore both auxiliary factors are units, and the j -th channel has exactly the T -valuation $r - 1$ of $[n]_u$.

At $j = r$, setting $T = 0$ turns the Gaussian coefficient into $\binom{n}{r}$, which is nonzero modulo n by the first paragraph. The phase $u^{r(r-1)/2}$ has constant coefficient one. This channel therefore has T -valuation zero.

Now insert the factor ε^j . For $j < r$, the total degree is at least $j + r - 1 \geq r$; at $j = 1$ it is exactly r . At $j = r$ the nonzero constant coefficient also has total degree r . Channels $j > r$ already have ε -degree greater than r . Hence

$$B_n \in \mathfrak{m}_n^r \setminus \mathfrak{m}_n^{r+1}, \quad \nu_{\mathfrak{m}}(B_n) = r.$$

For truncation depth $D < r$, a channel $j \leq D$ starts with T^{r-1} and is killed modulo T^D , while a channel $j > D$ is killed modulo ε^{D+1} . Conversely, if $r \leq D$, the $\varepsilon^r T^0$ coefficient survives. This proves the exact factor-coverage equivalence.

Hasse release of bidegree (a, b) maps $\mathfrak{m}^s$ into $\mathfrak{m}^{s-a-b}$. Therefore fewer than r total first-order releases cannot produce a nonzero ordinary residue from B_n . The two nonzero degree- r coefficients exhibited above show that r releases suffice. This proves the observability equality, including attainability rather than merely a lower bound. $\square$

Full compiler-path indistinguishability argument. Let a valuation-local circuit have nodes topologically ordered. Compare its execution on $x \in \mathfrak{m}^r$ with its execution on zero. Associate to each

node v the largest lowering cost L_v on a path from the input to v . We prove

$$y_v(x) - y_v(0) \in \mathfrak{m}^{r-L_v}$$

by induction.

At the input the difference is x . For addition, scalar multiplication by a unit, semilinear substitution and fixed Frobenius, the declared filtration preservation gives the step. For a product node with predecessor pairs (a, a') and (b, b') ,

$$ab - a'b' = (a - a')b + a'(b - b').$$

Each term has filtration at least the smaller predecessor bound; the product operation itself has lowering cost zero. A fixed polynomial is handled by telescoping each monomial. An order- s Hasse release lowers the exponent by at most s , exactly matching the cost assigned to that node.

If the total budget is less than r , then $r - L_v > 0$ at every ordinary branch observation. Corresponding values differ by an element of $\mathfrak{m}$ and therefore have identical ordinary residues. Conditional branches agree, so the two executions follow the same path and the induction remains valid after each branch. Their terminal ordinary outputs agree.

For sufficiently large X , Bertrand's postulate twice supplies primes

$$\frac{1}{4}\sqrt{X} < r < \frac{1}{2}\sqrt{X}, \quad r < s < 2r.$$

Then $n = rs < X$ and $\ell(n) = r$. A compiler of budget less than $\frac{1}{4}\sqrt{X} < r$ cannot distinguish B_n from zero. Section correctness requires different labels, giving the contradiction and the F036 bound.

The proof uses the circuit restrictions at two exact places: constants cannot branch on n independently of the supplied section, and a derivative of order s is charged s . Removing either restriction changes the compiler class and invalidates, rather than strengthens, the theorem. $\square$

A.20 Consolidated proof of the multiplication-history child object

Expanded proof of Theorems 53.1, 53.2 and 55.1. Fix prime p and $n \in (p, 2p)$. If n is composite, every prime factor is strictly less than p : a factor $r \geq p$ would have a complementary factor at least two and force $n \geq 2p$. Unique factorization gives a unique exponent vector on the accepted axes below p . Its total exponent is at least two. Conversely any such exponent vector has composite weight. This proves a bijection

$$\{\text{normalized parent exponent objects of corridor weight}\} \longleftrightarrow \{\text{composite corridor grades}\}.$$

Different word orderings of one exponent vector are not silently identified. They are related by the declared q-Weyl/PBW 2-cells, and their holonomy residual is retained until the coherence quotient proves future equivalence. The bijection concerns normalized exponent objects, not raw words.

Order bases of $\mathcal{C}_0$ by corridor grade and bases of $\mathcal{C}_1$ by their target weights and normalized source words. In these bases, the matrix of d has exactly one unit entry in each composite target row, no entry in a prime target row, and exactly one entry in every source column. After restricting the codomain to the composite target summand, this matrix is a permutation of an identity matrix.

Consequently

$$\ker d = 0, \quad \text{im } d = \bigoplus_{n \text{ composite}} \mathbb{Z}\langle \mathbf{Q}_n \rangle.$$

The cokernel is the complementary ordered prime summand. Prime powers and mixed composites use different source fibres, but both columns still land in the same composite boundary module.

On a source-attached composite basis vector, the map h restores the unique normalized source object retained by the forward construction. Thus

$$dh(\mathbf{Q}_n) = \mathbf{Q}_n$$

for a composite grade and $dh(\mathbf{Q}_q) = 0$ for a harmonic grade. On a source basis vector, forgetting and restoring its own retained provenance gives $hd = I$. Therefore, degree by degree,

$$dh + hd = I - P_{\mathcal{H}}.$$

Let q be the first grade on which $P_{\mathcal{H}}$ is nonzero. The strict prefix has no harmonic summand, so the identity restricts to $dh + hd = I$ there. At q , the surviving complete basis object satisfies $P_{\mathcal{H}}\mathbf{Q}_q = \mathbf{Q}_q$. These are the two components of the same-map packet. The cokernel calculation proves that the ordinary grade is prime, and corridor order proves that no smaller prime grade follows p .

The construction cannot be replaced by a contraction on ordinary integers. The ordinary weight map identifies raw ordered words that can differ by q-Weyl holonomy and vertical singular provenance. A purported shadow contraction would have to return one history for their common scalar, so it would be wrong on at least one future-distinguishable lift. In particular, removing the vertical source object for a prime power deletes the only column mapping to that target and changes the cokernel. This proves the algorithmic necessity statement for the declared complex. $\square$

Full harmonic-series and extension argument. In the formal series $\mathfrak{A}_p(\Omega)$, the coefficient of a composite grade n is

$$\mathbf{Q}_n - d(\mathbf{H}_{\alpha(n)} \boxtimes \mathbf{Q}_n),$$

which is zero in the cokernel because the second term is the named boundary whose image is the first. At a prime grade there is no source exponent object, so the coefficient remains $\mathbf{Q}_q$. The corridor grading is ordered, and hence the least exponent of Ω with nonzero cokernel coefficient selects the first complete harmonic object.

Only after that obstruction and its prefix certificate exist is a new primitive axis added to the history monoid. Freeness gives

$$\mathbf{M}_q \cong \mathbf{M}_p \oplus \mathbf{N}_{\mathbf{e}_q}.$$

Because $\mathbf{e}_q$ is not an input generator of $\mathbf{M}_p$, no source history can have used the future child axis to create its own obstruction. The extension binds the old state, child fibre and same-map packet, so it is a high-to-high morphism. Reading the weight of that already-created axis is the terminal projection and supplies no branch used by the construction. $\square$

A.21 Failure models for the three axiom families

The axioms define a model class and therefore are not proved from no assumptions. The following finite failure models show why each family appears in the Angel Prime contract. They are necessity witnesses for the declared semantics, not a claim of logical independence in every possible formal

language.

Failure model A: scalar collapse

Let the state carrier be only R and let every history word and residual map to zero. Ordinary embedding and projection are perfect, and ordinary addition/multiplication remain correct. Nevertheless states $(0, [\alpha], 0)$ and $(0, [\beta], 0)$ collapse. Multiplying by $(0, 0, [\gamma])$ can no longer distinguish the future words $[\alpha\gamma]$ and $[\beta\gamma]$. Thus a conservative scalar ring does not realize history-sensitive identity or future-sufficient execution.

Failure model B: history without a residual class

Take $R \oplus \mathfrak{h}_{\text{Hist}}$ and define division by a nonunit d to return only the truncating quotient. For $X = (7, 5[\alpha])$ and $d = 2$, this would return $(3, 2[\alpha])$. Multiplying back gives $(6, 4[\alpha])$, not X . Treating the missing $(1, [\alpha])$ as generic debug history does not restore the denominator-bound reconstruction contract. A separately typed residual class is necessary for exact HDIV in this chart.

Failure model C: complete values without causal order

Let states retain Ord/Hist/Sing but allow a terminal residue to be promoted back into a native operand without a new upload event. A program can then observe a Wilson bit, branch on it and synthesize a state whose history claims to precede that observation. The same terminal scalar may be correct, but the source-first proof is circular. Thus complete arithmetic coordinates alone do not enforce Angel causality; ingress, native execution and egress require disjoint typed stages.

Failure model D: observation-level interference

Let two branches end at

$$X = (0, [\alpha], 0), \quad Y = (0, [\beta], 0)$$

with weights $+1$ and -1 , and group branches by ordinary projection. The packet cancels because both projections are zero. After continuation by a singular token, however, a coefficient observer for $[\alpha\gamma]$ distinguishes the two futures. The canceled packet had a nonzero legal future. Interference must therefore group complete structural endpoints, not current ordinary shadows.

Failure model E: certificates without program identity

Suppose a factorial bridge certificate binds the result integer or residue but not the compact source program and native state seals. Two different requests with an accidental equal terminal residue can then share the displayed certificate. A later exact materializer or different modulus distinguishes them, but the bridge has already merged their authority. Binding argument, program, state, request and execution result prevents this substitution.

Proposition A.3 (Family-separation witness). *No one of the five failure models above satisfies all of the following at once:*

1. conservative ordinary arithmetic and complete multidimensional state;

2. *exact reconstructing arithmetic with history-distinguishable futures;*
3. *source-first native dynamics with non-authoritative observation;*
4. *exact structural path interference and certified factorial execution.*

The free path-algebra maze model satisfies all four on its declared finite chart.

Proof. Model A violates the future distinction in item two. Model B violates exact reconstruction. Model C violates source-first order in item three. Model D violates structural path exactness in item four. Model E violates certified factorial execution. The free path-algebra construction has a split ordinary summand, distinct nonsingular/singular word classes, typed stage morphisms, structural endpoint grouping and deterministic seals binding the five bridge identities, as proved in [Appendices A.15](#) and [A.16](#). $\square$

A.22 Theorem dependency and evidence boundary matrix

The matrix below makes explicit which statements are mathematical deductions, which require a chart assumption, and which are supported only as finite implementation conformance.

Statement	Mathematical dependencies	Executable obligation	Status
Three-sector ring laws	Associative history algebra and two-sided singular ideal.	Checked sparse convolution and overflow continuation.	Implemented.
Central HDIV reconstruction	Signed coefficient quotient/remainder convention.	Preserve quotient, residual and operands at boundaries.	Implemented.
Spectral HDIV	Semisimple idempotent chart with declared eigenvalues.	No universal lowering claimed.	Restricted theory.
Maze category	Deterministic replay and composable complete endpoints.	Opaque state, event and certificate carriers.	Implemented finite chart.
Class-quantum cancellation	Finite integer path module and structural endpoint equality.	Exact grouping; no floating rank.	Implemented.
Factorial principal jet	q-Pochhammer concatenation and local valuation proof.	Independent program/state certificate replay.	Implemented.
Factorial maze functor	Principal-jet theorem plus three-sector embedding.	Opaque upload/execute stages and five-identity bridge seal.	Implemented.
Wilson observation	Native rank binding, complement pairing and Wilson's theorem.	Composite-safe modular polynomial backend.	Implemented.
Cyclic closure	Character decomposition, exact-period valuation and transport.	Bounded dense/jet evaluator only.	Restricted reference.

Statement	Mathematical dependencies	Executable obligation	Status
Least-factor depth	Binomial divisibility and Gaussian channel recurrence.	Finite room identities in Volume V verifier.	Proved; verifier support.
F036 barrier	Least-factor depth, filtered compiler class and charged Hasse order.	No claim outside the compiler class.	Proved scoped barrier.
Tri-class child boundary	Frozen parent axes, source-generated histories and ordered corridor.	Same-map certificate and no target inversion.	Research verifier line.
Residual stability	$p < E < 2p$ and frozen canonical source-head law.	Verify every retained packet across child extension.	Verifier-only.
Zipper no-recomputation Horizon complexity	Stability plus depth-first paid-packet lifecycle. Disjoint deficient intervals and explicit machine costs.	Every node refined at most once. Work/live counters correspond to theorem terms.	Verifier-only. Theorem + finite audit.
$p^{1/2+o(1)}$ shadow	Exact horizon theorem plus external analytic estimates.	Must remain labelled amortized, not cold.	Boundary interpretation.
Cold $p^{o(1)}$ generator	Would require collective descriptor-native source morphism escaping F042.	No shipped or verifier implementation.	Open.

How to read the matrix

“Implemented” never means that finite tests prove the universal theorem. It means the theorem has a proof and the finite program has independent conformance evidence on the exercised domain. “Verifier-only” never means a public SDK API. “Open” never means that a pseudocode name has solved the interface.

B Proofs for the higher-dimensional Angel Prime atlas

This section proves the universal statements introduced only after the executable and Volume V main lines. None of the proofs uses finite probe agreement as a premise.

B.1 The procyclic inverse system and source atlas

Full proof of Theorem 73.2. For $d \mid m$, the assignment $S_m \mapsto S_d$ respects the relation $S_m^m = 1$ because

$$S_d^m = (S_d^d)^{m/d} = 1.$$

It therefore induces a surjective ring map $\rho_{m,d} : \mathbb{Z}[C_m] \rightarrow \mathbb{Z}[C_d]$. If $e \mid d \mid m$, both $\rho_{d,e}\rho_{m,d}$ and $\rho_{m,e}$ send the generator S_m to S_e , so the maps form an inverse system.

The family $(S_m)_m$ is compatible and hence defines $\hat{S}$. Ring operations in an inverse limit are

coordinatewise. Therefore

$$\pi_m \left(\prod_{k=1}^n (1 - \widehat{S}^k) \right) = \prod_{k=1}^n (1 - S_m^k) = P_{m,n},$$

and the identical argument gives the formula for $\widehat{J}_n$. Finally, the polynomial identity

$$1 - X^k = (1 - X)(1 + X + \cdots + X^{k-1}) = (1 - X)[k]_X$$

holds over $\mathbb{Z}[X]$. Multiplying it for $1 \leq k \leq n$ gives

$$\prod_{k=1}^n (1 - X^k) = (1 - X)^n \prod_{k=1}^n [k]_X = (-1)^n (1 - X)^n J_n(X).$$

Substitute $X = \widehat{S}$ to obtain the claimed identity in the inverse limit. $\square$

Full proof of Proposition 73.3. The self-room family uses the prefix $m - 1$ in room m . Let $d \mid m$ with $m > d$. Since $1 \leq d \leq m - 1$, the product K_m contains the factor $1 - S_m^d$. Its image under $\rho_{m,d}$ is

$$1 - S_d^d = 0,$$

so $\rho_{m,d}(K_m) = 0$. On the other hand, the primitive-projector theorem gives $K_d = d\Pi_d^{\text{prim}}$, which is nonzero in characteristic zero. Thus $(K_m)_m$ violates the compatibility equation required of an inverse limit point.

It is nevertheless a well-defined section of the family whose fibre over m consists of room- m prefix operators: it assigns the dependent value $(m, m - 1, K_m)$ to each base index. Remembering the prefix restores the missing type information. This proves both assertions. $\square$

The nested q-Pochhammer quotient tower used in Definition 74.1 is also well defined: the factorization $F_{n+1} = F_n(1 + Zu^n)$ implies $(F_{n+1}) \subseteq (F_n)$, so reduction modulo (F_n) is a canonical ring map from $B/(F_{n+1})$ to $B/(F_n)$, and these reductions compose by uniqueness.

B.2 Every coefficient of the factorial jet

Full proof of Theorem 75.2. For each $k \geq 1$,

$$\begin{aligned} 1 - (1 + T)^k &= - \sum_{h=1}^k \binom{k}{h} T^h \\ &= -kT \sum_{j=0}^{k-1} \frac{1}{k} \binom{k}{j+1} T^j = -kT A_k(T). \end{aligned}$$

Multiplication from $k = 1$ through n yields

$$Q_n(T) = (-1)^n \left(\prod_{k=1}^n k \right) T^n \prod_{k=1}^n A_k(T) = (-1)^n n! T^n B_n(T).$$

Therefore $a_{n,r} = (-1)^n n! b_{n,r}$.

To extract $b_{n,r}$, choose from the k th factor A_k a monomial of degree r_k . The total degree is r

exactly when $r_1 + \cdots + r_n = r$, and the selected coefficient is

$$\prod_{k=1}^n \frac{1}{k} \binom{k}{r_k + 1}.$$

Terms outside $0 \leq r_k \leq k - 1$ vanish. Summing over all weak compositions gives the displayed partition formula.

Since $B_{n+1} = B_n A_{n+1}$, coefficient convolution gives

$$b_{n+1,r} = \sum_{j=0}^{\min(r,n)} [T^{r-j}] B_n [T^j] A_{n+1} = \sum_{j=0}^{\min(r,n)} b_{n,r-j} \frac{1}{n+1} \binom{n+1}{j+1}.$$

It remains to derive the first two higher orders. The linear coefficient of A_k is $(k-1)/2$, so

$$b_{n,1} = \sum_{k=1}^n \frac{k-1}{2} = \frac{n(n-1)}{4}.$$

The quadratic coefficient of A_k is $(k-1)(k-2)/6$. Hence

$$b_{n,2} = \sum_{k=1}^n \frac{(k-1)(k-2)}{6} + \sum_{1 \leq i < j \leq n} \frac{(i-1)(j-1)}{4}.$$

The first sum is $n(n-1)(n-2)/18$. For the second, use

$$\sum_{i < j} (i-1)(j-1) = \frac{1}{2} \left(\left(\sum_{i=1}^n (i-1) \right)^2 - \sum_{i=1}^n (i-1)^2 \right).$$

Substituting the elementary sums and simplifying gives

$$b_{n,2} = \frac{n(n-1)(9n^2 - 5n - 26)}{288}.$$

Multiplication by $(-1)^n n!$ proves all three displayed coefficient formulas. Finally, integrality follows independently from the original product $Q_n(T) \in \mathbb{Z}[T]$; no divisibility claim is being inferred from the rational normalization alone. $\square$

For the Bell-polynomial organization, the formal logarithm is valid because $B_n(0) = 1$. The defining generating identity

$$\exp \left(\sum_{j \geq 1} \kappa_j \frac{T^j}{j!} \right) = \sum_{r \geq 0} Y_r(\kappa_1, \dots, \kappa_r) \frac{T^r}{r!}$$

then identifies coefficients directly.

B.3 Ramanujan coordinates of the primitive projector

Full proof of Theorem 76.2. Let $\zeta_m = e^{2\pi i/m}$. The Fourier idempotent for the character $S_m \mapsto \zeta_m^a$ is

$$e_a = \frac{1}{m} \sum_{j=0}^{m-1} \zeta_m^{-aj} S_m^j.$$

The primitive-period projector is the sum over $(a, m) = 1$:

$$\Pi_m^{\text{prim}} = \sum_{(a,m)=1} e_a = \frac{1}{m} \sum_{j=0}^{m-1} \left(\sum_{(a,m)=1} \zeta_m^{-aj} \right) S_m^j.$$

The inner sum is $c_m(-j) = c_m(j)$, because the reduced residue system is closed under negation. Multiplying by m and using $K_m = m\Pi_m^{\text{prim}}$ gives

$$K_m = \sum_{j=0}^{m-1} c_m(j) S_m^j.$$

For completeness, impose the condition $(a, m) = 1$ by Möbius inversion:

$$\mathbf{1}_{(a,m)=1} = \sum_{h|(a,m)} \mu(h).$$

Then

$$\begin{aligned} c_m(j) &= \sum_{a=1}^m e^{2\pi i a j / m} \sum_{h|(a,m)} \mu(h) \\ &= \sum_{h|m} \mu(h) \sum_{\substack{1 \leq a \leq m \\ h|a}} e^{2\pi i a j / m}. \end{aligned}$$

Writing $a = hb$, the inner geometric sum has value m/h if $(m/h) \mid j$ and zero otherwise. Hence

$$c_m(j) = \sum_{t|\gcd(m,j)} t \mu(m/t).$$

This standard divisor form is equivalent, with $g = \gcd(m, j)$, to

$$c_m(j) = \mu(m/g) \frac{\varphi(m)}{\varphi(m/g)};$$

if m/g is not squarefree both sides are zero, while the squarefree case follows by comparing prime factors.

For prime p , $c_p(0) = \varphi(p) = p - 1$ and $c_p(j) = \mu(p) = -1$ for $j \not\equiv 0 \pmod{p}$. A circulant with this row is $pI - \mathbf{1}\mathbf{1}^T$.

Conversely, if the row has that pure form, the corresponding matrix has rank $m - 1$. But $K_m = m\Pi_m^{\text{prim}}$ has rank $\varphi(m)$. Therefore $\varphi(m) = m - 1$, which for $m \geq 2$ forces m to be prime. This also proves the stated “if and only if.” $\square$

B.4 Laplacian, spectrum, and divisor support

Full proof of Theorem 77.1. The primitive-projector theorem diagonalizes K_m : it acts by m on the $\varphi(m)$ primitive characters and by zero on the remaining $m - \varphi(m)$ characters. Thus

$$\text{rank } K_m = \varphi(m), \quad \text{nullity } K_m = m - \varphi(m),$$

and its characteristic-zero spectrum is

$$\{0^{(m-\varphi(m))}, m^{(\varphi(m))}\}.$$

The equality $\varphi(m) = m - 1$ is equivalent to primality for $m \geq 2$. This proves the equivalence of items one, two, three and six.

By [Theorem 76.2](#), prime m gives $K_m = mI - \mathbf{1}\mathbf{1}^T$ and the pure row. Either matrix or row identity has rank $m - 1$, so its converse follows from the preceding paragraph. This proves items four and five.

Finally, the time-valuation support is

$$\{(d, m/d) : d \mid m\}.$$

It has only the two boundary divisors $1, m$ exactly when m has no proper nontrivial divisor, again exactly when m is prime. This proves item seven and closes the equivalence chain. For a prime, $mI - J$ is the standard complete-graph Laplacian by its diagonal degree $m - 1$ and off-diagonal edge weights -1 . $\square$

B.5 The graph/cyclotomic numerical bridge

Full proof of [Theorem 77.2](#). The Laplacian $K_p = pI - J$ has nonzero eigenvalue p with multiplicity $p - 1$. The matrix-tree theorem states that every principal cofactor equals the product of the nonzero Laplacian eigenvalues divided by the number of vertices:

$$\det K_p^{[i]} = \frac{p^{p-1}}{p} = p^{p-2}.$$

The same cofactor counts spanning trees, giving Cayley's value p^{p-2} .

For the cyclotomic side, $\Phi_p(x) = (x^p - 1)/(x - 1)$. The discriminant of a monic polynomial is, up to the standard sign, the product of its derivative at its roots. At a nontrivial p th root ζ ,

$$\Phi_p'(\zeta) = \frac{p\zeta^{p-1}}{\zeta - 1}.$$

Multiplying over the $p - 1$ nontrivial roots gives absolute value

$$\frac{p^{p-1}}{\left| \prod_{a=1}^{p-1} (\zeta_p^a - 1) \right|} = \frac{p^{p-1}}{p} = p^{p-2},$$

because the cyclotomic product has absolute value p . This is the absolute discriminant of $\mathbb{Q}(\zeta_p)$. The proof compares exact values only and constructs no canonical lattice isomorphism. $\square$

B.6 Collision of the infinitesimal and cyclic prime charts

Full proof of [Theorem 78.1](#). The principal coefficient is the case $n = p - 1$ of [Theorem 75.2](#):

$$[T^{p-1}]Q_{p-1} = (-1)^{p-1}(p-1)!.$$

Wilson's theorem gives $(p-1)! \equiv -1 \pmod{p}$. If p is odd, $(-1)^{p-1} = 1$; if $p = 2$, both 1 and -1 are the same residue. Hence the principal coefficient is -1 modulo p in every prime room.

The Ramanujan-shadow theorem gives the cyclic row $(p-1, -1, \dots, -1)$, and each coordinate reduces to -1 modulo p . Both statements arise by applying different chart homomorphisms to the common source F_{p-1} , followed by reduction to $\mathbb{F}_p$. No equality of their unreduced carriers is

needed. □

The corollary follows because Wilson's theorem characterizes primes and [Theorem 77.1](#) independently characterizes primes by the pure Laplacian. The certificates remain different because their chart maps and retained histories differ.

B.7 Higher tensor and feature observers

Full proof of Theorem 79.1. Tensoring the identity $K_m = m\Pi_m^{\text{prim}}$ r times gives

$$K_m^{\otimes r} = m^r (\Pi_m^{\text{prim}})^{\otimes r}.$$

Over a field, ranks multiply under tensor products, so

$$\text{rank } K_m^{\otimes r} = (\text{rank } K_m)^r = \varphi(m)^r.$$

For prime p , $K_p = pI - J$, hence reduction modulo p gives $K_p \equiv -J$. Tensoring commutes with reduction and

$$K_p^{\otimes r} \equiv (-J)^{\otimes r} \pmod{p}.$$

Every entry of $J^{\otimes r}$ is one, proving the entrywise value $(-1)^r$. The conclusion is scoped to this defined tensor-power observer. □

Full proof of Theorem 79.2. The map $d \mapsto (d, m/d)$ from divisors of m to time-valuation pairs is injective because its first coordinate is d . Therefore the support size of $\mathbb{T}_m$ is exactly the divisor count $\tau(m)$. It equals two exactly when the only divisors are 1 and m , which is equivalent to primality. In that case the multiplicities are $\varphi(1) = 1$ and $\varphi(m) = m - 1$.

If m is composite, each proper divisor $1 < d < m$ gives $1 < d < m$ and $1 < m/d < m$, an interior point distinct from every other divisor's point. The enhanced packet $\mathbb{G}_{m,N}$ retains the first coordinate d , so attaching a leading germ coefficient cannot identify or remove two different strata. □

B.8 Finite observers cannot recover the procyclic point

Full proof of Theorem 80.1. Let L be the least common multiple of the finitely many observed room sizes. The polynomial $X^L - 1$ evaluated at $\hat{S}$ defines an element $a = \hat{S}^L - 1$ of the inverse limit. In room m_i ,

$$\pi_{m_i}(a) = S_{m_i}^L - 1 = 0$$

because $m_i \mid L$. Choose a prime q not dividing L . Then S_q has exact order q , so $S_q^L \neq 1$ and $\pi_q(a) \neq 0$. Thus a is a nonzero inverse-limit element in the kernel of every observed projection. Any map factoring through those projections has the same value on a and zero and is not injective. □

B.9 Restricted spectral division

Full proof of Theorem 81.1. In the character idempotent basis, write

$$a = \sum_{\chi} a_{\chi} e_{\chi}, \quad b = \sum_{\chi} b_{\chi} e_{\chi}.$$

Then

$$bq = \sum_{b_\chi \neq 0} b_\chi \frac{a_\chi}{b_\chi} e_\chi = \sum_{b_\chi \neq 0} a_\chi e_\chi,$$

while

$$r = \sum_{b_\chi = 0} a_\chi e_\chi.$$

Adding gives $a = bq + r$, and multiplying r by b gives zero sectorwise.

If q' is supported only where $b_\chi \neq 0$ and $bq' = a - r$, then on each such sector $b_\chi q'_\chi = a_\chi$, forcing $q'_\chi = a_\chi/b_\chi = q_\chi$. If r' is kernel-supported and $a = bq + r'$, comparison on a zero sector forces $r'_\chi = a_\chi = r_\chi$. This proves uniqueness.

A room map that carries the declared idempotent splitting to the declared splitting and does not merge a zero eigenvalue with a nonzero one carries the two sectorwise formulas to the same formulas in the target. That is the stated restricted functoriality. Without this preservation hypothesis the spectral support may change, so no unrestricted claim follows. $\square$

B.10 Path-module interference and future safety

Full proof of Theorem 82.1. The free module construction makes addition of weighted paths literal module addition. The endpoint map is the unique linear map sending each basis path to its complete endpoint basis state. Therefore paths ending at the same state have their integer weights added, and a packet cancels structurally exactly when its endpoint sum is zero, that is, exactly when it lies in $\ker E_X$.

Let f be a common legal continuation. Postcomposition induces a linear map $f_* : \mathbb{Z}[\text{Path}_X] \rightarrow \mathbb{Z}[\text{Path}_X]$ on paths and a linear map on endpoint states. For every basis path $\gamma : X \rightarrow Y$,

$$E_X(f_*[\gamma]) = [f(Y)] = f_*E_X([\gamma]).$$

The square commutes on a basis and hence on the entire free module. It follows that $f_*(\ker E_X) \subseteq \ker E_X$: structural cancellation is stable under common continuation.

If one first applies ordinary projection, distinct complete endpoints can become equal. Their difference then lies in the kernel of the projected endpoint map but need not lie in $\ker E_X$. A later legal continuation that distinguishes the endpoints makes the projected cancellation invalid. Thus ordinary cancellation is future safe only when the projection is proved future sufficient for the language. $\square$

C Axiom, arithmetic, and naming migration ledger

This appendix records where the restored definitions came from and what was added to bind them to the executable SDK. The source is the exact attached file `Angel_Prime_Mathematical_Annals_Integrated_Volumes_I_IV.tex`, dated 20 August 2026. The source is preserved in the release archive under `paper/source_material/`. The present paper normalizes dispersed source material into one axiom and operation chapter; it does not silently attribute new SDK claims to the earlier volume.

The residual-horizon extension is derived from the separately supplied Volume V v5.9 strict research package. Its complete archive, TeX, claim boundary, traceability ledger and strict verifier are preserved under `paper/source_material/`. The factorial-maze bridge, by contrast, is new in

this revision and is explicitly marked as such.

The Angel naming account and the SN–OFCS chronology are authorial provenance reported for this revision. They are not documentary mathematical sources for an equation. In particular, the complete formula remembered as having appeared in the originating dream is no longer available for textual comparison. The present source family is therefore described as a rigorous reconstruction from the surviving structure and iteration, never as a verified verbatim transcription of the forgotten display.

Table 16: Source-to-annals migration map.

Integrated-volume source	Restored object	Present destination
A conservative three-class history arithmetic	Unitized algebra $R \oplus \mathfrak{h}$, ordinary embedding/projection, ghost-history ideal	Axioms 13.1 and 13.2 and Theorem 13.11
Primitive and singular history directions	$\text{Ord} \oplus \text{Hist} \oplus \text{Sing}$ decomposition and non-field warning	Axiom 13.1 and Equation (HMUL)
Why ordinary division by zero is rejected	No scalar totalization satisfying the classical inverse law	Axiom 13.6 and Definition 14.4
Totalized singular quotients	$a \oslash_A b = (ab^\#, aK_b)$, reconstruction, solution fibre, $a \oslash_A 0$	Definition 14.4 and Theorem 14.5
History-sensitive equality and information compression	Structural versus ordinary equality and future sufficiency	Axioms 13.3 and 13.7
Tri-sector primitive-history arithmetic	Ordinary, exact-period history, and inherited singular sectors	Axiom 13.1 ; cyclic specialization remains in Section 44
The exact Angel transition operator	Complete state-to-state causal evolution	Axiom 13.4 and Definition 15.1
The trivariate Angel field	Separation of iteration, ordinary position, and vertical history depth	Represented here by distinct maze address, epoch, and history coordinates in Definition 11.1 , Section 17 , and Table 22
Tri-class mixed arithmetic algebra	Complete three-sector state and totalized division summary	Sections 13 and 14
Deterministic local maps and quantum-like path packets	Formal weighted path family and sign-reversing involution	Axiom 13.8 , Definition 16.1 , and Theorem 16.4
Ringed causal connections and future equivalence	Path composition, holonomy, and causal quotient principle	Definitions 11.1, 15.2 and 15.3

Integrated-volume source	Restored object	Present destination
New SDK binding in this revision	Sparse HADD/HSUB/H-MUL/HDIV, maze state, functor action, structural interference, tensor-minor certificate	Section 17 and Theorem 17.1
New factorial bridge in this revision	Three-sector principal-jet chart, specialized factorial execution functor and two opaque post-execution consumers	Section 32 and Theorem 33.1
Volume V v5.9 strict source	History-Curvature normal form, horizon source-program stability, Causal Horizon Zipper, residual no-recomputation and amortized work theorem	Section 63 and Theorems 64.2 and 67.2
Author’s project history	Dream presentation of a structure and high-dimensional formula; later recognition of a prime formula after a workday lunch nap; complete display not retained	Section 11 ; explicitly excluded from mathematical premises
SN–OFCS precursor vocabulary	Singular-number carrier, Omega future continuation, typed zero-division payload and singular-space engineering	Section 12 and Theorem 14.7 ; historical mapping plus a new conservative-extension theorem
Surviving remembered skeleton	Structure and iteration, without a recoverable verbatim mother formula	q-Pochhammer/principal-jet/cyclic construction treated as the current proved reconstruction, not as uniquely recovered memory
Fold-multiplication recollection	Hierarchical interval multiplication as the first geometric view of factorial	ordered fold trees, contraction functor, nodewise jet signatures and the scoped fold–shadow collision in Sections 18 and 19

Migration honesty rule

An earlier definition is cited as mathematical provenance. A C++ type or test is called implemented only when it exists in the accompanying revision. The general spectral noncentral inverse, arbitrary-precision signed public chart, and native prime-birth functor are therefore retained as open interfaces rather than retroactively marked complete.

Memory/reconstruction boundary

Three claims must remain separate. First, the author recalls an originating episode and retained structural programme. Second, SN-OFCS documents the project's conceptual route toward singular payloads and future computation. Third, the present memoir defines concrete algebras, functors, source families and observers and proves theorems about them. Only the third layer carries mathematical proof authority. No match between the first and third layers is used to fill a missing hypothesis, and no experimental agreement proves that the lost formula was unique.

D Pseudocode for the executable path

D.1 Certified q-Pochhammer source

Listing 1: Publication-facing pseudocode for the compact source program.

```
Program BuildQpochProgram(BigNat n) {
  if (n == 0) return Program{UNIT};

  Program p{SEED_FACTOR};          // represented length = 1
  for (bit : remaining_bits_from_msb(n)) {
    p.push(DOUBLE_WITH_CONTEXT_SHIFT); // length a -> 2a
    if (bit == 1)
      p.push(APPEND_SHIFTED_SINGLETON); // 2a -> 2a+1
  }
  return SealAndCertify(n, p);
}
```

D.2 Native factorial state

Listing 2: Native factorial-state construction.

```
FactorialState AngelFactorial(BigNat n) {
  CertifiedProgram source = Certify(BuildQpochProgram(n));
  Interval interval = ExactInterval(0, n);
  PrincipalJet jet = BindPrincipalJet(
    source,
    interval,
    valuation = n,
    sign = parity(n));
  return CertifyLiveState(source, jet);
}
```

D.3 Joint external Wilson observer

Listing 3: Composite-safe joint time-space observation.

```
Observation JointWilsonObserve(NativeFactorialView native, Modulus m) {
  proof = IndependentlyReplayNativeFactorial(native);
  Require(proof.accepted && proof.coefficient == 1);
}
```

```

n = proof.native_rank;           // factor count comes from native state
Require(m == n + 1);             // m is modulus + consistency constraint
h = floor(n / 2);
w = ChooseBoundedNearSquareWidth(h);

P = ProductTreePolynomial({X+1, ..., X+w}, modulus=m);
half = 1 mod m;
if (h != 0) {
  half *= ConstantCoefficient(P); // P(0), no point leaf/Horner pass
  for (range : TwoSequentialRanges(1, floor(h/w))) {
    tree = BuildPointProductTreeOnDemand(range, stride=w, modulus=m);
    DescendRemaindersLeftThenRight(
      P, tree,
      on_value = [&](value) { half = half * value mod m; },
      release_completed_nodes = true);
  }
  half *= Product(floor(h/w)*w + 1, ..., h) mod m;
}

residue = half * half mod m;
if (h is odd) residue = -residue mod m;
if (n is odd) residue = residue * (h+1) mod m;

Require(NativeStateUnchanged(native));
return Observation{residue, prime = (residue == m-1)};
}

```

E Formal boundary invariants

The publication-facing protocol is summarized by the following invariants.

Invariant	Requirement
Known-input-only ingress	Upload receives only already-known input and a task specification. It cannot preload $n!$, a residue, a Wilson bit, answer-derived branch advice or an ordinary trace that reconstructs the answer.
Complete-state entry	Native execution consumes the complete certified state, never a residue, hash, view or projection as a replacement.
Native factor source	The modular consumer obtains its factor count from the independently verified native factorial rank; the candidate is only the modulus and the check $m = n + 1$.
High-to-high core	Every semantic step between ingress and egress has Angel-state inputs and outputs, including typed continuations.
Non-terminal factorial	The factorial result remains usable by native state operations and is not required to download immediately.
Single ordinary egress	Ordinary integers, residues and decisions leave only through the observation boundary.
No observation feedback	A residue, decision or resource-limit result cannot rewrite, merge, prune, select or advance the native state.

Invariant	Requirement
Derived-map non-authority	A displayed or transported observation is accepted only by comparison with the canonical closed state and cannot become an update to that state.
Equality separation	Structural equality, observer equality and future equivalence are distinct relations.
Separate ledgers	Upload, native execution, download binding, external evaluation and full bit materialization are reported separately; deterministic work and peak live space are both recorded.
No hidden materialization	The joint Wilson path stores neither the full factorial nor input-scale point/value arrays; materialized coordinate count is explicit.

F Computational evidence

This appendix contains all finite evidence used to assess the supplied implementation. It is intentionally outside the theorem narrative. A row marked *pass* means that one executable condition held for the stated toolchain and input campaign; it does not prove an infinite mathematical statement.

F.1 Evidence hierarchy

Table 18: Claim classes and their admissible support.

Claim class	Primary support	What the support cannot establish
Algebraic identity	theorem and proof	memory safety of a C++ lowering
Asymptotic bound	explicit model and derivation	a favorable finite timing curve outside that model
Implementation equivalence	independent differential oracle	correctness on all inputs
State firewall	types, negative compilation and runtime integrity checks	the algebraic identity being observed
Release reproducibility	hashes, commands and captured logs	mathematical novelty or optimality

F.2 Verification matrix

Table 19: Executable verification matrix.

Campaign	Acceptance condition	Release status
Debug/release equivalence	complete test transcripts are byte-identical at $-O0$ and $-O3$	PASS
Address and undefined behavior	all positive tests complete under ASAN and UBSAN; leak scope stated separately	PASS

Campaign	Acceptance condition	Release status
Negative compilation firewall	all fourteen prohibited ascent, shortcut, stage-skip or direct candidate-consumption programs fails to compile	PASS
Frozen-source verification	protected source and legacy public headers match their frozen hashes	PASS
Three-class arithmetic laws	HADD/HSUB are coordinatewise exact; HMUL is associative and distributive on the finite sparse model; singular words remain in a two-sided ideal	PASS
HDIV reconstruction	central-scalar quotient and typed residual reconstruct every tested numerator, including the zero-denominator packet	PASS
Causal functor laws	the empty program is identity and concatenated programs equal sequential execution in value, ordered events and state seal	PASS
Class-quantum histories	structural interference groups only complete equal endpoints; exact opposite weights cancel and equal ordinary shadows alone do not	PASS
Entanglement certificate	separable tensors have rank at most one over $\mathbb{Q}$ and a declared entangled packet supplies a nonzero exact 2×2 minor	PASS
Exact factorial oracle	sequential and independent product-tree big-integer derivations agree	PASS
Factorial-maze bridge	upload and execution preserve the sealed principal-jet core, bind the four uploaded history words plus one execution word, distinguish equal ordinary shadows with different factorial histories, and admit exact/joint consumers only after execution	PASS
Wilson differential campaign	legacy, native-bound, paired and independent exact residues agree on the continuous and structured domains	PASS
Polynomial-ring differential	transform/CRT products and fast monic remainders equal independent schoolbook references at composite and near-maximum word moduli	PASS
Parallel/serial transforms	per-prime parallel and serial schedules return identical polynomials	PASS
Derived-map tamper rejection	a modified primitive observation copy is rejected and the canonical state summary is unchanged	PASS
Joint finite Pareto gate	residue equality and all selected strict work/peak/coordinate inequalities hold on every frozen row	PASS
Wilson structural atlas	4,095 optimized packets agree with independent primality, Euler- φ and two-boundary divisor support	PASS

Campaign	Acceptance condition	Release status
Cyclic Ramanujan shadow	255 directly convolved rows are pure Laplacian exactly in the 54 prime rooms; all 54 prime coordinate folds give -1 modulo the room	PASS
Period-rank filtration	547 modular circulant ranks equal the exact-period divisor formula	PASS
Higher factorial jets	64 principal and first higher coefficient identities agree modulo the audit prime	PASS
Volume V v5.9 strict verifier	four frozen horizons emit 2,119 identical high-dimensional child weights in zipper and cold-reference modes; every residual packet passes the stability check and retained siblings are reused	PASS (separate research verifier)

F.3 Deterministic finite resource ledger

The work unit W is the published deterministic sum of charged ring, polynomial and scalar events. The peak quantity P is the conservative scratch-inclusive live-coefficient bound. The coordinate count C measures persistent scalar slots at the external schedule boundary. In the corrected baseline,

$$C_0 = s_0 + 1 + 2\lfloor n/s_0 \rfloor;$$

the two block-sized arrays are evaluation points and block values. Tail factors are streamed time events and are not stored slots.

Table 20: Deterministic Pareto sequence regenerated by the release build.

m	W_0	W_1	W_1/W_0	P_1/P_0	C_1/C_0	status
257	807	383	0.4746	0.3256	0.1837	yes
1021	2661	1251	0.4701	0.3333	0.1895	yes
4093	9570	4297	0.4490	0.3260	0.2147	yes
16381	35811	16068	0.4487	0.3285	0.2376	yes
65521	137686	61142	0.4441	0.3301	0.2360	yes
100003	208616	92350	0.4427	0.3281	0.2363	yes
250003	514444	226954	0.4412	0.3301	0.2360	yes
500009	1013841	449704	0.4436	0.1917	0.2362	yes
1000003	1527021	979902	0.6417	0.5280	0.2360	yes

F.4 Interpretation of the finite ratios

Every displayed row lies below the Pareto boundary 1 in the three recorded ratios. This establishes a reproducible strict improvement for those inputs under the stated counters. It does not establish $T_1 = o(T_0)$, $S_1 = o(S_0)$, or an information-theoretic lower bound. The formal body therefore states only the structural reduction from the n -scale block problem to the $h = \lfloor n/2 \rfloor$ scale and retains the same asymptotic classes.

`allocation_count` has a deliberately narrow meaning: it counts tracked outer product/remainder-schedule objects. It is not a process-wide interposition count for every allocator invocation.

Transform and reconstruction scratch is instead included in the conservative peak bound.

F.5 Polynomial and state-integrity adversaries

The direct polynomial test uses an independent schoolbook multiplication routine and an independent monic long-division routine. It forces the transform/CRT branch at a composite target modulus and at a modulus close to the largest 64-bit unsigned value. It then runs the same inputs with per-prime parallelism disabled and enabled. Equality is required coefficient by coefficient.

The observation-integrity adversary closes a canonical primitive observation, downloads a derived packet, and first verifies equality. It then flips one entry in a copied primitive column. Validation must fail on the copy, while the canonical state's order, cycle count, session and jet horizon remain unchanged. The test addresses authority direction: a displayed map is evidence read from closure, never a write source for closure.

F.6 Cyclic reference campaigns

The bounded cyclic reference implementation checks primitive-projector identities over orders 2 through 1024, normalized kernel identities over orders 2 through 256 in two finite audit fields, and representative prime, semiprime, prime-power and Carmichael orders. Bounded closure jets verify the time-valuation spectra and quotient transport on selected divisibility maps. These calculations test the reference evaluator; the spectral and valuation theorems are proved independently in [Appendices A.8 to A.10](#) and [A.12](#).

F.7 Higher-dimensional Wilson structural campaigns

The additive structural probe produces four complete raw tables under `results/`:

- `angel_structure_wilson_probe.csv`: 4,095 supplied candidates;
- `angel_structure_cyclic_shadow_probe.csv`: 255 cyclic rooms;
- `angel_structure_period_rank_probe.csv`: 547 prefix ranks; and
- `angel_structure_higher_jet_probe.csv`: 64 jet orders.

The first campaign preserves the public native-coordinate Wilson path. Trial primality, φ , divisor counts, direct cyclic convolution and Gaussian rank are computed only after the observation is frozen and never feed the state. All 4,961 primary records pass. Their aggregate ledger is `results/ANGEL_STRUCTURE_PROBE_SUMMARY.txt`.

The campaigns are not used as proofs of the procyclic, Ramanujan, Laplacian or higher-jet theorems. Those proofs appear in [Appendices A.5](#) and [B.1 to B.4](#).

F.8 Volume V residual-horizon research verifier

The separately frozen v5.9 verifier constructs the high-dimensional zipper before invoking its independent ordinary audit. The principal run has four horizon cases, 2,119 emitted critical edges, 1,068 refinements, 93,682 zipper tail inspections, 22,764 stability checks, 19,581 reused full packets and 382 peak live residual packets. Its cold source-compiler reference performs 14,720,643 tail

inspections on the same child-weight output, a finite saving of 14,626,961 inspections under the verifier’s exact counters. The $K = L, L^2, L^3$ runs have the same child-weight hash.

These numbers support the implemented lifecycle and the finite instance only. The general stability and amortized results are proved in [Appendices A.17](#) and [A.18](#). Source, claim boundary, traceability table, logs and sanitizer records are retained with the research input; the public factorial SDK does not export this verifier as an API. The exact rerun transcript is preserved at `results/VOLUME_V_V5_9_VERIFIER_COMPLETE.log`.

G Engineering audit and reproducibility

G.1 Audit findings and repairs

Table 21: Material findings from the code/manuscript audit.

Finding	Why it mattered	Repair in this revision
Coordinate undercount	the reconstructed baseline owns both a point array and a block-value array, but the old formula charged only one block-sized array and mixed streamed tail work into stored space	changed the definition to block coefficients plus twice the full-block count; tail factors remain charged as time events
Indirect transform coverage	transform/CRT multiplication and fast remainder were reached through a larger Wilson campaign but lacked independent local oracles	added deterministic differential tests against separately written schoolbook references
Parallel-path gap	the default per-prime parallel path had no direct equality test against the serial path	added serial/parallel coefficientwise equivalence as a release gate
Derived map without explicit validator	ordinary observation types were terminal, but a copied observation vector had no public comparison routine against canonical closure	added <code>validate_download_packet</code> and a tamper/immutability test
Ambiguous allocation label	the published counter could be read as every process heap allocation	documented its exact outer-schedule scope and charged transform scratch in the peak bound
Mixed publication roles	the original article interleaved theorem/proof, CI status, hashes, sanitizer facts and finite counters at one narrative level	expanded the expository body and moved every finite engineering claim to these appendices

Finding	Why it mattered	Repair in this revision
Missing unified Angel axiom layer	the source volumes contained history, singular quotient, causal and path-packet ingredients, but did not state one three-class axiom system or bind all four operations and functors to an executable public model	added ten explicit high-dimensional axioms, HADD/HSUB/HMUL/HDIV, native/history/ future-quotient functors, class-quantum history definitions, a relative model proof and a finite exact SDK realization
Missing factorial-to-maze bridge	the short edition proved a generic three-class algebra and later used a principal-jet factorial state without embedding the latter into the former	added a four-word uploaded chart, specialized high-to-high factorial functor, bridge certificate, two legal executed-state consumers, positive tests and two new negative-compilation barriers
Unexplained three-sector “dimension”	readers could mistake three semantic coordinates for a three-dimensional Euclidean claim and could not see why the split was forced	constructed the free path-algebra maze category, separated sector number from linear rank, proved a scoped minimal semantic-separation theorem and documented every maze coordinate’s duty
Terminal-only Angel Prime story	factorial/Wilson decided a supplied candidate but did not explain how history could participate in later prime-state transitions	integrated the verifier-only Volume V v5.9 residual-horizon theorem with its F041/F042 barriers, exact amortized ledger and surviving cold interface
Compressed and inaccurate origin account	the short naming paragraph replaced the author’s actual recollection by a different two-iteration anecdote, omitted SN–OFCS and did not distinguish the forgotten formula from the current reconstruction	recorded the dream/lunch-nap recognition account as authorial provenance, added the SN–OFCS migration ledger, proved conservative typed zero division, explained why factorial was the first tractable probe, and kept every recollection outside theorem premises
Binary-program diagram	two append edges were drawn as impossible direct jumps $2 \mapsto 5$ and $10 \mapsto 21$	restored the required intermediate doubles $2 \mapsto 4$ and $10 \mapsto 20$

G.2 Mathematics-to-code correspondence

Table 22: Selected formal roles and implementation surfaces.

Formal role	Implementation surface	Audited contract
Binary q-Pochhammer source	CompactQpochProgram and certificate	seed/double/append recurrence and mutation rejection
Factorial interval state	FactorialInterval	endpoint, valuation, parity and principal-coefficient bindings
Factorial maze chart	Coordinate	argument/program/state/request history binding, independent native replay, bridge certificate and unchanged sealed principal-jet core
Executed factorial consumers	factorial_maze::Specification, opaque uploaded/executed states and execute	only the executed carrier is admitted; neither observation converts back to a maze state
Read-only native Wilson coordinate	project_jointly and derive_exact	rank and coefficient are independently replayed before observation
Composite-safe polynomial chart	NativeFactorialView	schoolbook or exact multi-prime transform, followed by integer CRT reconstruction
Terminal residue observation	CompositeSafe	no conversion or operator path back into native execution
Canonical/derived observation comparison	PolynomialRing	complete observation and causal-cut equality; no reverse assignment
Three-class arithmetic	JointWilsonObservation	ordered sparse history words, singular-ideal retention, checked coefficients and typed quotient/residual reconstruction
Native and class-quantum functors	high::TriClassValue, hadd, hsub, hmul, hdiv	identity/composition, structural interference/cancellation and nonzero-minor entanglement certificates
Volume V residual horizon	high::NativeFunctor, ClassQuantumFunctor, TensorHistory	source-first packets, stable horizons, single refinement, certificate transport and post-freeze audit
	separate v5.9 strict theory-engineering verifier	

G.3 Reproduction protocol

From the archive root, the principal release command is

```
./build_and_test.sh
```

It builds debug, optimized and sanitizer variants; compares positive-test transcripts; runs the deterministic probes; requires negative compilation; verifies frozen source and public API compatibility; and writes the final summary under `results/BUILD_AND_TEST_COMPLETE.log`.

A CMake route is also supplied:

```
cmake -S . -B build -DANGEL_BUILD_TESTS=ON
cmake --build build --parallel
ctest --test-dir build --output-on-failure
```

For exact reproduction, the compiler identity and any unavailable sanitizer scope must be read from the archived logs rather than inferred from a generic word such as *tested*.

G.4 Known limitations

- The optimized modular chart is fixed-word; a general L -bit modulus lowering remains unproved and unimplemented.
- The external coordinate remains $\Theta(\sqrt{n})$.
- The finite Pareto campaign is not an asymptotic theorem.
- The cyclic implementation is a bounded reference evaluator, not the compact native compiler proposed by the structural theory.
- The Volume V Causal Horizon Zipper is a separate strict verifier, not an exported module in the factorial SDK; its parent-room interface is frozen and addressable.
- Leak detection is skipped in the restricted execution environment and is not reported as a pass.
- Wilson observation decides a supplied candidate; it is not a competitive general primality routine. The Volume V chart emits causal children only over a paid anchor horizon and is not a $p^{o(1)}$ cold next-prime compiler.
- Generalized HDIV for arbitrary noncentral denominators and unbounded coefficients remains outside the finite public realization.
- Class-quantum history is an exact deterministic path algebra; it supplies no physical Born rule, unitary evolution or quantum-hardware claim.

Audit conclusion

Within the reviewed scope, the implemented arithmetic agrees with independent oracles and preserves the declared state boundary. The material corrections were accounting, direct test coverage, observation-map authority, a missing unified three-class arithmetic/functor layer, the factorial-maze bridge, and publication architecture. No reviewed counterexample invalidated the core q-Pochhammer, principal-jet, complement-pairing, Wilson, cyclic-kernel, closure-valuation, quotient-transport, or scoped residual-horizon identities.

H Notation index

Symbol	Meaning
$F_n(Z, u)$	finite q-Pochhammer section $\prod_{j=0}^{n-1} (1 + Zu^j)$
$\mathfrak{T}_A(R)$	Angel three-sector value module $R \oplus \mathfrak{h}_{\text{Hist}} \oplus \mathfrak{s}$
$X \boxplus_A Y, X \boxminus_A Y$	exact high-dimensional addition and subtraction across Ord/Hist/Sing
$X \boxtimes_A Y$	ordered high-dimensional multiplication with singular two-sided-ideal retention
$X \oslash_A Y$	typed high-dimensional division packet: regular quotient, unresolved residual and reconstruction certificate
$\mathcal{E}, \mathcal{H}, \mathcal{Q}_{\mathcal{L}}$	native execution, history and future-quotient functors
Ψ, Θ	class-quantum path superposition and coupled tensor history
$\Phi[a, b]$	factorial interval $\prod_{k=a+1}^b k$
$\mathcal{I}_n$	uploaded Angel input state with ordinary coordinate n
$\mathcal{Q}_n$	certified factorial state indexed by length n , with ordinary denotation $n!$

Symbol	Meaning
$\mathfrak{F}(J_n)$	three-class factorial maze chart with coefficient one, four uploaded history words and zero successful-chart singular payload
$\mathcal{F}_1$	specialized certified high-to-high factorial execution functor
AFac	native factorial-state operator
$\equiv_{\text{str}}, \equiv_{\text{obs}}, \equiv_{\text{fut}}$	structural, observational and future equivalence
$P(X)$	external block polynomial $\prod_{i=1}^s (X + i)$
A_m	integral cyclic group ring $\mathbb{Z}[C_m]$
V_d	exact-period- d character sector after characteristic-zero extension
$J_{m,n}$	normalized cyclic q-factorial action $(-1)^n \prod_{k=1}^n [k]_S$
K_m	primitive-period operator $\prod_{k=1}^{m-1} (I - S^k)$
$\mathcal{F}_{m,N}$	N -cycle self-closure germ
$\tau_m(d), \nu_m(d)$	first-annihilation time and self-closure depth on V_d
$\mathcal{H}(p, E)$	Volume V anchor horizon $(p, E]$ with $p < E < 2p$
$\mathcal{P}_{p,E}$	paid canonical source program frozen at the anchor parent
$\mathcal{Z}_{p,E}$	K -ary Causal Horizon Zipper of retained paid packets
Σ_q^{res}	state-dependent future-valid residual source payload
$\rho_{p \Rightarrow q}^E$	curvature-normalized direct horizon transport

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